



US 20250283625A1

(19) **United States**

(12) **Patent Application Publication**
SHINDE et al.

(10) **Pub. No.: US 2025/0283625 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **COST SAVINGS FROM FAULT PREDICTION AND DIAGNOSIS**

(71) Applicant: **Tyco Fire & Security GmbH**,
Neuhausen am Rheinfall (CH)

(72) Inventors: **Priti SHINDE**, Vambori (IN);
Kathiresan RAJAGOPAL, Kalyan
West (IN); **Abu Bakr KHAN**, Franklin,
WI (US); **Young M. Lee**, Old
Westbury, NY (US)

(73) Assignee: **Tyco Fire & Security GmbH**,
Neuhausen am Rheinfall (CH)

(21) Appl. No.: **19/212,270**

(22) Filed: **May 19, 2025**

Related U.S. Application Data

(63) Continuation of application No. 18/242,976, filed on Sep. 6, 2023, now Pat. No. 12,305,874, which is a continuation of application No. 17/318,877, filed on May 12, 2021, now Pat. No. 11,859,846, which is a continuation-in-part of application No. 16/198,456, filed on Nov. 21, 2018, now Pat. No. 11,474,485, said application No. 17/318,877 is a continuation-in-part of application No. 16/198,377, filed on Nov. 21, 2018, now Pat. No. 11,604,441, said application No. 17/318,877 is a continuation-in-part of application No. 16/198,416, filed on Nov. 21, 2018, now Pat. No. 11,531,310.

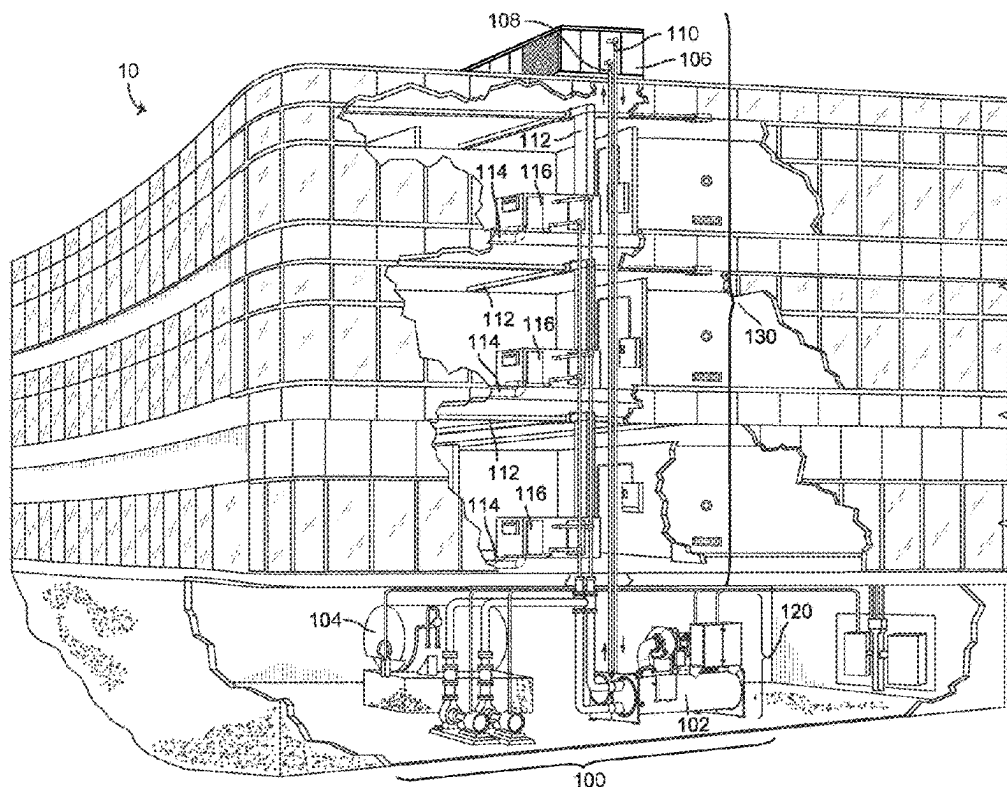
(60) Provisional application No. 62/685,618, filed on Jun. 15, 2018, provisional application No. 62/685,618, filed on Jun. 15, 2018, provisional application No. 62/685,618, filed on Jun. 15, 2018.

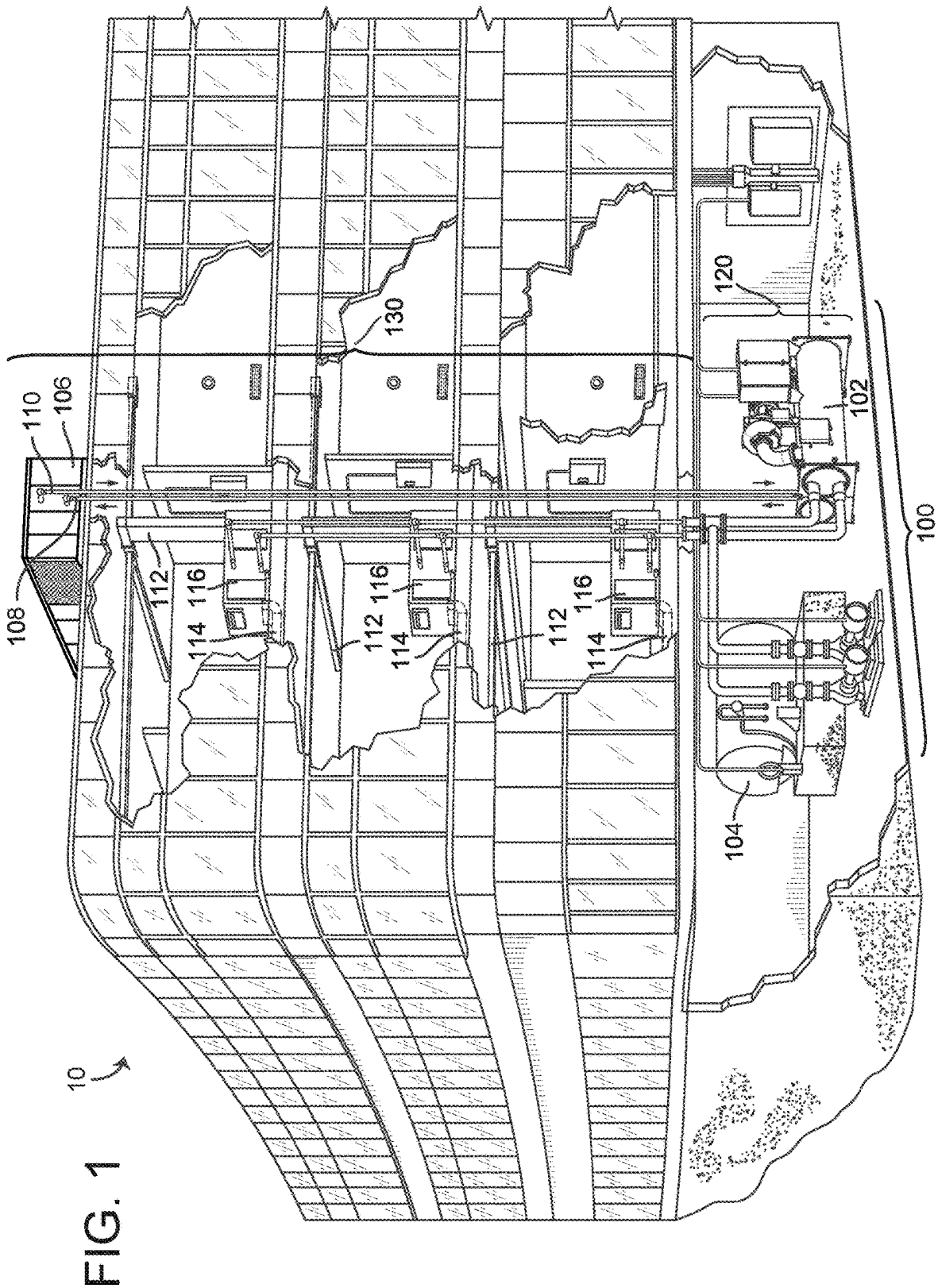
Publication Classification

(51) **Int. Cl.**
F24F 11/63 (2018.01)
G05B 19/042 (2006.01)
G06N 20/10 (2019.01)
G06Q 50/06 (2024.01)
(52) **U.S. Cl.**
CPC **F24F 11/63** (2018.01); **G05B 19/042**
(2013.01); **G06N 20/10** (2019.01); **G06Q**
50/06 (2013.01)

(57) ABSTRACT

A prediction system for a building including a processing circuit including a processor and memory, the memory having instructions stored thereon that, when executed by the processor, cause the processing circuit to receive data relating to a plurality of components, the data indicating performance of the plurality of components, generate, based on the received data, a univariate prediction model and a multivariate prediction model, generate, using the received data, one or more predicted operational parameters for the plurality of components corresponding to a future time period, and execute at least one of the univariate prediction model or the multivariate prediction model on the one or more predicted operational parameters to predict a fault associated with at least one of the plurality of components to occur during the future time period.





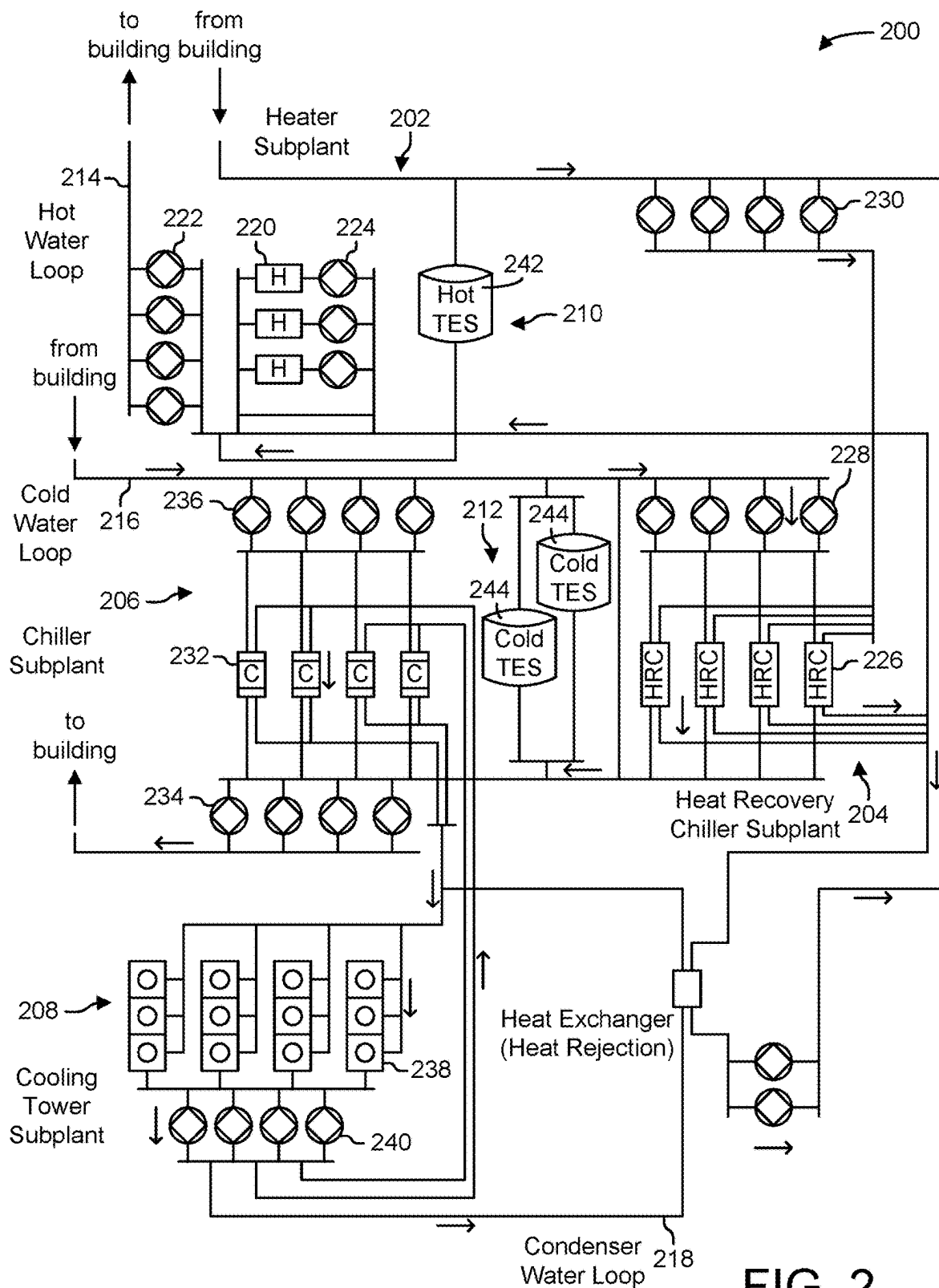
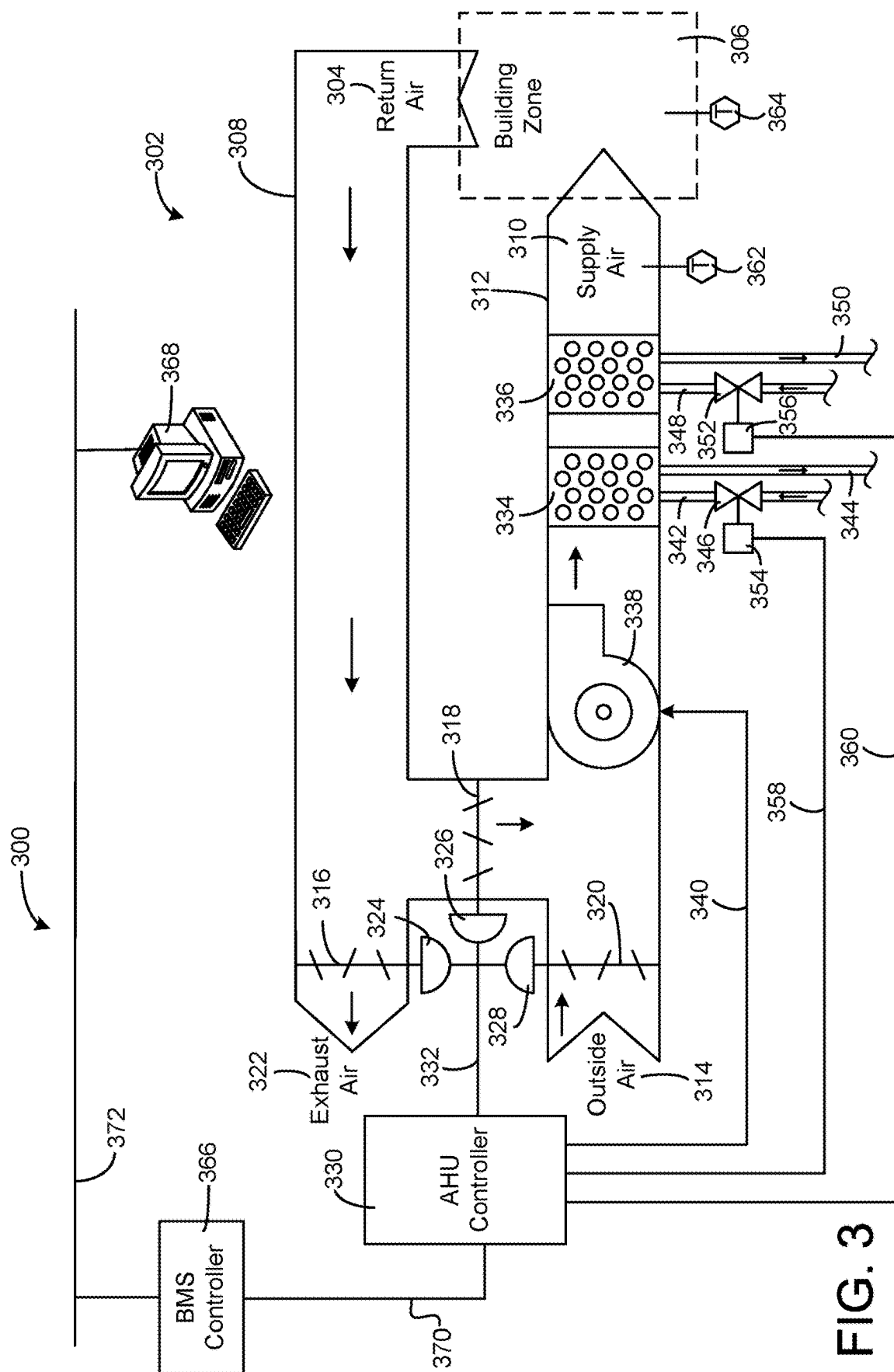


FIG. 2



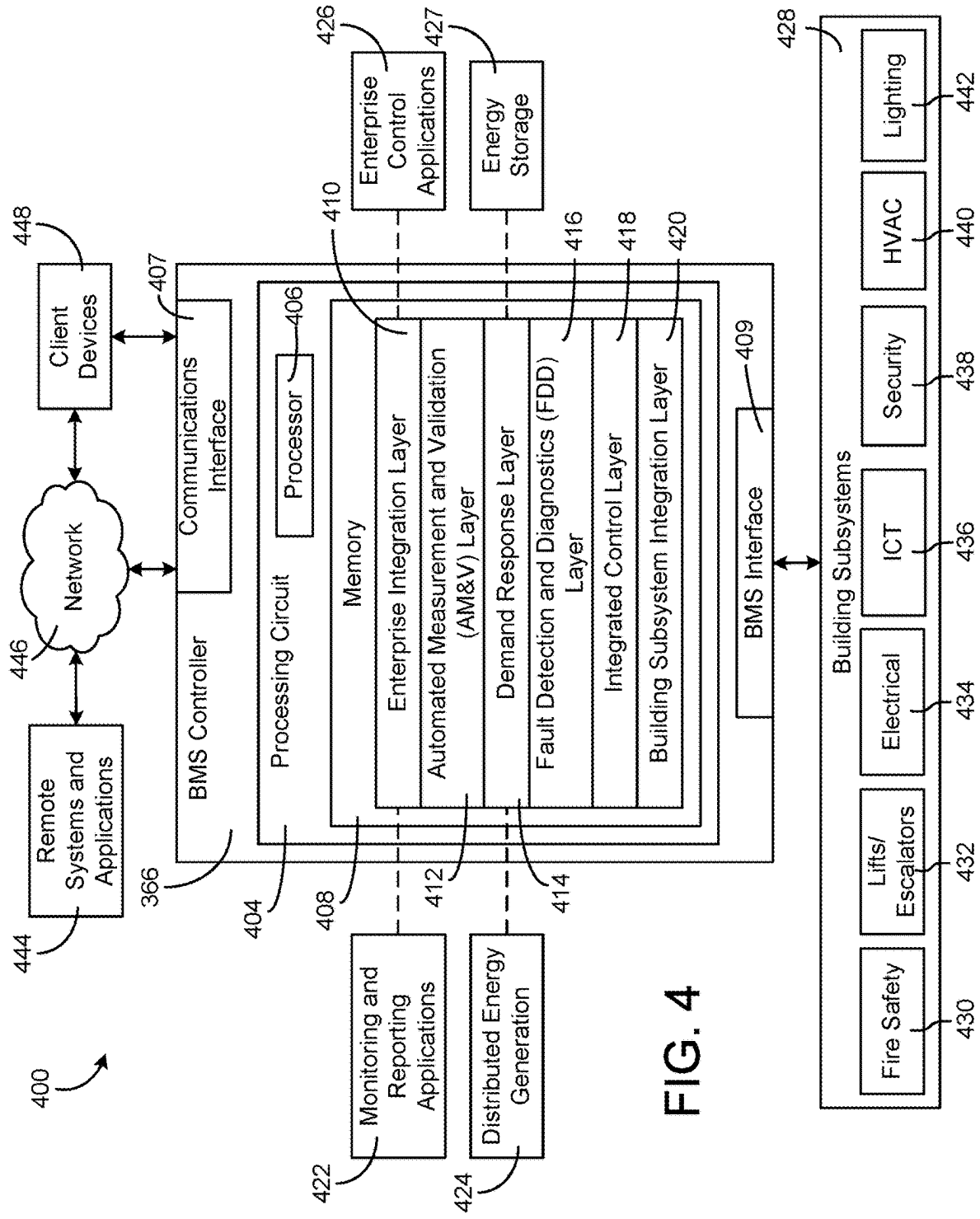


FIG. 4

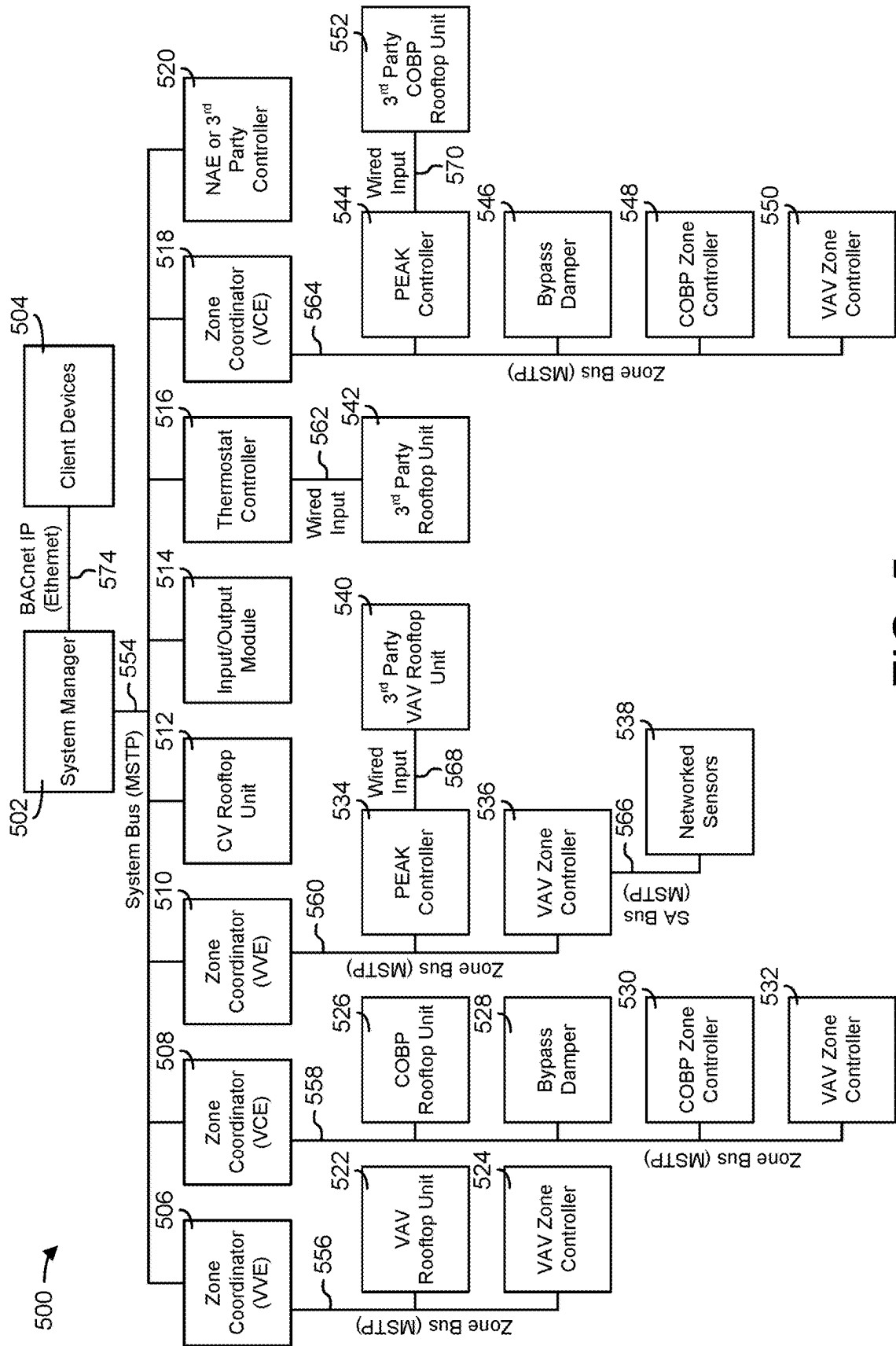


FIG. 5

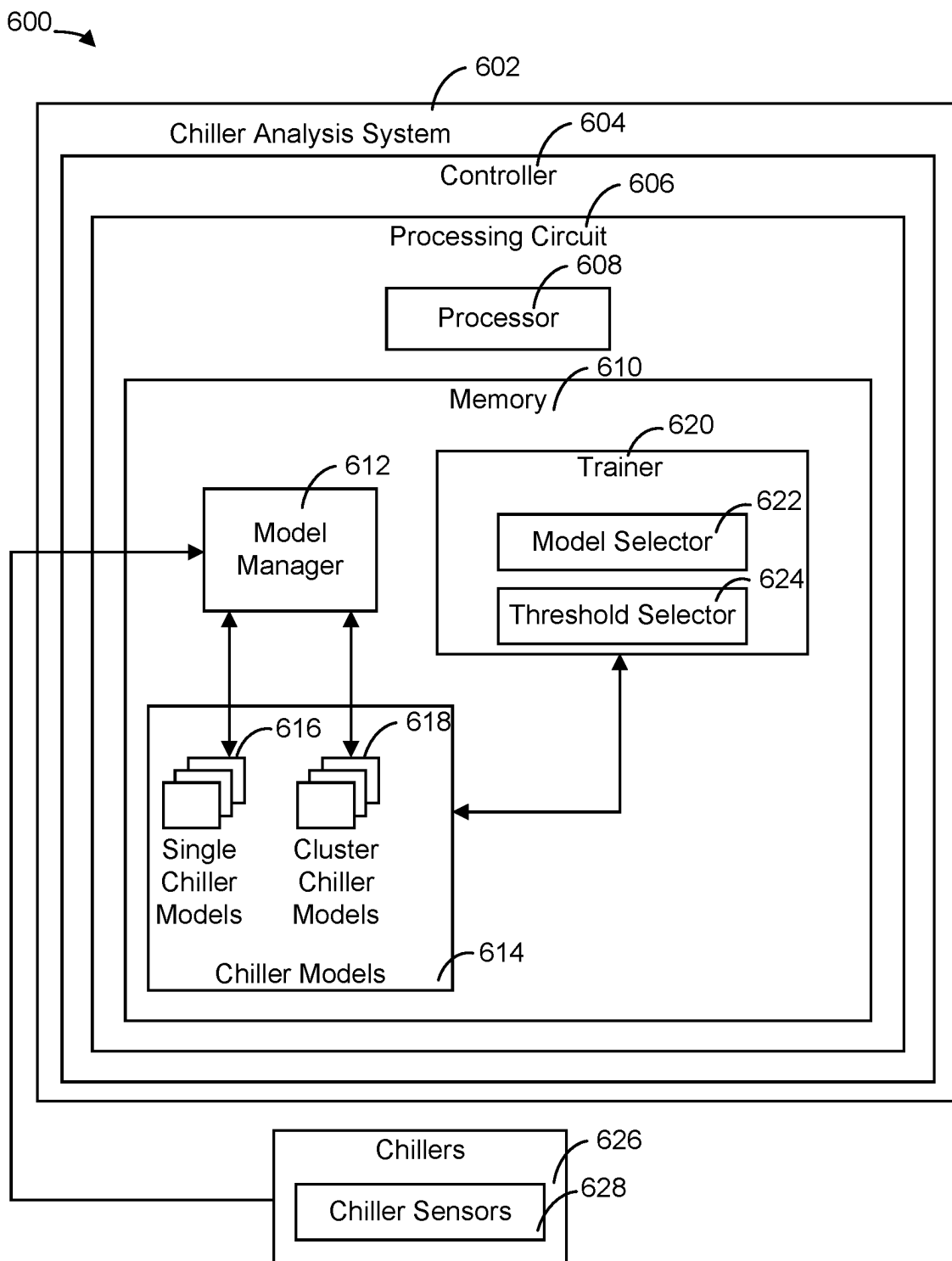


FIG. 6

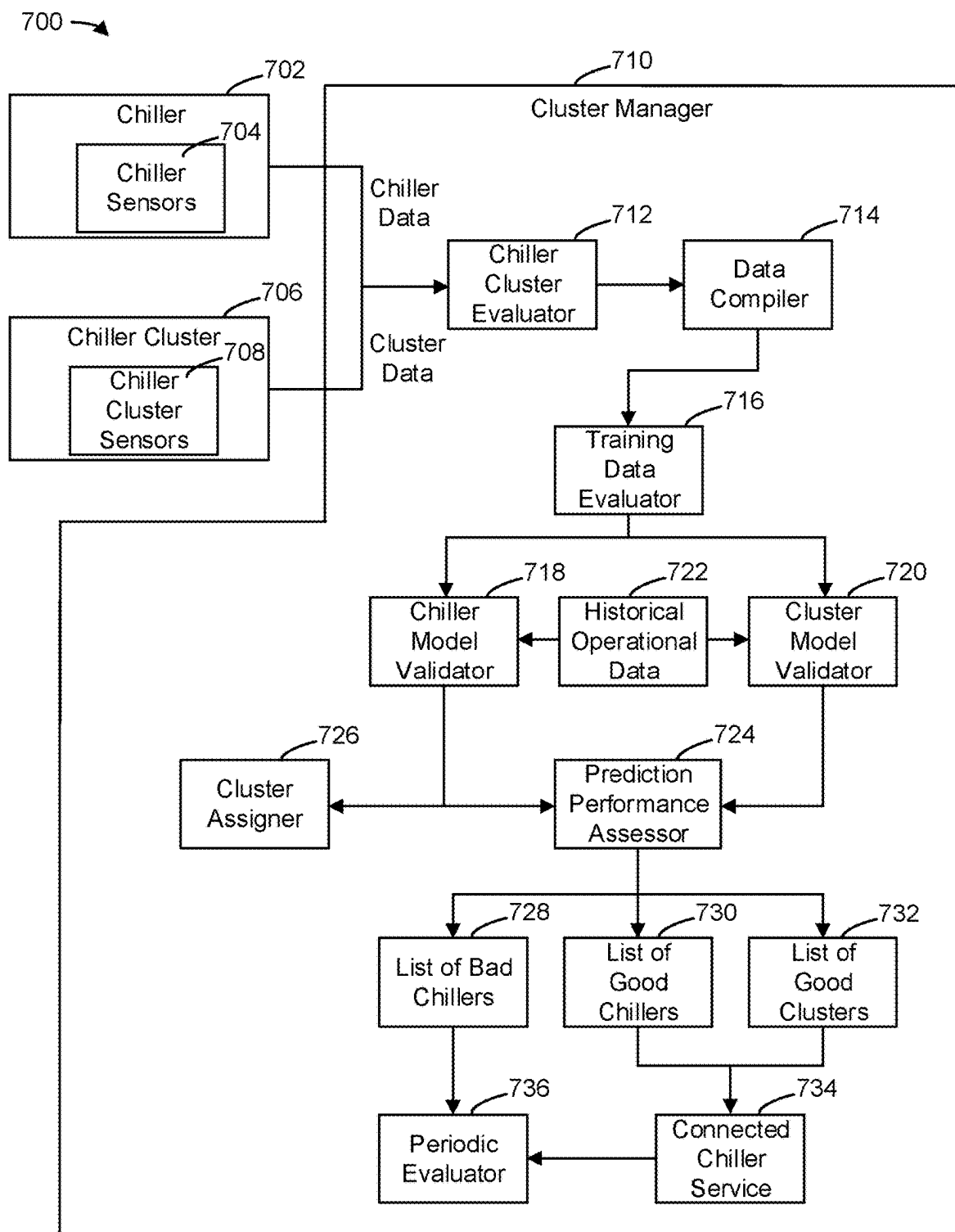


FIG. 7A

740 →

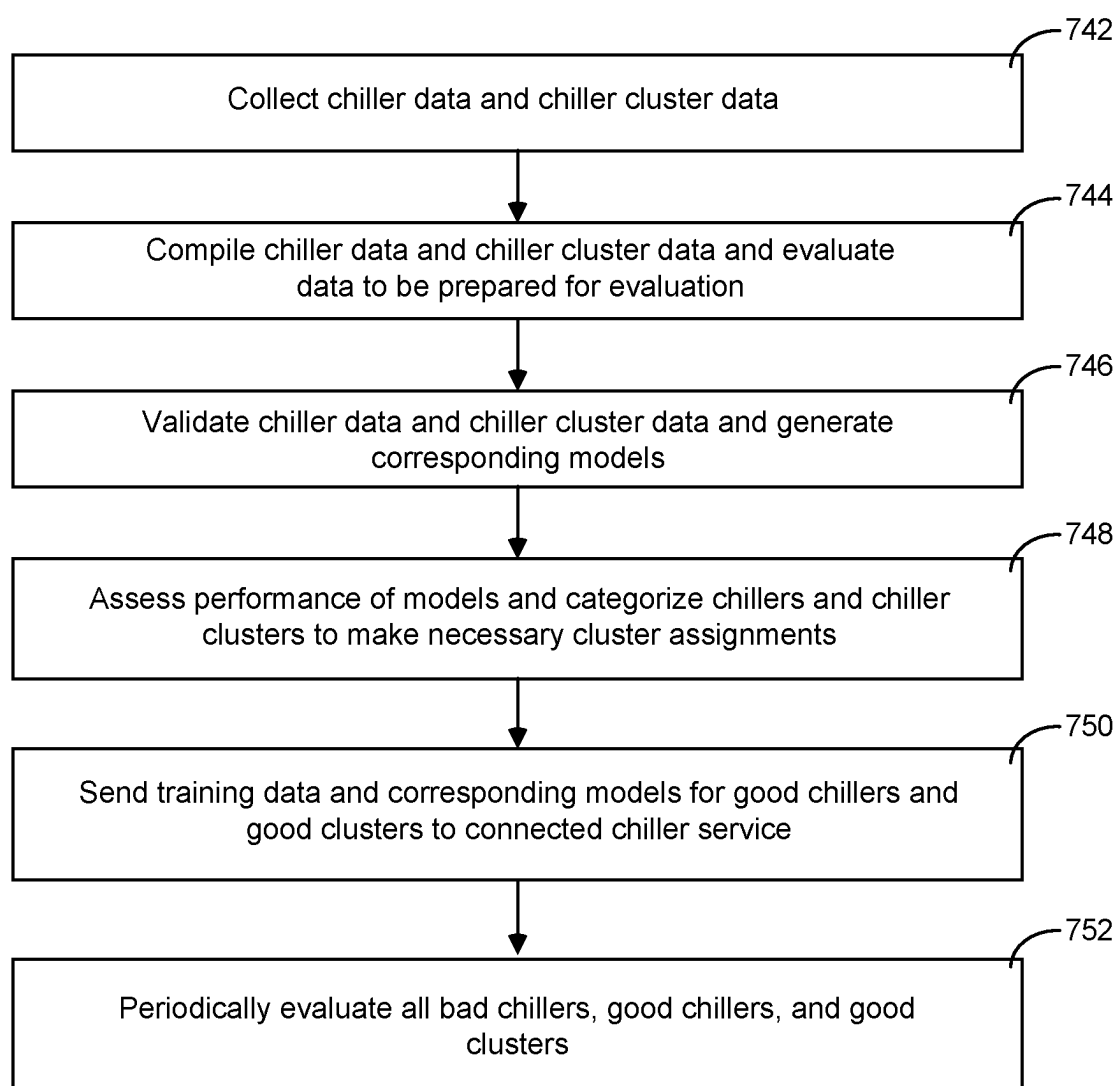
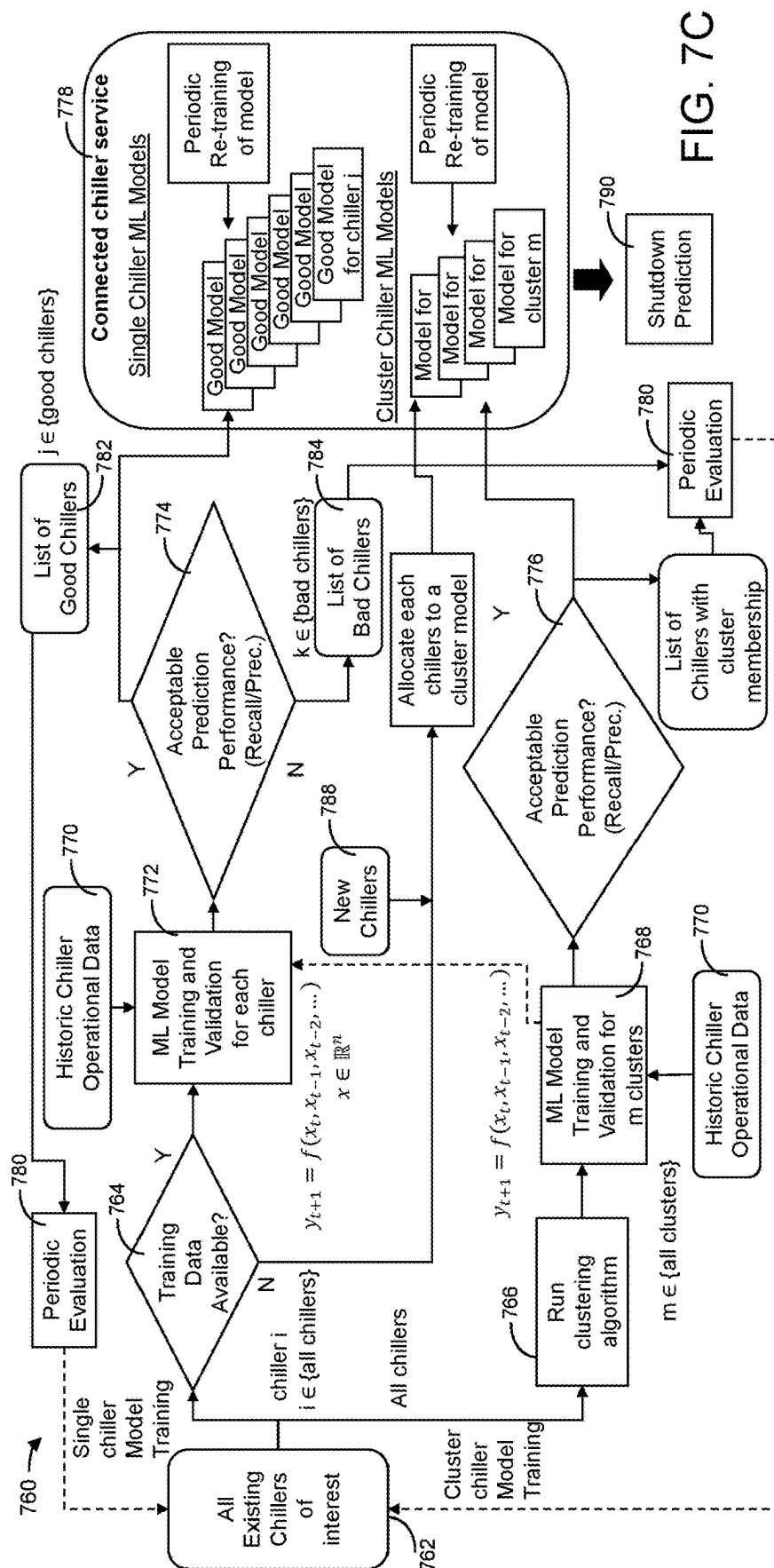


FIG. 7B



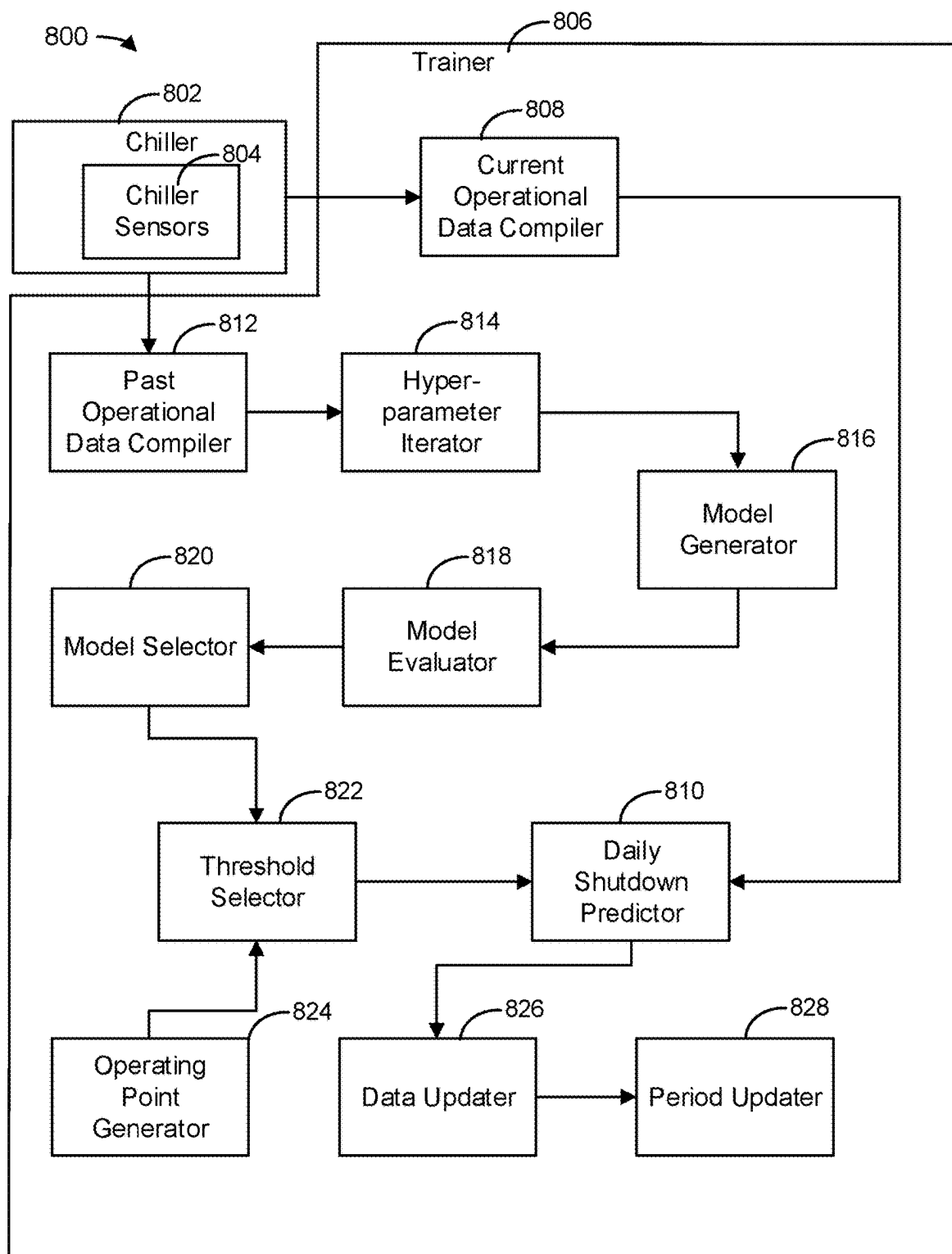


FIG. 8A

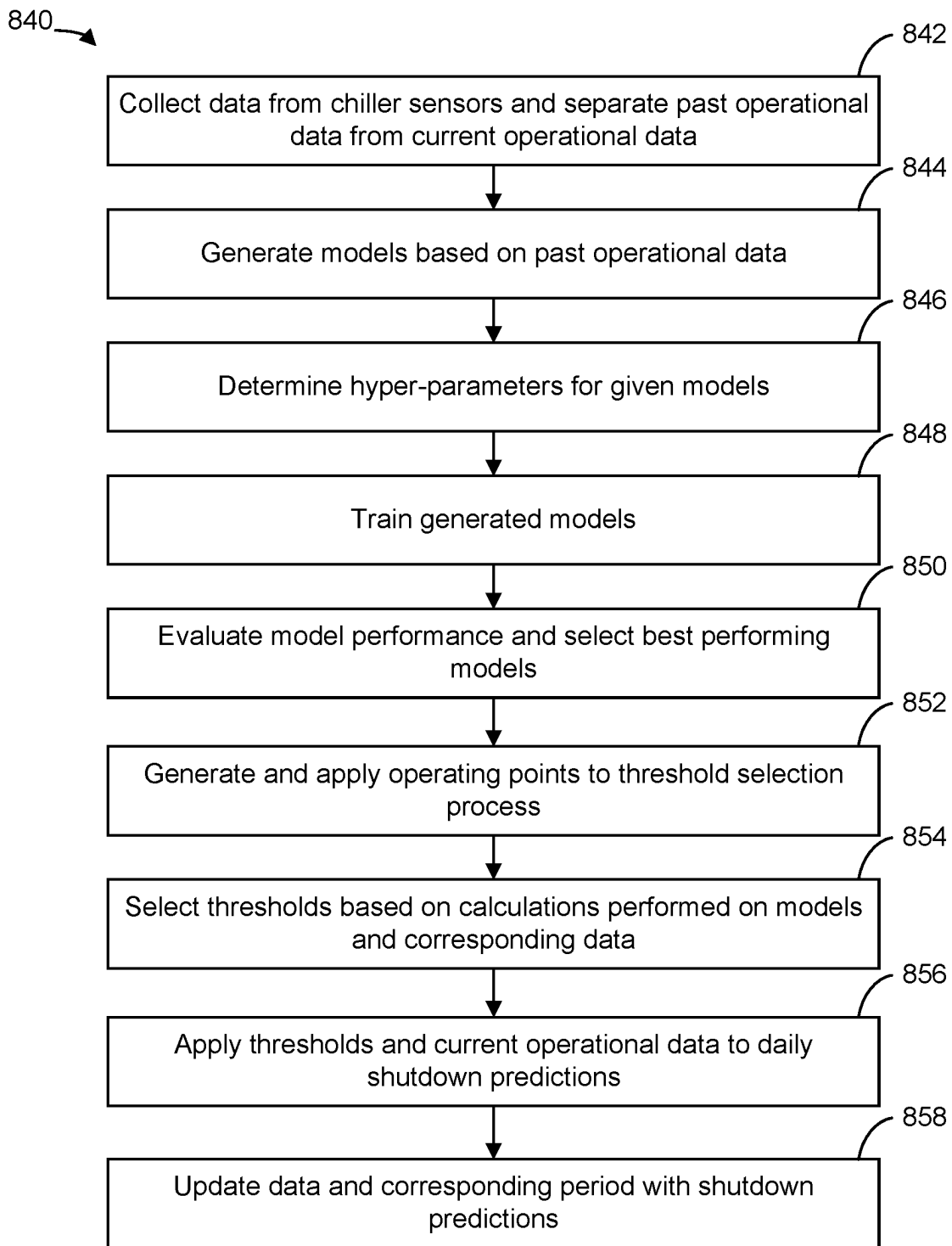
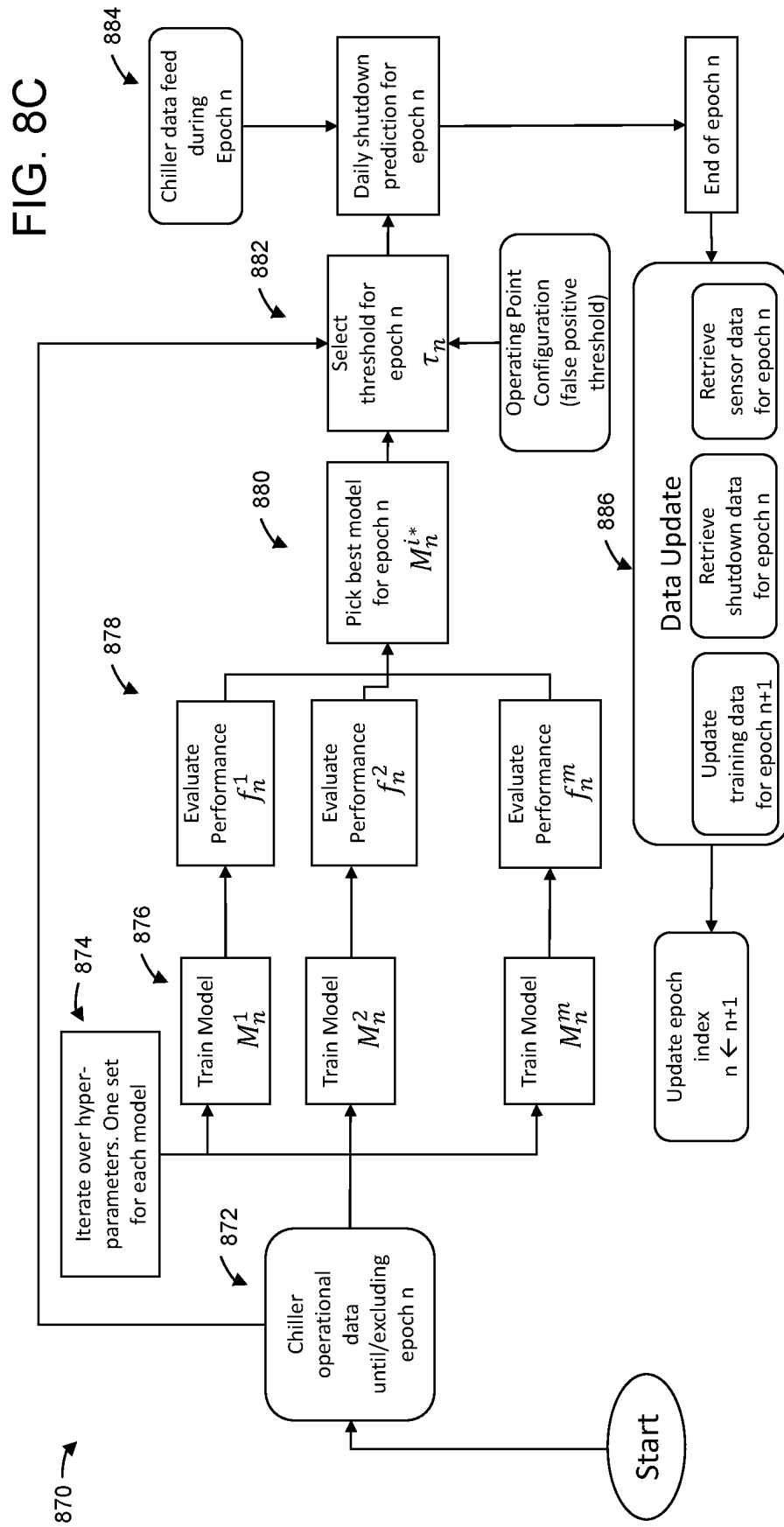


FIG. 8B



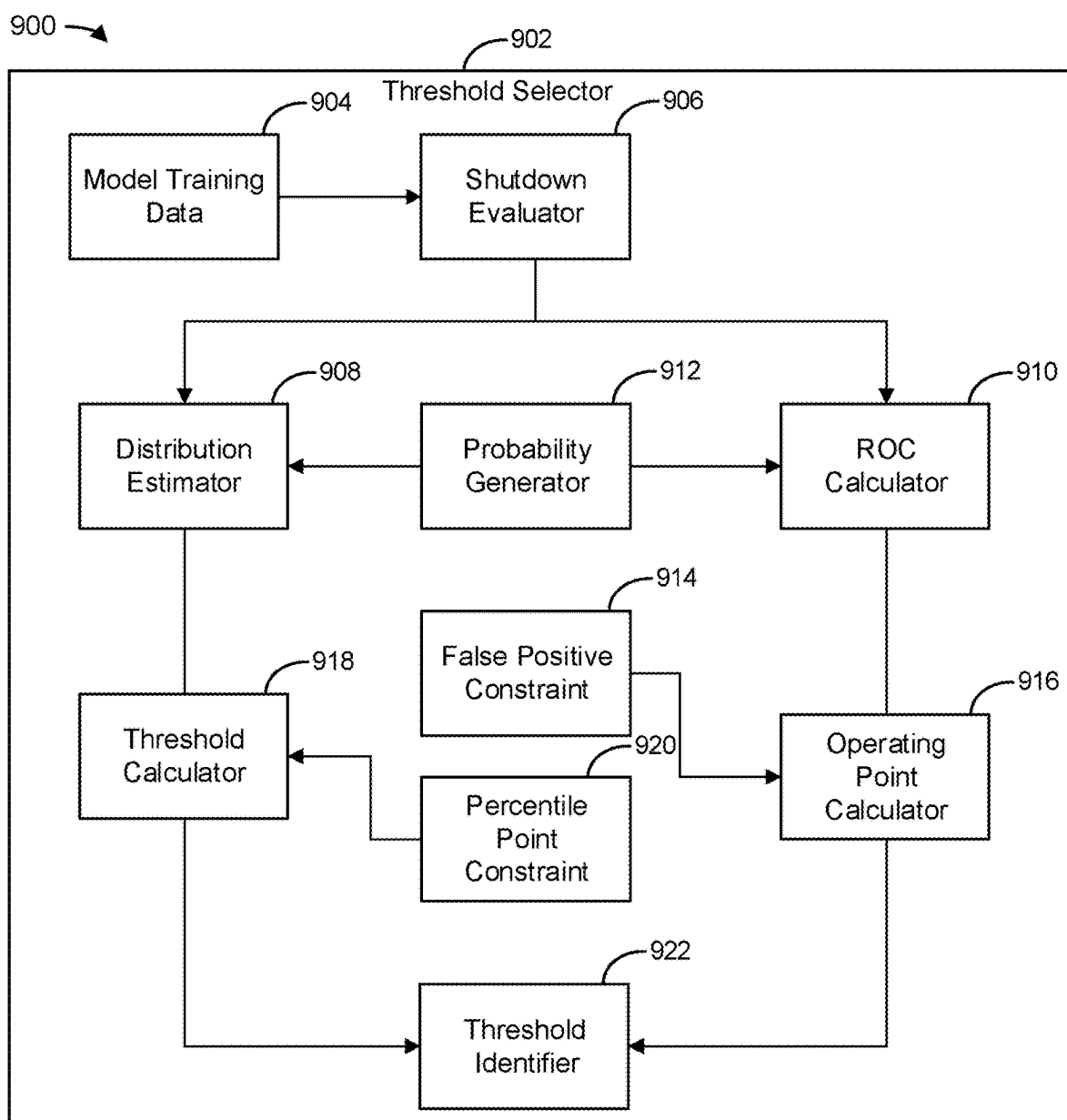


FIG. 9A

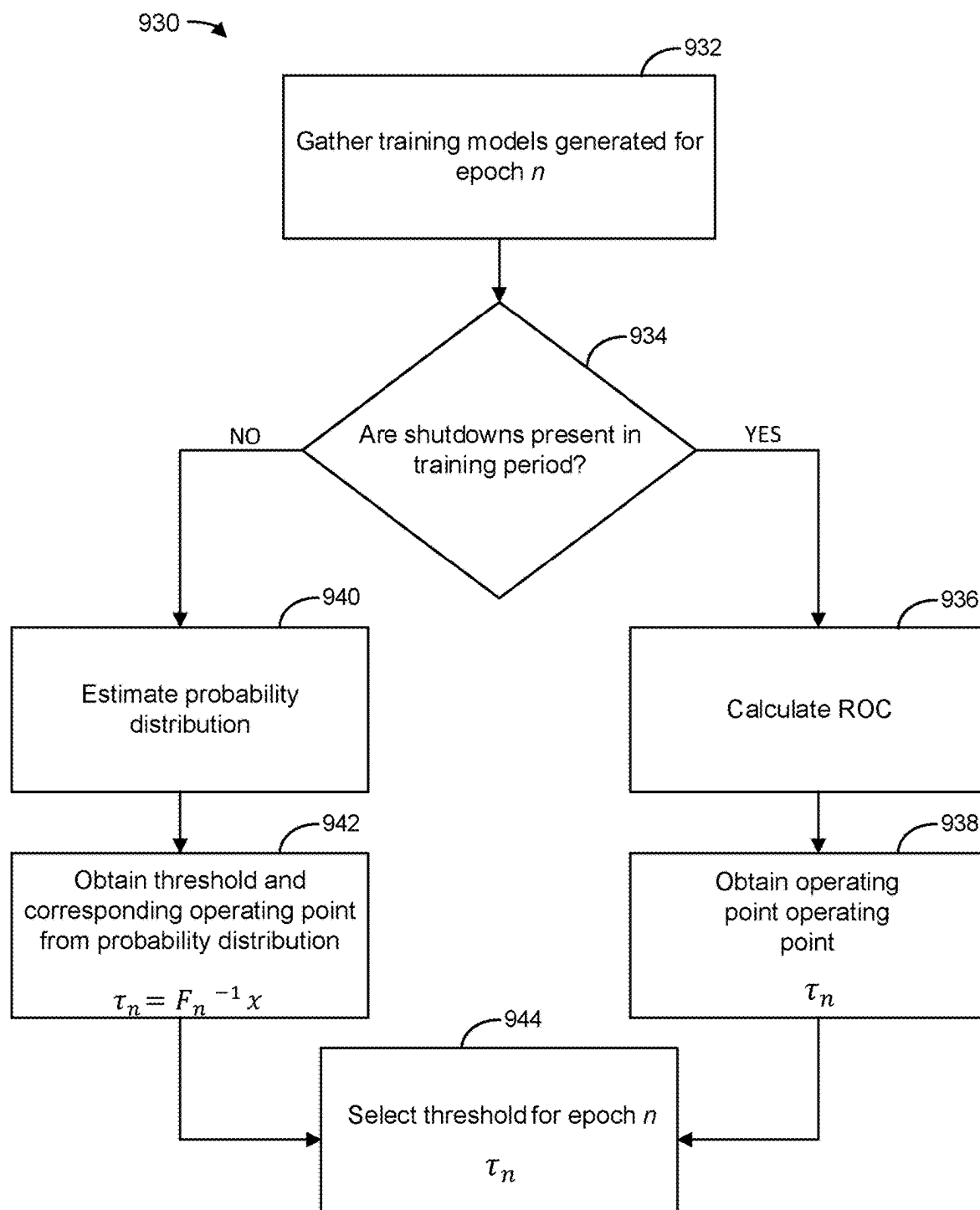
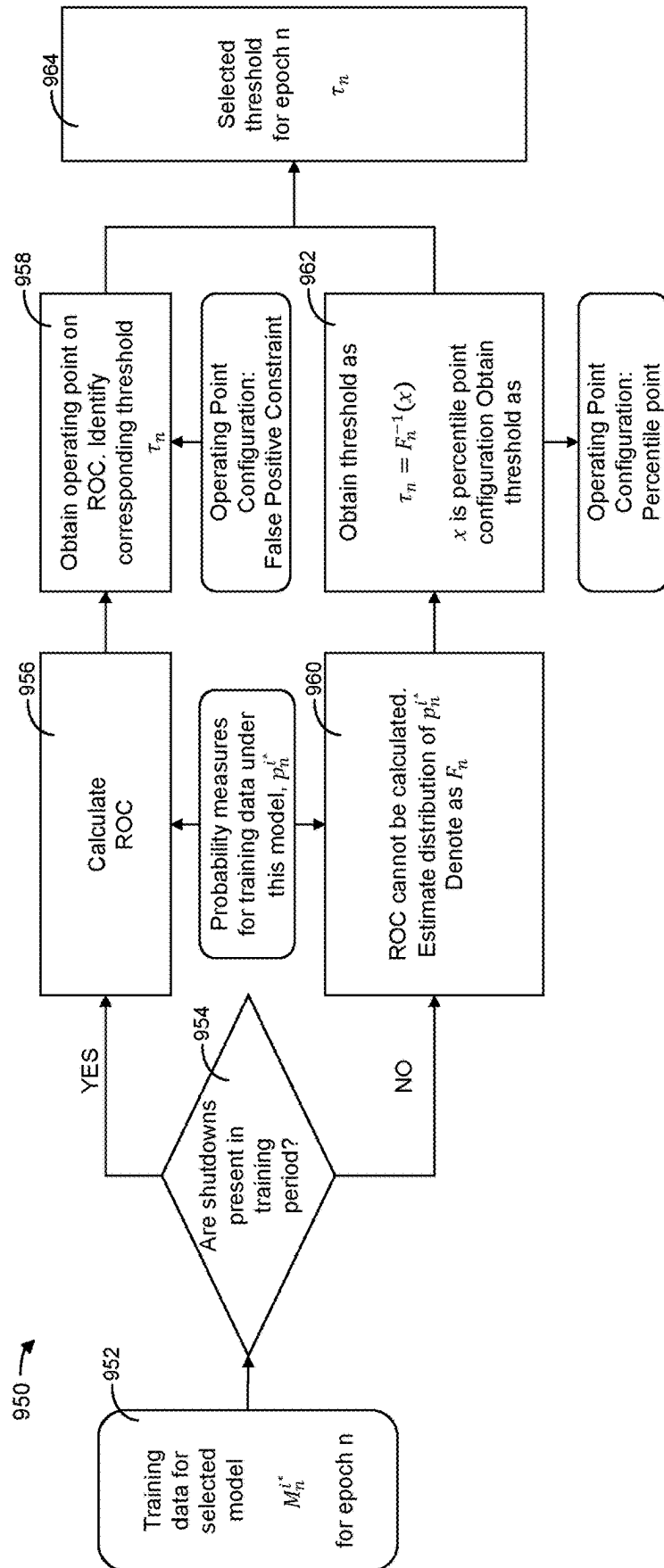


FIG. 9B



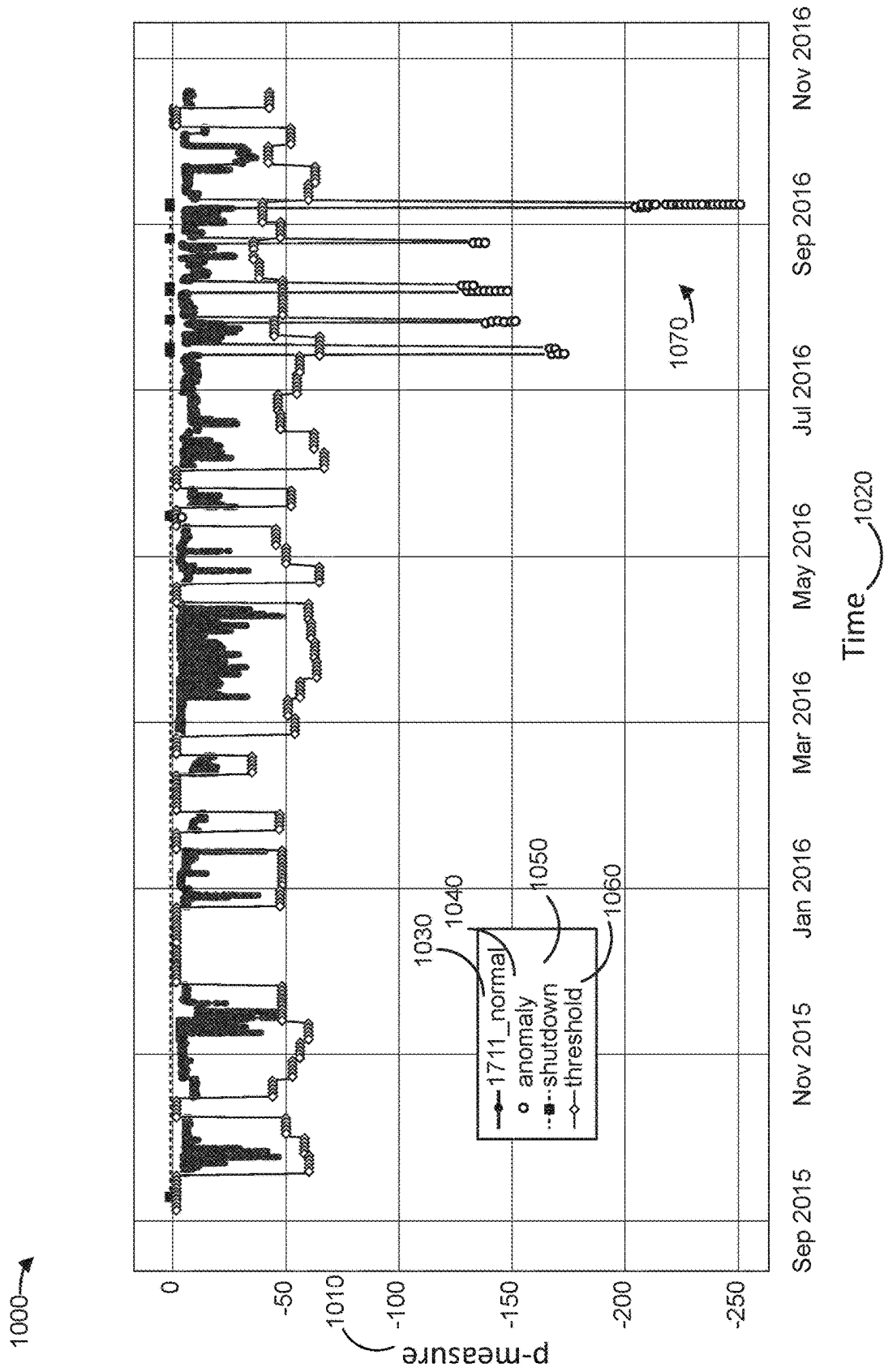


FIG. 10

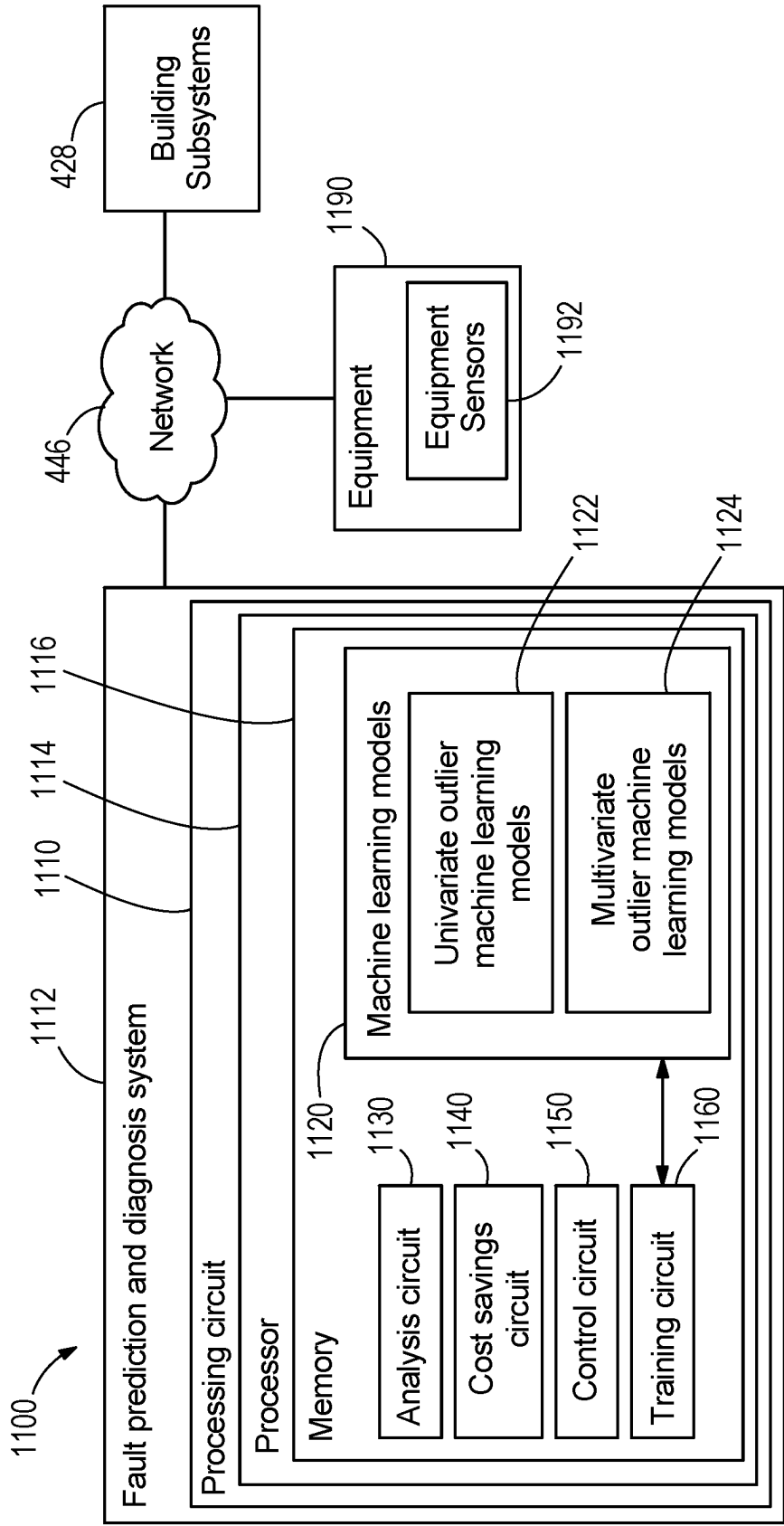


FIG. 11

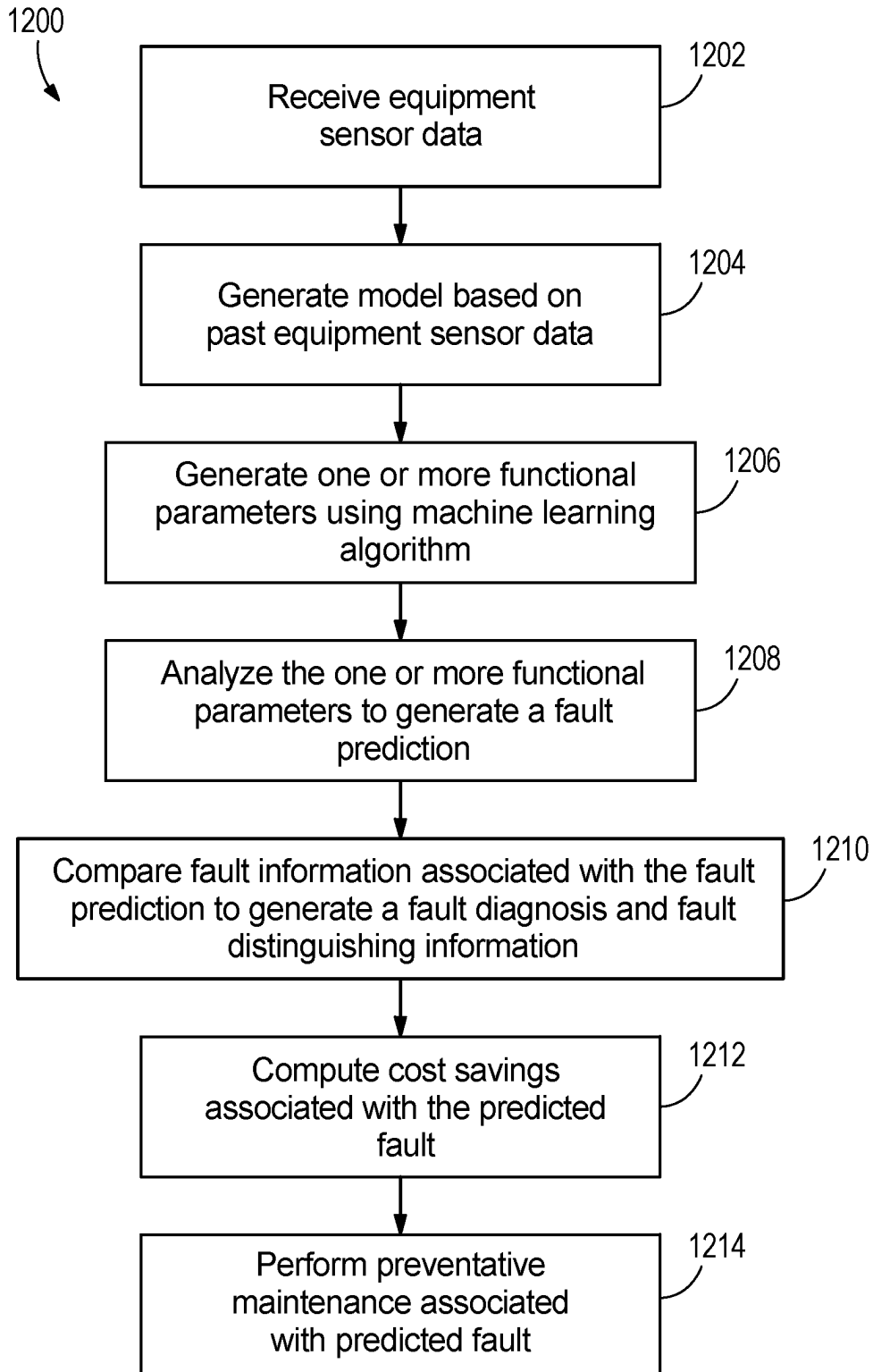


FIG. 12

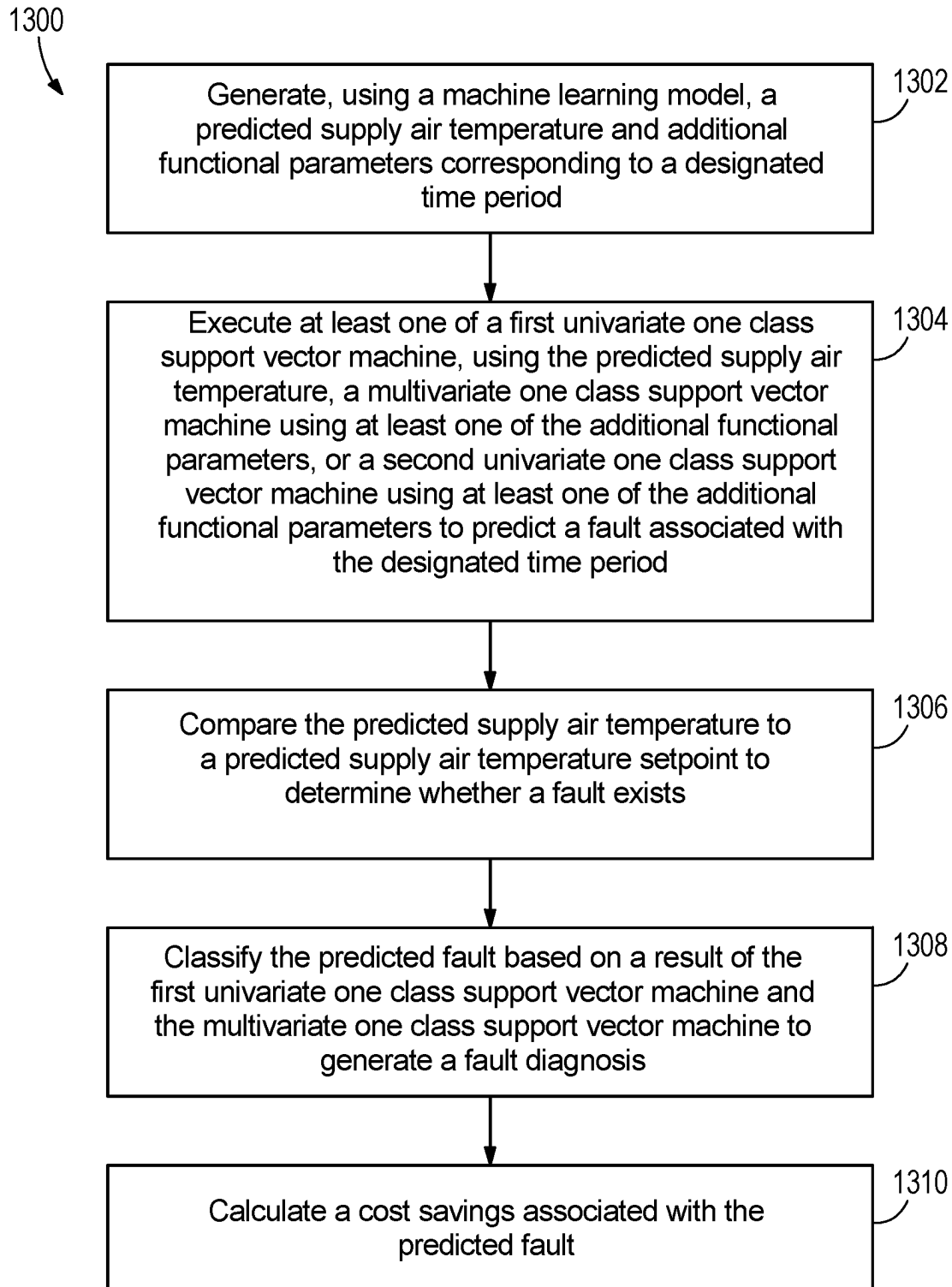


FIG. 13

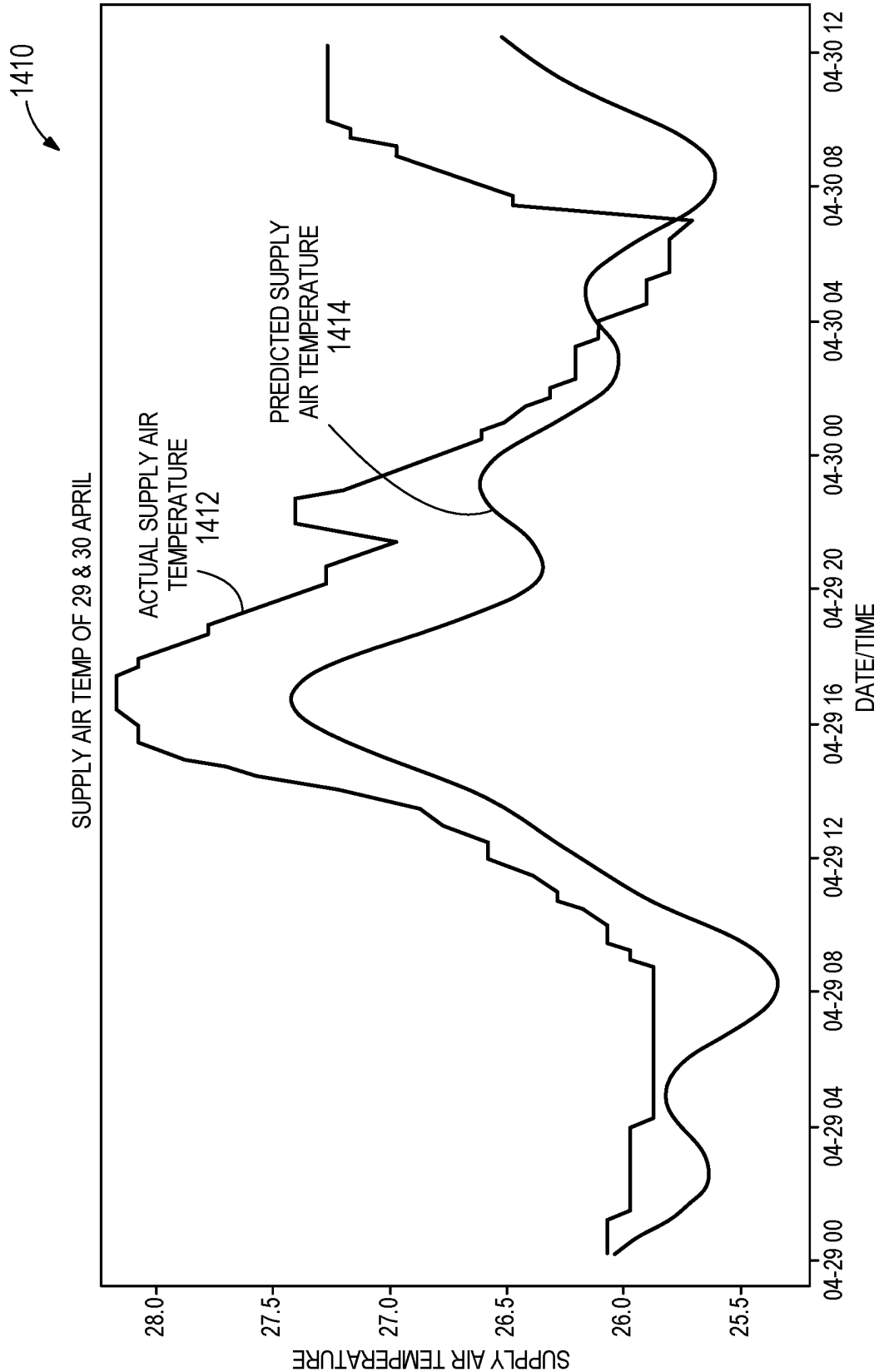


FIG. 14A

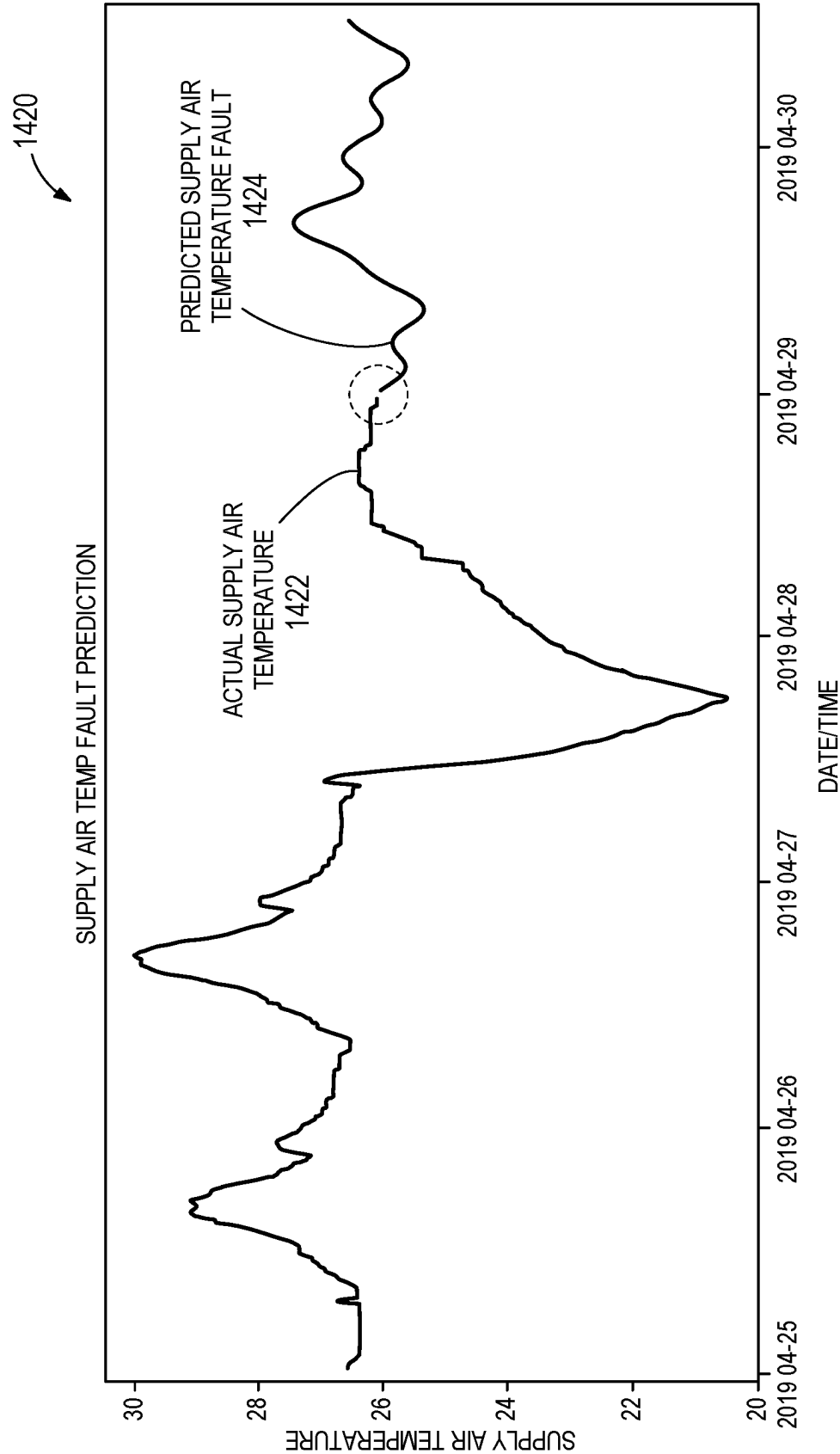


FIG. 14B

COST SAVINGS FROM FAULT PREDICTION AND DIAGNOSIS

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS

[0001] This application is a continuation of U.S. patent application Ser. No. 18/242,976, filed Sep. 6, 2023, which is a continuation of U.S. patent application Ser. No. 17/318,877, filed on May 12, 2021, which is a continuation-in-part of U.S. patent application Ser. No. 16/198,456 filed on Nov. 21, 2018, U.S. patent application Ser. No. 16/198,416 filed on Nov. 21, 2018, and U.S. patent application Ser. No. 16/198,377 filed on Nov. 21, 2018, all of which claim priority to U.S. Provisional Patent Application No. 62/685,618 filed on Jun. 15, 2018, the entire disclosures of each of which are incorporated by reference herein.

BACKGROUND

[0002] The present disclosure relates generally to chillers operating in conjunction with HVAC systems. The present disclosure relates more particularly to predicting chiller faults using models trained with machine learning and deep learning.

[0003] Chillers are often found in buildings and are components of HVAC systems. Chillers are subject to faults, which can cause unplanned shutdowns due to safety and other concerns. More specifically, chiller shutdowns may cause loss of efficiency, as well as damage to other expensive HVAC equipment during a shutdown. It is desirable to predict chiller shutdowns prior to shutdowns occurring.

[0004] Chiller faults are often unexpected and difficult to predict. Various factors may cause a chiller fault including overuse, required maintenance, safety concerns and environmental conditions, among other possible factors. With many factors capable of influencing sudden chiller faults, predicting future chiller failure is challenging.

SUMMARY

[0005] One implementation of the present disclosure is a heating, ventilation, and air conditioning (HVAC) fault prediction system for a building including a processing circuit including a processor and memory, the memory having instructions stored thereon that, when executed by the processor, cause the processing circuit to receive HVAC data relating to a plurality of HVAC components, the HVAC data indicating performance of the plurality of HVAC components, generate, based on the received HVAC data, a univariate prediction model and a multivariate prediction model, generate, using the received HVAC data, one or more predicted operational parameters for the plurality of HVAC components corresponding to a future time period, and execute at least one of the univariate prediction model or the multivariate prediction model on the one or more predicted operational parameters to predict a HVAC fault associated with at least one of the plurality of HVAC components to occur during the future time period.

[0006] In some embodiments, the instructions further cause the processing circuit to calculate an amount of energy consumption associated with the predicted HVAC fault based at least in part on the one or more predicted operational parameters. In some embodiments, the instructions further cause the processing circuit to calculate a cost savings associated with the predicted HVAC fault based on

the amount of energy consumption calculated. In some embodiments, the instructions further cause the processing circuit to display the predicted HVAC fault to a user based on the calculated cost savings associated with the predicted HVAC fault, where predicted HVAC faults with high cost savings are emphasized over predicted HVAC faults with low cost savings. In some embodiments, the instructions further cause the processing circuit to perform an action comprising at least one of (i) generate a suggestion for a user regarding how to prevent the predicted HVAC fault, the suggestion including a cause of the predicted HVAC fault or (ii) automatically generate a work order ticket including an indication of a piece of equipment associated with the predicted HVAC fault. In some embodiments, the instructions further cause the processing circuit to compare a cost associated with the action to a cost associated with the predicted HVAC fault to determine whether to generate the suggestion or automatically generate the work order ticket. In some embodiments, the instructions further cause the processing circuit to populate a fault diagnosis matrix using the univariate prediction model and the multivariate prediction model and map one or more entries of the fault diagnosis matrix to a fault diagnosis to determine a cause of the predicted HVAC fault. In some embodiments, the instructions further cause the processing circuit to compare a predicted setpoint of the one or more predicted operational parameters to a predicted operational parameter associated with the predicted setpoint of the one or more predicted operational parameters to classify the predicted HVAC fault as at least one of a high zone temperature fault or a low zone temperature fault.

[0007] Another implementation of the present disclosure is one or more non-transitory computer-readable storage mediums having instructions stored thereon that, when executed by a processor, cause the processor to receive data relating to a plurality of heating, ventilation, and air conditioning (HVAC) components, the data indicating performance of the plurality of HVAC components, generate, based on the received data, a univariate prediction model and a multivariate prediction model, generate, using the received data, one or more predicted operational parameters for the plurality of HVAC components corresponding to a future time period, execute at least one of the univariate prediction model or the multivariate prediction model on the one or more predicted operational parameters to predict a HVAC fault associated with at least one of the plurality of HVAC components to occur during the future time period.

[0008] In some embodiments, the instructions further cause the processor to calculate an amount of energy consumption associated with the predicted HVAC fault based at least in part on the one or more predicted operational parameters. In some embodiments, the instructions further cause the processor to calculate a cost savings associated with the predicted HVAC fault based on the amount of energy consumption calculated. In some embodiments, the instructions further cause the processor to display the predicted HVAC fault to a user based on the calculated cost savings associated with the predicted HVAC fault, where predicted HVAC faults with high cost savings are emphasized over predicted HVAC faults with low cost savings. In some embodiments, the instructions further cause the processor to perform an action comprising at least one of (i) generate a suggestion for a user regarding how to prevent the predicted HVAC fault, the suggestion including a cause of

the predicted HVAC fault or (ii) automatically generate a work order ticket including an indication of a piece of equipment associated with the predicted HVAC fault. In some embodiments, the instructions further cause the processor to compare a cost associated with the action to a cost associated with the predicted HVAC fault to determine whether to generate the suggestion or automatically generate the work order ticket. In some embodiments, the instructions further cause the processor to populate a fault diagnosis matrix using the univariate prediction model and the multivariate prediction model and map one or more entries of the fault diagnosis matrix to a fault diagnosis to determine a cause of the predicted HVAC fault. In some embodiments, the instructions further cause the processor to compare a predicted setpoint of the one or more predicted operational parameters to a predicted operational parameter associated with the predicted setpoint of the one or more predicted operational parameters to classify the predicted HVAC fault as at least one of a high zone temperature fault or a low zone temperature fault.

[0009] Another implementation of the present disclosure is a building management system (BMS), comprising a processing circuit including a processor and memory, the memory having instructions stored thereon that, when executed by the processor, cause the processing circuit to receive data relating to a plurality of heating, ventilation, and air conditioning (HVAC) components, the data indicating performance of the plurality of HVAC components, generate, based on the received data, a univariate prediction model and a multivariate prediction model, generate, using the received data, one or more predicted operational parameters for the plurality of HVAC components corresponding to a future time period, execute at least one of the univariate prediction model or the multivariate prediction model on the one or more predicted operational parameters to predict a HVAC fault associated with at least one of the plurality of HVAC components to occur during the future time period, and calculate an amount of energy consumption associated with the predicted HVAC fault based at least in part on the one or more predicted operational parameters.

[0010] In some embodiments, the instructions further cause the processing circuit to calculate a cost savings associated with the predicted HVAC fault based on the amount of energy consumption calculated. In some embodiments, the instructions further cause the processing circuit to display the predicted HVAC fault to a user based on the calculated cost savings associated with the predicted HVAC fault, where predicted HVAC faults with high cost savings are emphasized over predicted HVAC faults with low cost savings. In some embodiments, the instructions further cause the processing circuit to perform an action comprising at least one of (i) generate a suggestion for a user regarding how to prevent the predicted HVAC fault, the suggestion including a cause of the predicted HVAC fault or (ii) automatically generate a work order ticket including an indication of a piece of equipment associated with the predicted HVAC fault. In some embodiments, the data includes chiller data indicating performance of a plurality of chillers, wherein the univariate prediction model and the multivariate prediction model include a single chiller prediction model and a cluster chiller prediction model, and wherein predicting the HVAC fault includes predicting a chiller fault with at least one of the single chiller prediction model or the cluster chiller prediction model.

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. 1 is a drawing of a building equipped with a HVAC system, according to some embodiments.

[0012] FIG. 2 is a schematic diagram of a waterside system which can be used in conjunction with the building of FIG. 1, according to some embodiments.

[0013] FIG. 3 is a schematic diagram of an airside system which can be used in conjunction with the building of FIG. 1, according to some embodiments.

[0014] FIG. 4 is a block diagram of a building management system (BMS) which can be used to monitor and control the building of FIG. 1, according to some embodiments.

[0015] FIG. 5 is a block diagram of another BMS which can be used to monitor and control the building of FIG. 1, according to some embodiments.

[0016] FIG. 6 is a block diagram of a chiller analysis system for generating single chiller models and cluster chiller models with a model manager and a trainer including a threshold selector and a model selector, according to an exemplary embodiment.

[0017] FIG. 7A is a block diagram of the model manager illustrated in FIG. 6 shown in greater detail, according to an exemplary embodiment.

[0018] FIG. 7B is a flow diagram of a process of generating the single chiller models and the cluster chiller models illustrated in FIG. 6 that can be performed by the model manager illustrated in FIG. 7A, according to an exemplary embodiment.

[0019] FIG. 7C is another flow diagram of the process of FIG. 7B, according to an exemplary embodiment.

[0020] FIG. 8A is a block diagram of the trainer including the threshold selector and the model selector illustrated in FIG. 6 shown in greater detail, according to an exemplary embodiment.

[0021] FIG. 8B is a flow diagram of a process of training models, selecting models, and determining thresholds for the selected models that can be performed by the trainer illustrated in FIG. 8A, according to an exemplary embodiment.

[0022] FIG. 8C is another flow diagram of the process illustrated in FIG. 8B, according to an exemplary embodiment.

[0023] FIG. 9A is a block diagram of a threshold system that may be used in conjunction with one or more chillers, according to an exemplary embodiment.

[0024] FIG. 9B is a flow diagram illustrating the system of FIG. 9A in greater detail, according to an exemplary embodiment.

[0025] FIG. 9C is another flow diagram of the process illustrated in FIG. 9B, according to an exemplary embodiment.

[0026] FIG. 10 is a chart illustrating p-measures of equipment data applied to a selected threshold, according to an exemplary embodiment.

[0027] FIG. 11 is a block diagram of a fault prediction and diagnosis system for identifying faults and generating cost savings, according to an exemplary embodiment.

[0028] FIG. 12 is a flow diagram illustrating a method of generating fault predictions and computing cost savings associated with the fault predictions, according to an exemplary embodiment.

[0029] FIG. 13 is a flow diagram illustrating a method of determining the presence of a fault and classifying faults, according to an exemplary embodiment.

[0030] FIG. 14A is a graph illustrating predicted air temperatures generated by the fault prediction and diagnosis system of FIG. 11, according to an exemplary embodiment.

[0031] FIG. 14B is a graph illustrating a predicted air temperature fault generated by the fault prediction and diagnosis system of FIG. 14B, according to an exemplary embodiment.

DETAILED DESCRIPTION

[0032] Referring now generally to the Figures, systems and methods of cost savings from fault prediction are presented. In various embodiments, a first machine learning model is generated by training a machine learning system such as a neural network on historical operating data. For example, the machine learning system may perform regression analysis on historical operating parameters such as CO₂ levels (e.g., measured in parts-per-million (PPM)), CO₂ setpoints (e.g., measured in PPM), cooling valve setpoints (e.g., measured as a percentage), cooling valve levels (e.g., measured as a percentage), filter statuses (e.g., indicating whether a filter alarm exists), heating valve setpoints (e.g., measured in ° C., etc.), outside air humidity (e.g., measured as a percentage of room humidity, etc.), outside air temperature (e.g., measured in ° C., etc.), outside air damper setpoint (e.g., measured as a percentage), outside air damper levels (e.g., measured as a percentage), return air humidity (e.g., measured as a percentage of room humidity, etc.), return air temperature (e.g., measured in ° C., etc.), return air damper setpoints (e.g., measured as a percentage), supply air humidity (e.g., measured as a percentage of room humidity), supply air temperature (e.g., measured in ° C., etc.), supply air static pressure setpoints (e.g., measured in Pascals, etc.), supply air static pressure levels (e.g., measured in Pascals, etc.), supply air temperature setpoints (e.g., measured in ° C., etc.), and/or supply fan frequency (e.g., measured in Hz, etc.). Additionally or alternatively, it should be understood that other machine learning systems are possible. For example, decision trees, support-vector machines, Bayesian networks, regression trees, linear regression, and/or the like may be used.

[0033] In various embodiments, the first machine learning model generates predicted functional parameters for a space and/or building. For example, the first machine learning model may ingest historical functional parameters (e.g., from the last two weeks, etc.) and may generate predicted functional parameters such as predicted supply air temperature for a future period such as the next two days. In various embodiments, one or more additional machine learning models are executed on the historical functional parameters and/or the predicted functional parameters to identify predicted faults. For example, a univariate one class support vector machine (SVM) may be executed on a predicted supply air temperature to generate a fault prediction. As another example, a multivariate one class SVM may be executed on a number of predicted functional parameters to generate a fault prediction. As yet another example, a univariate one class SVM may be executed on one or more predicted functional parameters to generate a fault prediction. In various embodiments, the one or more additional machine learning models are configured to predict faults associated with a building management system (BMS) and/or a component thereof (e.g., such as an HVAC system, etc.). For example, the one or more additional machine learning models may be trained on historical fault information relat-

ing to a HVAC system such as historical functional parameters associated with the HVAC system, any faults that occurred, and/or the root cause of the faults.

[0034] In various embodiments, the systems and methods of the present disclosure classify predicted faults. For example, the one or more additional machine learning models may generate one or more fault predictions and/or associated context data such as root cause information and the systems and methods of the present disclosure may analyze this information to classify one or more predicted faults. In various embodiments, the one or more predicted faults may be classified as high zone temperature faults or low zone temperature faults. For example, if a predicted supply air temperature is greater than a predicted supply air temperature setpoint then a corresponding predicted fault may be classified as a high zone temperature fault.

[0035] In various embodiments, the systems and methods of the present disclosure diagnose predicted faults. For example, the systems and methods of the present disclosure may analyze the one or more fault predictions and/or the associated context data generated by the one or more additional machine learning models to diagnose one or more predicted faults. In various embodiments, the predicted faults are diagnosed according to one or more categories. For example, the predicted faults may be diagnosed as no fault, supply air temperature sensor fault, other than supply air temperature fault, fault of supply air temperature because of other feeding equipment, and/or the like. In various embodiments, context data is used to diagnose predicted faults. For example, systems and methods of the present disclosure may analyze a time range associated with a predicted fault, one or more pieces of equipment that may impact the predicted fault, the outside air temperature feeding the HVAC system, any related faults (e.g., a fault indicating a malfunctioning damper, etc.), and/or the like.

[0036] In various embodiments, the systems and methods of the present disclosure may calculate cost savings associated with predicted faults. For example, the systems and methods of the present disclosure may measure an amount of energy consumption associated with the predicted fault (e.g., an amount of excess energy consumption that would have occurred if the fault had not be predicted and averted before it occurred, etc.) and may calculate a cost associated with the energy consumption (e.g., in dollars, etc.). In various embodiments, systems and methods of the present disclosure automatically perform one or more actions to prevent and/or fix the predicted faults. For example, a work order ticket may be automatically generated to fix a piece of equipment causing a predicted fault.

Overview

[0037] Chillers are generally critical components of HVAC equipment in buildings and can consume about half of building energy consumption. As such, it is desirable to maintain chillers properly so as to ensure optimal functionality as well as ideal performance. In some instances even temporary loss of a chiller can lead to substantial losses including financial losses, losses of related HVAC equipment due to a malfunctioning or non-functioning chiller, as well as other potential losses including those associated with inefficient or incomplete operation. Such substantial losses from chiller malfunction and/or non-function such as may be

experienced from a chiller shutdown present a desire for predicting chiller events such as shutdowns before they happen.

[0038] A chiller is equipped with numerous sensors capable of collecting a variety of data from chillers in real time in accordance with some embodiments. In some instances, this can aid in scheduling maintenance in advance of any unexpected chiller events, such as shutdowns. Predicting faults (events including chiller shutdowns, such as safety shutdowns) prior to occurring through analysis of chiller data can allow for preventative maintenance, thus optimizing the performance of chillers and preventing costly losses including unplanned maintenance costs and damage to expensive equipment. In order to predict chiller faults, IoT (Internet of Things) learning and deep learning accurately predict in advance (with service lead time) future faults using historic data collected by connected chiller sensors in some embodiments. For example, with regard to machine learning Multivariate Gaussian Modeling (MVG) and Graphical Gaussian Modeling (GGM) may be implemented. Deep learning techniques may include Long Short Term Memory (LSTM) with autoencoder, as well as Variational Autoencoder (VAE) methods. Ultimately, connected chiller sensors and associated connected chillers can enable the collection of chiller operation data in real time and allow for analysis thereof.

[0039] However, the use of machine learning and deep learning methods and modeling to predict chiller faults can require that the models be carefully trained to make accurate predictions. As such, historic training data is used in the training process for machine learning and deep learning models. More specifically, the quality of historic training data available and hyper-parameter tuning of machine learning and deep learning models may impact the accuracy of fault projections. Machine learning and deep learning models can be developed for each chiller (i.e., individual chillers) and/or for clustered chillers (i.e., groups of chillers). Machine learning can be trained on groups of chillers since newly commissioned chillers may not have adequate individual historic data to train machine learning and deep learning models for predictions of the individual chiller. With regard to machine learning and deep learning models, prediction accuracy can be variable and for some models may be good enough to deploy the model, in which case the model may be applied to other chillers and/or used for training purposes. In the event that machine learning and deep learning models are not good enough to be deployed, the model may be discarded with new machine learning and deep learning models generated based on newly collected data.

[0040] Machine learning and deep learning models may implement anomaly detection, which is a machine learning and deep learning technique that can identify patterns for normal states, and then compare patterns with new observations. In the event of a new pattern being observed that is significantly different, this may be classified as an anomaly, or fault in some embodiments. An anomaly detection model can capture underlying probability distributions of non-anomalous (i.e., normal) data in a space that may be different from that of anomalous data. With regard to machine learning and deep learning models, each model has a set of hyper-parameters that may be tuned to ensure model accuracy. As such, developing an effective combination of machine learning and deep learning models with correct

hyper-parameters can be critical to model accuracy. Thus, multiple models may be trained using different sets of hyper-parameters, where a selection may then be made as to which combination may produce the most accurate predictions. For anomaly detection models, an anomaly threshold may be set in the underlying probability distribution to distinguish normal behavior from abnormal behavior. In some instances, this may allow for correct separation of true negative results from true positive results, for example. As such, developing anomaly detection in machine learning and deep learning models with an effective threshold may allow for models to perform optimally and output the most accurate predictions.

Building and HVAC System

[0041] Referring particularly to FIG. 1, a perspective view of a building 10 is shown. Building 10 is served by a BMS. A BMS is, in general, a system of devices configured to control, monitor, and manage equipment in or around a building or building area. A BMS can include, for example, a HVAC system, a security system, a lighting system, a fire alerting system, any other system that is capable of managing building functions or devices, or any combination thereof.

[0042] The BMS that serves building 10 includes a HVAC system 100. HVAC system 100 can include a plurality of HVAC devices (e.g., heaters, chillers, air handling units, pumps, fans, thermal energy storage, etc.) configured to provide heating, cooling, ventilation, or other services for building 10. For example, HVAC system 100 is shown to include a waterside system 120 and an airside system 130. Waterside system 120 may provide a heated or chilled fluid to an air handling unit of airside system 130. Airside system 130 may use the heated or chilled fluid to heat or cool an airflow provided to building 10. An exemplary waterside system and airside system which can be used in HVAC system 100 are described in greater detail with reference to FIGS. 2-3.

[0043] HVAC system 100 is shown to include a chiller 102, a boiler 104, and a rooftop air handling unit (AHU) 106. Waterside system 120 may use boiler 104 and chiller 102 to heat or cool a working fluid (e.g., water, glycol, etc.) and may circulate the working fluid to AHU 106. In various embodiments, the HVAC devices of waterside system 120 can be located in or around building 10 (as shown in FIG. 1) or at an offsite location such as a central plant (e.g., a chiller plant, a steam plant, a heat plant, etc.). The working fluid can be heated in boiler 104 or cooled in chiller 102, depending on whether heating or cooling is required in building 10. Boiler 104 may add heat to the circulated fluid, for example, by burning a combustible material (e.g., natural gas) or using an electric heating element. Chiller 102 may place the circulated fluid in a heat exchange relationship with another fluid (e.g., a refrigerant) in a heat exchanger (e.g., an evaporator) to absorb heat from the circulated fluid. The working fluid from chiller 102 and/or boiler 104 can be transported to AHU 106 via piping 108.

[0044] AHU 106 may place the working fluid in a heat exchange relationship with an airflow passing through AHU 106 (e.g., via one or more stages of cooling coils and/or heating coils). The airflow can be, for example, outside air, return air from within building 10, or a combination of both. AHU 106 may transfer heat between the airflow and the working fluid to provide heating or cooling for the airflow.

For example, AHU 106 can include one or more fans or blowers configured to pass the airflow over or through a heat exchanger containing the working fluid. The working fluid may then return to chiller 102 or boiler 104 via piping 110.

[0045] Airside system 130 may deliver the airflow supplied by AHU 106 (i.e., the supply airflow) to building 10 via air supply ducts 112 and may provide return air from building 10 to AHU 106 via air return ducts 114. In some embodiments, airside system 130 includes multiple variable air volume (VAV) units 116. For example, airside system 130 is shown to include a separate VAV unit 116 on each floor or zone of building 10. VAV units 116 can include dampers or other flow control elements that can be operated to control an amount of the supply airflow provided to individual zones of building 10. In other embodiments, airside system 130 delivers the supply airflow into one or more zones of building 10 (e.g., via supply ducts 112) without using intermediate VAV units 116 or other flow control elements. AHU 106 can include various sensors (e.g., temperature sensors, pressure sensors, etc.) configured to measure attributes of the supply airflow. AHU 106 may receive input from sensors located within AHU 106 and/or within the building zone and may adjust the flow rate, temperature, or other attributes of the supply airflow through AHU 106 to achieve setpoint conditions for the building zone.

Waterside System

[0046] Referring now to FIG. 2, a block diagram of a waterside system 200 is shown, according to some embodiments. In various embodiments, waterside system 200 may supplement or replace waterside system 120 in HVAC system 100 or can be implemented separate from HVAC system 100. When implemented in HVAC system 100, waterside system 200 can include a subset of the HVAC devices in HVAC system 100 (e.g., boiler 104, chiller 102, pumps, valves, etc.) and may operate to supply a heated or chilled fluid to AHU 106. The HVAC devices of waterside system 200 can be located within building 10 (e.g., as components of waterside system 120) or at an offsite location such as a central plant.

[0047] In FIG. 2, waterside system 200 is shown as a central plant having a plurality of subplants 202-212. Subplants 202-212 are shown to include a heater subplant 202, a heat recovery chiller subplant 204, a chiller subplant 206, a cooling tower subplant 208, a hot thermal energy storage (TES) subplant 210, and a cold thermal energy storage (TES) subplant 212. Subplants 202-212 consume resources (e.g., water, natural gas, electricity, etc.) from utilities to serve thermal energy loads (e.g., hot water, cold water, heating, cooling, etc.) of a building or campus. For example, heater subplant 202 can be configured to heat water in a hot water loop 214 that circulates the hot water between heater subplant 202 and building 10. Chiller subplant 206 can be configured to chill water in a cold water loop 216 that circulates the cold water between chiller subplant 206 and building 10. Heat recovery chiller subplant 204 can be configured to transfer heat from cold water loop 216 to hot water loop 214 to provide additional heating for the hot water and additional cooling for the cold water. Condenser water loop 218 may absorb heat from the cold water in chiller subplant 206 and reject the absorbed heat in cooling tower subplant 208 or transfer the absorbed heat to hot water loop 214. Hot

TES subplant 210 and cold TES subplant 212 may store hot and cold thermal energy, respectively, for subsequent use.

[0048] Hot water loop 214 and cold water loop 216 may deliver the heated and/or chilled water to air handlers located on the rooftop of building 10 (e.g., AHU 106) or to individual floors or zones of building 10 (e.g., VAV units 116). The air handlers push air past heat exchangers (e.g., heating coils or cooling coils) through which the water flows to provide heating or cooling for the air. The heated or cooled air can be delivered to individual zones of building 10 to serve thermal energy loads of building 10. The water then returns to subplants 202-212 to receive further heating or cooling.

[0049] Although subplants 202-212 are shown and described as heating and cooling water for circulation to a building, it is understood that any other type of working fluid (e.g., glycol, CO₂, etc.) can be used in place of or in addition to water to serve thermal energy loads. In other embodiments, subplants 202-212 may provide heating and/or cooling directly to the building or campus without requiring an intermediate heat transfer fluid. These and other variations to waterside system 200 are within the teachings of the present disclosure.

[0050] Each of subplants 202-212 can include a variety of equipment configured to facilitate the functions of the subplant. For example, heater subplant 202 is shown to include a plurality of heating elements 220 (e.g., boilers, electric heaters, etc.) configured to add heat to the hot water in hot water loop 214. Heater subplant 202 is also shown to include several pumps 222 and 224 configured to circulate the hot water in hot water loop 214 and to control the flow rate of the hot water through individual heating elements 220. Chiller subplant 206 is shown to include a plurality of chillers 232 configured to remove heat from the cold water in cold water loop 216. Chiller subplant 206 is also shown to include several pumps 234 and 236 configured to circulate the cold water in cold water loop 216 and to control the flow rate of the cold water through individual chillers 232.

[0051] Heat recovery chiller subplant 204 is shown to include a plurality of heat recovery heat exchangers 226 (e.g., refrigeration circuits) configured to transfer heat from cold water loop 216 to hot water loop 214. Heat recovery chiller subplant 204 is also shown to include several pumps 228 and 230 configured to circulate the hot water and/or cold water through heat recovery heat exchangers 226 and to control the flow rate of the water through individual heat recovery heat exchangers 226. Cooling tower subplant 208 is shown to include a plurality of cooling towers 238 configured to remove heat from the condenser water in condenser water loop 218. Cooling tower subplant 208 is also shown to include several pumps 240 configured to circulate the condenser water in condenser water loop 218 and to control the flow rate of the condenser water through individual cooling towers 238.

[0052] Hot TES subplant 210 is shown to include a hot TES tank 242 configured to store the hot water for later use. Hot TES subplant 210 may also include one or more pumps or valves configured to control the flow rate of the hot water into or out of hot TES tank 242. Cold TES subplant 212 is shown to include cold TES tanks 244 configured to store the cold water for later use. Cold TES subplant 212 may also include one or more pumps or valves configured to control the flow rate of the cold water into or out of cold TES tanks 244.

[0053] In some embodiments, one or more of the pumps in waterside system 200 (e.g., pumps 222, 224, 228, 230, 234, 236, and/or 240) or pipelines in waterside system 200 include an isolation valve associated therewith. Isolation valves can be integrated with the pumps or positioned upstream or downstream of the pumps to control the fluid flows in waterside system 200. In various embodiments, waterside system 200 can include more, fewer, or different types of devices and/or subplants based on the particular configuration of waterside system 200 and the types of loads served by waterside system 200.

Airside System

[0054] Referring now to FIG. 3, a block diagram of an airside system 300 is shown, according to some embodiments. In various embodiments, airside system 300 may supplement or replace airside system 130 in HVAC system 100 or can be implemented separate from HVAC system 100. When implemented in HVAC system 100, airside system 300 can include a subset of the HVAC devices in HVAC system 100 (e.g., AHU 106, VAV units 116, ducts 112-114, fans, dampers, etc.) and can be located in or around building 10. Airside system 300 may operate to heat or cool an airflow provided to building 10 using a heated or chilled fluid provided by waterside system 200.

[0055] In FIG. 3, airside system 300 is shown to include an economizer-type air handling unit (AHU) 302. Economizer-type AHUs vary the amount of outside air and return air used by the air handling unit for heating or cooling. For example, AHU 302 may receive return air 304 from building zone 306 via return air duct 308 and may deliver supply air 310 to building zone 306 via supply air duct 312. In some embodiments, AHU 302 is a rooftop unit located on the roof of building 10 (e.g., AHU 106 as shown in FIG. 1) or otherwise positioned to receive both return air 304 and outside air 314. AHU 302 can be configured to operate exhaust air damper 316, mixing damper 318, and outside air damper 320 to control an amount of outside air 314 and return air 304 that combine to form supply air 310. Any return air 304 that does not pass through mixing damper 318 can be exhausted from AHU 302 through exhaust damper 316 as exhaust air 322.

[0056] Each of dampers 316-320 can be operated by an actuator. For example, exhaust air damper 316 can be operated by actuator 324, mixing damper 318 can be operated by actuator 326, and outside air damper 320 can be operated by actuator 328. Actuators 324-328 may communicate with an AHU controller 330 via a communications link 332. Actuators 324-328 may receive control signals from AHU controller 330 and may provide feedback signals to AHU controller 330. Feedback signals can include, for example, an indication of a current actuator or damper position, an amount of torque or force exerted by the actuator, diagnostic information (e.g., results of diagnostic tests performed by actuators 324-328), status information, commissioning information, configuration settings, calibration data, and/or other types of information or data that can be collected, stored, or used by actuators 324-328. AHU controller 330 can be an economizer controller configured to use one or more control algorithms (e.g., state-based algorithms, extremum seeking control (ESC) algorithms, proportional-integral (PI) control algorithms, proportional-integral-derivative (PID) control algorithms, model predictive

control (MPC) algorithms, feedback control algorithms, etc.) to control actuators 324-328.

[0057] Still referring to FIG. 3, AHU 302 is shown to include a cooling coil 334, a heating coil 336, and a fan 338 positioned within supply air duct 312. Fan 338 can be configured to force supply air 310 through cooling coil 334 and/or heating coil 336 and provide supply air 310 to building zone 306. AHU controller 330 may communicate with fan 338 via communications link 340 to control a flow rate of supply air 310. In some embodiments, AHU controller 330 controls an amount of heating or cooling applied to supply air 310 by modulating a speed of fan 338.

[0058] Cooling coil 334 may receive a chilled fluid from waterside system 200 (e.g., from cold water loop 216) via piping 342 and may return the chilled fluid to waterside system 200 via piping 344. Valve 346 can be positioned along piping 342 or piping 344 to control a flow rate of the chilled fluid through cooling coil 334. In some embodiments, cooling coil 334 includes multiple stages of cooling coils that can be independently activated and deactivated (e.g., by AHU controller 330, by BMS controller 366, etc.) to modulate an amount of cooling applied to supply air 310.

[0059] Heating coil 336 may receive a heated fluid from waterside system 200 (e.g., from hot water loop 214) via piping 348 and may return the heated fluid to waterside system 200 via piping 350. Valve 352 can be positioned along piping 348 or piping 350 to control a flow rate of the heated fluid through heating coil 336. In some embodiments, heating coil 336 includes multiple stages of heating coils that can be independently activated and deactivated (e.g., by AHU controller 330, by BMS controller 366, etc.) to modulate an amount of heating applied to supply air 310.

[0060] Each of valves 346 and 352 can be controlled by an actuator. For example, valve 346 can be controlled by actuator 354 and valve 352 can be controlled by actuator 356. Actuators 354-356 may communicate with AHU controller 330 via communications links 358-360. Actuators 354-356 may receive control signals from AHU controller 330 and may provide feedback signals to controller 330. In some embodiments, AHU controller 330 receives a measurement of the supply air temperature from a temperature sensor 362 positioned in supply air duct 312 (e.g., downstream of cooling coil 334 and/or heating coil 336). AHU controller 330 may also receive a measurement of the temperature of building zone 306 from a temperature sensor 364 located in building zone 306.

[0061] In some embodiments, AHU controller 330 operates valves 346 and 352 via actuators 354-356 to modulate an amount of heating or cooling provided to supply air 310 (e.g., to achieve a setpoint temperature for supply air 310 or to maintain the temperature of supply air 310 within a setpoint temperature range). The positions of valves 346 and 352 affect the amount of heating or cooling provided to supply air 310 by cooling coil 334 or heating coil 336 and may correlate with the amount of energy consumed to achieve a desired supply air temperature. AHU 330 may control the temperature of supply air 310 and/or building zone 306 by activating or deactivating coils 334-336, adjusting a speed of fan 338, or a combination of both.

[0062] Still referring to FIG. 3, airside system 300 is shown to include a building management system (BMS) controller 366 and a client device 368. BMS controller 366 can include one or more computer systems (e.g., servers, supervisory controllers, subsystem controllers, etc.) that

serve as system level controllers, application or data servers, head nodes, or master controllers for airside system 300, waterside system 200, HVAC system 100, and/or other controllable systems that serve building 10. BMS controller 366 may communicate with multiple downstream building systems or subsystems (e.g., HVAC system 100, a security system, a lighting system, waterside system 200, etc.) via a communications link 370 according to like or disparate protocols (e.g., LON, BACnet, etc.). In various embodiments, AHU controller 330 and BMS controller 366 can be separate (as shown in FIG. 3) or integrated. In an integrated implementation, AHU controller 330 can be a software module configured for execution by a processor of BMS controller 366.

[0063] In some embodiments, AHU controller 330 receives information from BMS controller 366 (e.g., commands, setpoints, operating boundaries, etc.) and provides information to BMS controller 366 (e.g., temperature measurements, valve or actuator positions, operating statuses, diagnostics, etc.). For example, AHU controller 330 may provide BMS controller 366 with temperature measurements from temperature sensors 362-364, equipment on/off states, equipment operating capacities, and/or any other information that can be used by BMS controller 366 to monitor or control a variable state or condition within building zone 306.

[0064] Client device 368 can include one or more human-machine interfaces or client interfaces (e.g., graphical user interfaces, reporting interfaces, text-based computer interfaces, client-facing web services, web servers that provide pages to web clients, etc.) for controlling, viewing, or otherwise interacting with HVAC system 100, its subsystems, and/or devices. Client device 368 can be a computer workstation, a client terminal, a remote or local interface, or any other type of user interface device. Client device 368 can be a stationary terminal or a mobile device. For example, client device 368 can be a desktop computer, a computer server with a user interface, a laptop computer, a tablet, a smartphone, a PDA, or any other type of mobile or non-mobile device. Client device 368 may communicate with BMS controller 366 and/or AHU controller 330 via communications link 372.

Building Management Systems

[0065] Referring now to FIG. 4, a block diagram of a building management system (BMS) 400 is shown, according to some embodiments. BMS 400 can be implemented in building 10 to automatically monitor and control various building functions. BMS 400 is shown to include BMS controller 366 and a plurality of building subsystems 428. Building subsystems 428 are shown to include a building electrical subsystem 434, an information communication technology (ICT) subsystem 436, a security subsystem 438, a HVAC subsystem 440, a lighting subsystem 442, a lift/escalators subsystem 432, and a fire safety subsystem 430. In various embodiments, building subsystems 428 can include fewer, additional, or alternative subsystems. For example, building subsystems 428 may also or alternatively include a refrigeration subsystem, an advertising or signage subsystem, a cooking subsystem, a vending subsystem, a printer or copy service subsystem, or any other type of building subsystem that uses controllable equipment and/or sensors to monitor or control building 10. In some embodi-

ments, building subsystems 428 include waterside system 200 and/or airside system 300, as described with reference to FIGS. 2-3.

[0066] Each of building subsystems 428 can include any number of devices, controllers, and connections for completing its individual functions and control activities. HVAC subsystem 440 can include many of the same components as HVAC system 100, as described with reference to FIGS. 1-3. For example, HVAC subsystem 440 can include a chiller, a boiler, any number of air handling units, economizers, field controllers, supervisory controllers, actuators, temperature sensors, and other devices for controlling the temperature, humidity, airflow, or other variable conditions within building 10. Lighting subsystem 442 can include any number of light fixtures, ballasts, lighting sensors, dimmers, or other devices configured to controllably adjust the amount of light provided to a building space. Security subsystem 438 can include occupancy sensors, video surveillance cameras, digital video recorders, video processing servers, intrusion detection devices, access control devices and servers, or other security-related devices.

[0067] Still referring to FIG. 4, BMS controller 366 is shown to include a communications interface 407 and a BMS interface 409. Interface 407 may facilitate communications between BMS controller 366 and external applications (e.g., monitoring and reporting applications 422, enterprise control applications 426, remote systems and applications 444, applications residing on client devices 448, etc.) for allowing user control, monitoring, and adjustment to BMS controller 366 and/or subsystems 428. Interface 407 may also facilitate communications between BMS controller 366 and client devices 448. BMS interface 409 may facilitate communications between BMS controller 366 and building subsystems 428 (e.g., HVAC, lighting security, lifts, power distribution, business, etc.).

[0068] Interfaces 407, 409 can be or include wired or wireless communications interfaces (e.g., jacks, antennas, transmitters, receivers, transceivers, wire terminals, etc.) for conducting data communications with building subsystems 428 or other external systems or devices. In various embodiments, communications via interfaces 407, 409 can be direct (e.g., local wired or wireless communications) or via a communications network 446 (e.g., a WAN, the Internet, a cellular network, etc.). For example, interfaces 407, 409 can include an Ethernet card and port for sending and receiving data via an Ethernet-based communications link or network. In another example, interfaces 407, 409 can include a Wi-Fi transceiver for communicating via a wireless communications network. In another example, one or both of interfaces 407, 409 can include cellular or mobile phone communications transceivers. In some embodiments, communications interface 407 is a power line communications interface and BMS interface 409 is an Ethernet interface. In other embodiments, both communications interface 407 and BMS interface 409 are Ethernet interfaces or are the same Ethernet interface.

[0069] Still referring to FIG. 4, BMS controller 366 is shown to include a processing circuit 404 including a processor 406 and memory 408. Processing circuit 404 can be communicably connected to BMS interface 409 and/or communications interface 407 such that processing circuit 404 and the various components thereof can send and receive data via interfaces 407, 409. Processor 406 can be implemented as a general purpose processor, an application

specific integrated circuit (ASIC), one or more field programmable gate arrays (FPGAs), a group of processing components, or other suitable electronic processing components.

[0070] Memory **408** (e.g., memory, memory unit, storage device, etc.) can include one or more devices (e.g., RAM, ROM, Flash memory, hard disk storage, etc.) for storing data and/or computer code for completing or facilitating the various processes, layers and modules described in the present application. Memory **408** can be or include volatile memory or non-volatile memory. Memory **408** can include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described in the present application. According to some embodiments, memory **408** is communicably connected to processor **406** via processing circuit **404** and includes computer code for executing (e.g., by processing circuit **404** and/or processor **406**) one or more processes described herein.

[0071] In some embodiments, BMS controller **366** is implemented within a single computer (e.g., one server, one housing, etc.). In various other embodiments BMS controller **366** can be distributed across multiple servers or computers (e.g., that can exist in distributed locations). Further, while FIG. 4 shows applications **422** and **426** as existing outside of BMS controller **366**, in some embodiments, applications **422** and **426** can be hosted within BMS controller **366** (e.g., within memory **408**).

[0072] Still referring to FIG. 4, memory **408** is shown to include an enterprise integration layer **410**, an automated measurement and validation (AM&V) layer **412**, a demand response (DR) layer **414**, a fault detection and diagnostics (FDD) layer **416**, an integrated control layer **418**, and a building subsystem integration layer **420**. Layers **410-420** can be configured to receive inputs from building subsystems **428** and other data sources, determine optimal control actions for building subsystems **428** based on the inputs, generate control signals based on the optimal control actions, and provide the generated control signals to building subsystems **428**. The following paragraphs describe some of the general functions performed by each of layers **410-420** in BMS **400**.

[0073] Enterprise integration layer **410** can be configured to serve clients or local applications with information and services to support a variety of enterprise-level applications. For example, enterprise control applications **426** can be configured to provide subsystem-spanning control to a graphical user interface (GUI) or to any number of enterprise-level business applications (e.g., accounting systems, user identification systems, etc.). Enterprise control applications **426** may also or alternatively be configured to provide configuration GUIs for configuring BMS controller **366**. In yet other embodiments, enterprise control applications **426** can work with layers **410-420** to optimize building performance (e.g., efficiency, energy use, comfort, or safety) based on inputs received at interface **407** and/or BMS interface **409**.

[0074] Building subsystem integration layer **420** can be configured to manage communications between BMS controller **366** and building subsystems **428**. For example, building subsystem integration layer **420** may receive sensor data and input signals from building subsystems **428** and provide output data and control signals to building subsystems

428. Building subsystem integration layer **420** may also be configured to manage communications between building subsystems **428**. Building subsystem integration layer **420** translate communications (e.g., sensor data, input signals, output signals, etc.) across a plurality of multi-vendor/multi-protocol systems.

[0075] Demand response layer **414** can be configured to optimize resource usage (e.g., electricity use, natural gas use, water use, etc.) and/or the monetary cost of such resource usage in response to satisfy the demand of building **10**. The optimization can be based on time-of-use prices, curtailment signals, energy availability, or other data received from utility providers, distributed energy generation systems **424**, from energy storage **427** (e.g., hot TES **242**, cold TES **244**, etc.), or from other sources. Demand response layer **414** may receive inputs from other layers of BMS controller **366** (e.g., building subsystem integration layer **420**, integrated control layer **418**, etc.). The inputs received from other layers can include environmental or sensor inputs such as temperature, carbon dioxide levels, relative humidity levels, air quality sensor outputs, occupancy sensor outputs, room schedules, and the like. The inputs may also include inputs such as electrical use (e.g., expressed in kWh), thermal load measurements, pricing information, projected pricing, smoothed pricing, curtailment signals from utilities, and the like.

[0076] According to some embodiments, demand response layer **414** includes control logic for responding to the data and signals it receives. These responses can include communicating with the control algorithms in integrated control layer **418**, changing control strategies, changing setpoints, or activating/deactivating building equipment or subsystems in a controlled manner. Demand response layer **414** may also include control logic configured to determine when to utilize stored energy. For example, demand response layer **414** may determine to begin using energy from energy storage **427** just prior to the beginning of a peak use hour.

[0077] In some embodiments, demand response layer **414** includes a control module configured to actively initiate control actions (e.g., automatically changing setpoints) which minimize energy costs based on one or more inputs representative of or based on demand (e.g., price, a curtailment signal, a demand level, etc.). In some embodiments, demand response layer **414** uses equipment models to determine an optimal set of control actions. The equipment models can include, for example, thermodynamic models describing the inputs, outputs, and/or functions performed by various sets of building equipment. Equipment models may represent collections of building equipment (e.g., sub-plants, chiller arrays, etc.) or individual devices (e.g., individual chillers, heaters, pumps, etc.).

[0078] Demand response layer **414** may further include or draw upon one or more demand response policy definitions (e.g., databases, XML files, etc.). The policy definitions can be edited or adjusted by a user (e.g., via a graphical user interface) so that the control actions initiated in response to demand inputs can be tailored for the user's application, desired comfort level, particular building equipment, or based on other concerns. For example, the demand response policy definitions can specify which equipment can be turned on or off in response to particular demand inputs, how long a system or piece of equipment should be turned off, what setpoints can be changed, what the allowable set point

adjustment range is, how long to hold a high demand setpoint before returning to a normally scheduled setpoint, how close to approach capacity limits, which equipment modes to utilize, the energy transfer rates (e.g., the maximum rate, an alarm rate, other rate boundary information, etc.) into and out of energy storage devices (e.g., thermal storage tanks, battery banks, etc.), and when to dispatch on-site generation of energy (e.g., via fuel cells, a motor generator set, etc.).

[0079] Integrated control layer 418 can be configured to use the data input or output of building subsystem integration layer 420 and/or demand response layer 414 to make control decisions. Due to the subsystem integration provided by building subsystem integration layer 420, integrated control layer 418 can integrate control activities of the subsystems 428 such that the subsystems 428 behave as a single integrated supersystem. In some embodiments, integrated control layer 418 includes control logic that uses inputs and outputs from a plurality of building subsystems to provide greater comfort and energy savings relative to the comfort and energy savings that separate subsystems could provide alone. For example, integrated control layer 418 can be configured to use an input from a first subsystem to make an energy-saving control decision for a second subsystem. Results of these decisions can be communicated back to building subsystem integration layer 420.

[0080] Integrated control layer 418 is shown to be logically below demand response layer 414. Integrated control layer 418 can be configured to enhance the effectiveness of demand response layer 414 by enabling building subsystems 428 and their respective control loops to be controlled in coordination with demand response layer 414. This configuration may advantageously reduce disruptive demand response behavior relative to conventional systems. For example, integrated control layer 418 can be configured to assure that a demand response-driven upward adjustment to the setpoint for chilled water temperature (or another component that directly or indirectly affects temperature) does not result in an increase in fan energy (or other energy used to cool a space) that would result in greater total building energy use than was saved at the chiller.

[0081] Integrated control layer 418 can be configured to provide feedback to demand response layer 414 so that demand response layer 414 checks that constraints (e.g., temperature, lighting levels, etc.) are properly maintained even while demanded load shedding is in progress. The constraints may also include setpoint or sensed boundaries relating to safety, equipment operating limits and performance, comfort, fire codes, electrical codes, energy codes, and the like. Integrated control layer 418 is also logically below fault detection and diagnostics layer 416 and automated measurement and validation layer 412. Integrated control layer 418 can be configured to provide calculated inputs (e.g., aggregations) to these higher levels based on outputs from more than one building subsystem.

[0082] Automated measurement and validation (AM&V) layer 412 can be configured to verify that control strategies commanded by integrated control layer 418 or demand response layer 414 are working properly (e.g., using data aggregated by AM&V layer 412, integrated control layer 418, building subsystem integration layer 420, FDD layer 416, or otherwise). The calculations made by AM&V layer 412 can be based on building system energy models and/or equipment models for individual BMS devices or subsystems.

For example, AM&V layer 412 may compare a model-predicted output with an actual output from building subsystems 428 to determine an accuracy of the model.

[0083] Fault detection and diagnostics (FDD) layer 416 can be configured to provide on-going fault detection for building subsystems 428, building subsystem devices (i.e., building equipment), and control algorithms used by demand response layer 414 and integrated control layer 418. FDD layer 416 may receive data inputs from integrated control layer 418, directly from one or more building subsystems or devices, or from another data source. FDD layer 416 may automatically diagnose and respond to detected faults. The responses to detected or diagnosed faults can include providing an alert message to a user, a maintenance scheduling system, or a control algorithm configured to attempt to repair the fault or to work-around the fault.

[0084] FDD layer 416 can be configured to output a specific identification of the faulty component or cause of the fault (e.g., loose damper linkage) using detailed subsystem inputs available at building subsystem integration layer 420. In other exemplary embodiments, FDD layer 416 is configured to provide “fault” events to integrated control layer 418 which executes control strategies and policies in response to the received fault events. According to some embodiments, FDD layer 416 (or a policy executed by an integrated control engine or business rules engine) may shut-down systems or direct control activities around faulty devices or systems to reduce energy waste, extend equipment life, or assure proper control response.

[0085] FDD layer 416 can be configured to store or access a variety of different system data stores (or data points for live data). FDD layer 416 may use some content of the data stores to identify faults at the equipment level (e.g., specific chiller, specific AHU, specific terminal unit, etc.) and other content to identify faults at component or subsystem levels. For example, building subsystems 428 may generate temporal (i.e., time-series) data indicating the performance of BMS 400 and the various components thereof. The data generated by building subsystems 428 can include measured or calculated values that exhibit statistical characteristics and provide information about how the corresponding system or process (e.g., a temperature control process, a flow control process, etc.) is performing in terms of error from its setpoint. These processes can be examined by FDD layer 416 to expose when the system begins to degrade in performance and alert a user to repair the fault before it becomes more severe.

[0086] Referring now to FIG. 5, a block diagram of another building management system (BMS) 500 is shown, according to some embodiments. BMS 500 can be used to monitor and control the devices of HVAC system 100, waterside system 200, airside system 300, building subsystems 428, as well as other types of BMS devices (e.g., lighting equipment, security equipment, etc.) and/or HVAC equipment.

[0087] BMS 500 provides a system architecture that facilitates automatic equipment discovery and equipment model distribution. Equipment discovery can occur on multiple levels of BMS 500 across multiple different communications buses (e.g., a system bus 554, zone buses 556-560 and 564, sensor/actuator bus 566, etc.) and across multiple different communications protocols. In some embodiments, equipment discovery is accomplished using active node tables,

which provide status information for devices connected to each communications bus. For example, each communications bus can be monitored for new devices by monitoring the corresponding active node table for new nodes. When a new device is detected, BMS 500 can begin interacting with the new device (e.g., sending control signals, using data from the device) without user interaction.

[0088] Some devices in BMS 500 present themselves to the network using equipment models. An equipment model defines equipment object attributes, view definitions, schedules, trends, and the associated BACnet value objects (e.g., analog value, binary value, multistate value, etc.) that are used for integration with other systems. Some devices in BMS 500 store their own equipment models. Other devices in BMS 500 have equipment models stored externally (e.g., within other devices). For example, a zone coordinator 508 can store the equipment model for a bypass damper 528. In some embodiments, zone coordinator 508 automatically creates the equipment model for bypass damper 528 or other devices on zone bus 558. Other zone coordinators can also create equipment models for devices connected to their zone busses. The equipment model for a device can be created automatically based on the types of data points exposed by the device on the zone bus, device type, and/or other device attributes. Several examples of automatic equipment discovery and equipment model distribution are discussed in greater detail below.

[0089] Still referring to FIG. 5, BMS 500 is shown to include a system manager 502; several zone coordinators 506, 508, 510 and 518; and several zone controllers 524, 530, 532, 536, 548, and 550. System manager 502 can monitor data points in BMS 500 and report monitored variables to various monitoring and/or control applications. System manager 502 can communicate with client devices 504 (e.g., user devices, desktop computers, laptop computers, mobile devices, etc.) via a data communications link 574 (e.g., BACnet IP, Ethernet, wired or wireless communications, etc.). System manager 502 can provide a user interface to client devices 504 via data communications link 574. The user interface may allow users to monitor and/or control BMS 500 via client devices 504.

[0090] In some embodiments, system manager 502 is connected with zone coordinators 506-510 and 518 via a system bus 554. System manager 502 can be configured to communicate with zone coordinators 506-510 and 518 via system bus 554 using a master-slave token passing (MSTP) protocol or any other communications protocol. System bus 554 can also connect system manager 502 with other devices such as a constant volume (CV) rooftop unit (RTU) 512, an input/output module (IOM) 514, a thermostat controller 516 (e.g., a TEC5000 series thermostat controller), and a network automation engine (NAE) or third-party controller 520. RTU 512 can be configured to communicate directly with system manager 502 and can be connected directly to system bus 554. Other RTUs can communicate with system manager 502 via an intermediate device. For example, a wired input 562 can connect a third-party RTU 542 to the thermostat controller 516, which connects to system bus 554.

[0091] System manager 502 can provide a user interface for any device containing an equipment model. Devices such as zone coordinators 506-510 and 518 and thermostat controller 516 can provide their equipment models to system manager 502 via system bus 554. In some embodiments,

system manager 502 automatically creates equipment models for connected devices that do not contain an equipment model (e.g., IOM 514, third party controller 520, etc.). For example, system manager 502 can create an equipment model for any device that responds to a device tree request. The equipment models created by system manager 502 can be stored within system manager 502. System manager 502 can then provide a user interface for devices that do not contain their own equipment models using the equipment models created by system manager 502. In some embodiments, system manager 502 stores a view definition for each type of equipment connected via system bus 554 and uses the stored view definition to generate a user interface for the equipment.

[0092] Each zone coordinator 506-510 and 518 can be connected with one or more of zone controllers 524, 530-532, 536, and 548-550 via zone buses 556, 558, 560, and 564. Zone coordinators 506-510 and 518 can communicate with zone controllers 524, 530-532, 536, and 548-550 via zone busses 556-560 and 564 using a MSTP protocol or any other communications protocol. Zone busses 556-560 and 564 can also connect zone coordinators 506-510 and 518 with other types of devices such as variable air volume (VAV) RTUs 522 and 540, changeover bypass (COBP) RTUs 526 and 552, bypass dampers 528 and 546, and PEAK controllers 534 and 544.

[0093] Zone coordinators 506-510 and 518 can be configured to monitor and command various zoning systems. In some embodiments, each zone coordinator 506-510 and 518 monitors and commands a separate zoning system and is connected to the zoning system via a separate zone bus. For example, zone coordinator 506 can be connected to VAV RTU 522 and zone controller 524 via zone bus 556. Zone coordinator 508 can be connected to COBP RTU 526, bypass damper 528, COBP zone controller 530, and VAV zone controller 532 via zone bus 558. Zone coordinator 510 can be connected to PEAK controller 534 and VAV zone controller 536 via zone bus 560. Zone coordinator 518 can be connected to PEAK controller 544, bypass damper 546, COBP zone controller 548, and VAV zone controller 550 via zone bus 564.

[0094] A single model of zone coordinator 506-510 and 518 can be configured to handle multiple different types of zoning systems (e.g., a VAV zoning system, a COBP zoning system, etc.). Each zoning system can include a RTU, one or more zone controllers, and/or a bypass damper. For example, zone coordinators 506 and 510 are shown as Verasys VAV engines (VVEs) connected to VAV RTUs 522 and 540, respectively. Zone coordinator 506 is connected directly to VAV RTU 522 via zone bus 556, whereas zone coordinator 510 is connected to a third-party VAV RTU 540 via a wired input 568 provided to PEAK controller 534. Zone coordinators 508 and 518 are shown as Verasys COBP engines (VCEs) connected to COBP RTUs 526 and 552, respectively. Zone coordinator 508 is connected directly to COBP RTU 526 via zone bus 558, whereas zone coordinator 518 is connected to a third-party COBP RTU 552 via a wired input 570 provided to PEAK controller 544.

[0095] Zone controllers 524, 530-532, 536, and 548-550 can communicate with individual BMS devices (e.g., sensors, actuators, etc.) via sensor/actuator (SA) busses. For example, VAV zone controller 536 is shown connected to networked sensors 538 via SA bus 566. Zone controller 536 can communicate with networked sensors 538 using a

MSTP protocol or any other communications protocol. Although only one SA bus **566** is shown in FIG. **5**, it should be understood that each zone controller **524**, **530-532**, **536**, and **548-550** can be connected to a different SA bus. Each SA bus can connect a zone controller with various sensors (e.g., temperature sensors, humidity sensors, pressure sensors, light sensors, occupancy sensors, etc.), actuators (e.g., damper actuators, valve actuators, etc.) and/or other types of controllable equipment (e.g., chillers, heaters, fans, pumps, etc.).

[0096] Each zone controller **524**, **530-532**, **536**, and **548-550** can be configured to monitor and control a different building zone. Zone controllers **524**, **530-532**, **536**, and **548-550** can use the inputs and outputs provided via their SA busses to monitor and control various building zones. For example, a zone controller **536** can use a temperature input received from networked sensors **538** via SA bus **566** (e.g., a measured temperature of a building zone) as feedback in a temperature control algorithm. Zone controllers **524**, **530-532**, **536**, and **548-550** can use various types of control algorithms (e.g., state-based algorithms, extremum seeking control (ESC) algorithms, proportional-integral (PI) control algorithms, proportional-integral-derivative (PID) control algorithms, model predictive control (MPC) algorithms, feedback control algorithms, etc.) to control a variable state or condition (e.g., temperature, humidity, airflow, lighting, etc.) in or around building **10**.

Adaptive Training and Deployment of Single Chiller and Clustered Chiller Fault Detection Models

[0097] Referring now to FIG. **6**, a system **600** is shown, according to an exemplary embodiment. The system **600** is shown to include a chiller analysis system **602** configured to generate prediction of equipment shutdowns. Chiller analysis system **602** is shown to communicate with chillers **626**, which include chiller sensors **628**. Chillers **626** may be one or multiple chillers, e.g., chiller **102** as described with reference to FIG. **1**. Chiller sensors **628** can be positioned on, within, and/or adjacent to chillers **626**, according to some embodiments. Further, chiller sensors **628** can be configured to collect a variety of data including usage time, efficiency metrics, input and output quantities, as well as other data. According to some embodiments, chiller sensors **628** can be configured to store and/or communicate collected chiller data. In some embodiments, chillers **626** can also be configured to store and/or communicate collected chiller data from chiller sensors **628**. Chiller analysis system **602** can receive performance data from chillers **626** and generate fault prediction models for the chillers and utilize the fault prediction models to determine if a fault will occur in the future for chillers **626**. Chiller analysis system **602** may not be limited to performing fault predictions for chillers and can also be configured to perform fault prediction for type of building equipment, e.g., air handler unit **106** as described with reference to FIG. **1**, boiler **104** as described with reference to FIG. **1**, etc.

[0098] Chiller analysis system **602** is shown to include a controller **604**, according to some embodiments. In some embodiments, controller **604** is shown to include a processing circuit **606**, a processor **608**, and a memory **610**. Controller **604** can include one or more computer systems (e.g., servers, supervisory controllers, subsystem controllers, etc.) that serve as system level controllers, application or data servers, head nodes, or master controllers for chiller analysis

system **602**. Controller **604** may communicate with multiple downstream systems or subsystems. Controller **604** can also be in communication with processing circuit **606**, processor **608**, and/or memory **610**, according to some embodiments.

[0099] Chiller analysis system **602** is shown to include processing circuit **606** which can be included in controller **604**, according to some embodiments. Processing circuit **606** is shown to include processor **608** and memory **610** in some embodiments. Processing circuit **606** can allow and enable communication between controller **604** and memory **610**, as well as other possible components that may be included in some embodiments. Processor **608** can be implemented as a general-purpose processor, an application specific integrated circuit (ASIC), one or more field programmable gate arrays (FPGAs), a group of processing components, or other suitable electronic processing components.

[0100] Chiller analysis system **602** is shown to include memory **610**, according to some embodiments. Memory **610** (e.g., memory, memory unit, storage device, etc.) can include one or more devices (e.g., RAM, ROM, Flash memory, hard disk storage, etc.) for storing data and/or computer code for completing or facilitating the various processes, layers and modules described in the present application. Memory **610** can be or include volatile memory or non-volatile memory. Memory **610** can include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described in the present application. According to some embodiments, memory **610** is communicably connected to processor **608** via processing circuit **606** and includes computer code for executing (e.g., by processing circuit **606** and/or processor **608**) one or more processes described herein.

[0101] Memory **610** is shown to include a model manager **612**, a trainer **620**, and chiller models **614**. Model manager **612** can be configured to manage chiller models **614**. Chiller models **614** are shown to include multiple single chiller models **616** and multiple cluster chiller models **618**. Model manager **612** can be configured to manage single chiller models **616** and/or cluster chiller models **618**. Chiller analysis system **602** may also vary according to desired preferences and specifications of a user and/or operator, according to some embodiments. For example, in some embodiments chiller analysis system **602** may be configured to perform certain tasks and functions periodically, with the intervals for which periodic tasks and functions are performed variable according to embodiment and user preferences. Further, some embodiments may be configured to accommodate a user and/or operator who desires various evaluations, analyses and re-training of models more frequently (i.e. shorter intervals) than another user and/or operator. As such, chiller analysis system **602** may be configured differently according to embodiment in order to perform a variety of functions with many configurations variable according to user and/or operator specifications, depending on the embodiment.

[0102] Model manager **612** is shown to be in communication with single chiller models **616** as well as cluster chiller models **618**, according to some embodiments. Model manager **612** may be configured to manage single chiller models **616** and cluster chiller models **618** at the same time, according to some embodiments. Further, model manager **612** may also be configured to perform certain analyses and evaluations for single chiller models **616** and cluster chiller models **618**. For example, model manager **612** may be

configured to—by means of user and/or operator preference, or other circumstance—periodically evaluate single chiller models **616** and/or cluster chiller models **618**, according to some embodiments. Additionally, model manager **612** may also be configured to make determinations as to the result of certain analyses and/or evaluations that may be performed with relation to single chiller models **616** and cluster models **618**, according to some embodiments. For example, upon completion of an evaluation of single chiller models **616** and/or cluster chiller models **618**, model manager **612** may determine if one or more of single chiller models **616** and/or cluster chiller models **618** may require modification, maintenance, and/or any other attention, according to some embodiments.

[0103] Chiller models **614** is shown to include single chiller models **616** and cluster chiller models **618**, according to some embodiments. Single chiller models **616** and cluster chiller models **618** are shown to be in communication with model manager **612**, as seen in some embodiments. Single chiller models **616** and cluster chiller models **618** can relate to chillers **626** and/or other similar components, according to some embodiments. Additionally, single chiller models **616** and cluster chiller models **618** can be built using data collected chillers **626**. Both single chiller models **616** and cluster chiller models **618** may be present within chiller models **614**. Further, chiller models **614** can also contain solely single chiller models **616** or solely cluster models **618**, according to some embodiments. Chiller models **614** can be in communication with model manager **612**. Further, chiller models **614** can communicate single chiller models **616** and/or cluster chiller models **618** to model manager **612** in some embodiments. Such communication of single chiller models **616** and/or cluster chiller models **618** may be scheduled, determined according to specific events that may occur, initiated by a user and/or operator, or a variety of other options, according to some embodiments.

[0104] Memory **610** is also shown to include trainer **620**, according to some embodiments. Trainer **620** is shown to be in communication with chiller models **614**, which can include single chiller models **616** and/or cluster chiller models **618**, according to some embodiments. As such, communication between trainer **620** and chiller models **614** can include the communication of single chiller models **616** and/or cluster chiller models **618**, according to some embodiments. Further, trainer **620** is shown to include a model selector **622** and a threshold selector **624**, according to some embodiments. Model selector **622** and threshold selector **624** can be in communication with chiller models **614**, as model selector **622** and threshold selector **624** are components configured within trainer **620**, which can communicate with chiller models **614**. Further, model selector **622** and threshold selector **624** can perform operations on single chiller models **616** and/or cluster chiller models **618**, according to some embodiments and user and/or operator preference.

[0105] Model selector **622** can perform operations involving single chiller models **616** and/or cluster chiller models **618**, according to some embodiments. For example, in some embodiments trainer **620** may receive single chiller models **616** and/or cluster chiller models **618**. Model selector **622** may subsequently evaluate, analyze, or perform other similar operations on and relating to single chiller models **616** and/or cluster chiller models **618** received by trainer **620**. Model selector can also be configured to evaluate and/or

otherwise analyze single chiller models **616** and/or cluster chiller models **618** according to various desired parameters and/or specifications. For example, the methods and techniques implemented by model selector **622** may vary based on the source and/or contents of single chiller models **616** and/or cluster chiller models **618**. Further, model selector **622** can be configured to operate differently according to user and/or operator specifications. For example, model selector **622** can be configured to select models from single chiller models **616** and/or cluster chiller models **618** that prioritize factors, parameters or variables desired by a user and/or operator such as efficiency, input or output quantities, as well as other possible factors.

[0106] Threshold selector **624** can perform operations on both single chiller models **616** as well as cluster chiller models **618**, according to some embodiments. As a component of trainer **620** which is shown to be in communication with single chiller models **614**, threshold selector **624** can communicate with single chiller models **616** and cluster chiller models **618**, which are components of chiller models **614**. In some embodiments, threshold selector **624** is configured to evaluate single chiller models **616** and/or cluster chiller models **618** and, through the implementation of one or more analysis techniques, select a desirable threshold for each corresponding model of single chiller models **616** and/or cluster chiller models **618**. The selection process of threshold selector **624** can in some embodiments include obtaining an operating point, which can incorporate user and/or operator preference. Further, threshold selector **624** can be configured to perform operations on, analyze and otherwise interact with single chiller models **616** and/or cluster chiller models **618** in various ways in order to select a threshold for one or more models. Additionally, threshold selector can be configured to operate the same or differently with regard to single chiller models **616** and cluster chiller models **618**. Threshold selector **624** can also be configured to implement different analysis techniques and/or other operations in the selection of thresholds for single chiller models **616** than for cluster chiller models **618**. For example, in some embodiments threshold selector **624** may be configured to incorporate user and/or operator preferences, such as a desire to optimize specific metrics, parameters and functions, differently than in some embodiments. Further, in the instance that a user and/or operator desired a threshold that optimized efficiency, threshold selector **624** can be configured to accommodate such a desire.

Single Chiller and Cluster Chiller Fault Detection Modeling

[0107] Referring now to FIG. 7A, a system for model management **700** is shown. System **700** is shown to include a chiller **702** and chiller sensors **704**, as well as a chiller cluster **706** which includes chiller cluster sensors **708**. Chiller **702** and chiller cluster **706** can be the same and/or similar to chiller **626s**, which is described in reference to FIG. 6. Similarly, both chiller sensors **704** and chiller cluster sensors **708** can be the same and/or similar to chiller sensors **628** which are described in reference to FIG. 6.

[0108] Chiller **702** and chiller cluster **706** are shown to communicate chiller data and cluster data, respectively, to a cluster manager **710**. In some embodiments, chiller data communicated to cluster manager **710** may include data collected from and processed by chiller sensors **704**. Chiller data may also pertain to chiller **702** as a whole, may be directed at a specific component of chiller **702**, and/or may

relate to other systems that may be working in conjunction with chiller 702. Similar to the chiller data communicated by chiller 702, cluster data communicated by chiller cluster 706 may include data collected by chiller cluster sensors 708. Cluster data may include data collected from and processed by chiller cluster sensors 708, and may correspond to components of one or more chillers, individual chillers, and/or chiller cluster 706 as a whole. Both chiller data and cluster data communicated to cluster manager 710 may vary according to some embodiments. For example, chiller data and cluster data communicated to cluster manager 710 may include input and/or output quantities, efficiency metrics, and/or past data for chiller 702 and/or chiller cluster 706, according to some embodiments.

[0109] Model manager 710 can receive both chiller data and cluster data from chiller 702 and chiller cluster 706 and the corresponding chiller sensors 704 and chiller cluster sensors 708. Model manager 710 can generate models through various means upon receiving chiller data and cluster data. Further, model manager 710 can be configured to be in communication with more components than chiller 702 and chiller cluster 706. In some embodiments, model manager 710 can be in communication with a plurality of equipment similar to chiller 702 and/or may also be in communication with a plurality of equipment clusters similar to chiller cluster 706. Model manager 710 can be configured to evaluate received chiller data and/or cluster data and evaluate and compile said data accordingly. For example, model manager 710 can be configured to sort and subsequently compile chiller data from cluster data in some embodiments or can also be configured to sort received data according to other variables such as data contents, time period, and data source, among other variables. Upon compiling chiller data and cluster data, model manager 710 can be configured to analyze, sort, and otherwise process the received data in a variety of ways. For example, model manager 710 may analyze received chiller data differently than received chiller cluster data. Additionally, model manager 710 can be configured to validate received data. In some embodiments, model manager 710 can generate models for chiller data and cluster data using the same or similar methods and techniques, while in some embodiments the methods and techniques for generating chiller models may be different than those for generating cluster models. Model manager 710 can also assess performance of the designated models, which can be a process that can be performed the same for chiller and/or cluster models or can be performed using different methods. In some embodiments, chiller models may be assessed, and upon completion of assessment, can be assigned to a cluster. Further, cluster models can also be assessed in some embodiments. Model manager 710 can label both chillers and clusters as good chillers, bad chillers, or good clusters, according to some embodiments. Good chillers can also be considered accurately predicting chillers, and good models can also be considered accurately predicting models, as models generated by such chillers can produce accurate predictions, according to some embodiments. In some embodiments, good chillers (or accurately predicting chillers) can produce data used in the generation of good chiller models (or accurately predicting chiller models). Similarly, good clusters can, in some embodiments, include clusters of which collected data can be used to generate models that can accurately predict future activity of chillers and/or clusters. Good clusters can also be considered accu-

ately predicting clusters and good cluster models can also be considered accurately predicting cluster models. Additionally, bad chillers can include chillers of which collected data cannot be used to generate models that can accurately predict future chiller activity. Bad chillers can also be considered fault chillers, as models generated by such chillers may not perform without fault. Model manager 710 can then deploy models that have been determined to perform sufficiently, with all models also being re-evaluated periodically. Model manager can be configured to accommodate user and/or operator preferences, such as optimizing specific parameters such as efficiency or output, among other parameters.

[0110] Model manager 710 is shown to include a chiller cluster evaluator 712. Chiller cluster evaluator 712 is shown to be in communication with chiller 702 and chiller cluster 706, and receive chiller data and cluster data, respectively. Chiller cluster evaluator 712 may be in communication with a plurality of equipment similar to chiller 702 and/or a plurality of equipment clusters similar to chiller cluster 706 in some embodiments. Chiller cluster evaluator 712 may perform a variety of functions in some embodiments. For example, chiller cluster evaluator 712 may evaluate inputted data from chiller 702 and/or chiller cluster 706 and determine which data corresponds to chiller 702 and/or similar, and which data corresponds to chiller cluster 706 and/or similar, among other functions. Chiller cluster evaluator 712 may perform other functions that evaluate data based on origin in some embodiments. For example, chiller cluster evaluator 712 may evaluate inputted chiller data and chiller cluster data in order to evaluate other variables associated with inputted data, such as geographic location and time, among other possible variables. Chiller cluster evaluator 712 may also evaluate chiller data and chiller cluster data in order to store, sort, and/or otherwise process said data in order to prepare the data for further evaluation and analysis. Additionally, chiller cluster evaluator 712 is shown to be in communication with a data compiler 714, according to some embodiments. Communication between chiller cluster evaluator 712 and data compiler 714 can include, for example, sorted data from both chiller 702 and chiller cluster 706 and/or other similar components.

[0111] Model manager 710 is shown to include data compiler 714. Data compiler 714 is shown to be in communication with chiller cluster evaluator 712, as well as a training data evaluator 716, according to some embodiments. Data compiler 714 can compile data communicated to and evaluated by chiller cluster evaluator 712 that can be received from chiller 702 and/or chiller cluster 706, according to some embodiments. Data compiler 714 can also prepare data that may include chiller data and cluster data outputted by chiller 702 and chiller cluster 706 for analysis by training data evaluator 716. For example, in some embodiments data compiler 714 can compile data from chiller 702 and chiller cluster 706 differently and/or separately in order to facilitate evaluation of said data by training data evaluator 716. Additionally, data compiler 714 can compile data from multiple sources. For example, data compiler 714 may compile data from chiller 702, as well as other equipment that may be similar to chiller 702, and/or may also compile data from chiller cluster 706 as well as other equipment clusters that may be similar to chiller cluster 706, according to some embodiments.

[0112] Model manager 710 is shown to include training data evaluator 716, according to an. Training data evaluator 716 is shown to be in communication with data compiler 714, as well as a chiller model validator 718 and a cluster model validator 720, according to some embodiments. In some embodiments, training data evaluator 716 can receive an input from data compiler 714. Data compiler 714 can input data that has been collected and compiled from chiller 702 and/or other similar equipment, as well as data that may have been collected and compiled from chiller cluster 706 and/or other similar equipment clusters. Training data evaluator 716 can evaluate data inputted from data compiler 714 in a variety of ways. For example, training data evaluator 716 can determine which data received from data compiler 714 corresponds to chiller 702 and/or other similar equipment, and which data corresponds to chiller cluster 706 and/or similar equipment clusters. Training data evaluator 716 can also prepare data received from data compiler 714 to be communicated to chiller model validator 718 and cluster model validator 720, according to some embodiments. For example, training data evaluator 716 can evaluate data received from data compiler 714 in order to prepare data initially collected by chiller 702 and/or other similar equipment to be communicated to chiller model validator 718 and can also prepare data initially collected by chiller cluster 706 and/or other similar equipment clusters to be communicated to cluster model validator 720, according to some embodiments.

[0113] Model manager 710 is shown to include chiller model validator 718 and cluster model validator 720, according to some embodiments. Additionally, both chiller model validator 718 and cluster model validator 720 can be in communication with and receive an input from training data evaluator 716 and can also be in communication with historical operational data 722. Chiller model validator 718 and cluster model validator 720 can operate in a variety of ways. For example, chiller model validator 718 can receive data from training data evaluator 716 that has been evaluated and determined to have originated from chiller 702 and/or other similar equipment. Chiller model validator 718 can then compare data from chiller 702 and/or other similar equipment to historical operational data 722, which may include data previously collected from chiller 702 and/or other similar equipment. Chiller model validator 718 can then determine the validity of data received from training data evaluator 716, which may include determining if said data received from training data evaluator 716 accurately represents activity of chiller 702 and/or other similar equipment. Similarly, cluster model validator 720 can receive data from training data evaluator 716 that has been evaluated and determined to have originated from chiller cluster 706 and/or other similar equipment clusters. Cluster model validator 720 can then compare data from chiller cluster 706 and/or other similar equipment clusters to historical operational data 722, which can include data previously collected from chiller cluster 706 and/or other similar equipment clusters. In some embodiments, cluster model validator 720 can then determine the validity of data received from training data evaluator 716, which can include determining if said data received from training data evaluator 716 accurately represents activity of chiller cluster 706 and/or other similar equipment clusters. It should also be noted that chiller model validator 718 may receive data from training data evaluator 716 that may originate from chiller 702

and/or a plurality of similar equipment, and cluster model validator 720 may similarly receive training data from training data evaluator 716 that may originate from chiller cluster 706 and/or a plurality of similar equipment clusters, according to some embodiments. It should also be noted that historical operational data 722 can include data originating from chiller 702 and/or a plurality of other similar equipment and can also include data originating from chiller cluster 706 and/or a plurality of other similar equipment clusters. Historical operational data 722 can vary in some embodiments, and parameters of historical operational data 722 including quantity and time period can also vary according to some embodiments.

[0114] Model manager 710 is shown to include a prediction performance assessor 724, according to some embodiments. Prediction performance assessor 724 is shown to be in communication with chiller model validator 718 as well as cluster model validator 720, according to some embodiments. Prediction performance assessor 724 can receive an input from both chiller model validator 718 as well as cluster model validator 720. Prediction performance assessor 724 can also assess the ability of the chiller data and cluster data validated by chiller model validator 718 and cluster model validator 720 to predict events that may occur for chiller 702 and/or other similar equipment, as well as chiller cluster 706 and/or other similar equipment clusters. For example, some chiller data may be validated by chiller model validator 718, which may indicate that the data and corresponding model is a valid representation of normal activity of chiller 702 and/or other similar equipment. Prediction performance assessor 724 may then assess the ability of the data and the corresponding (now validated) model to predict various events known to occur with chiller 702 and/or other similar equipment. In some embodiments, it may be advantageous to predict a shutdown of chiller 702 and/or other similar equipment prior to the shutdown occurring, in which case prediction performance assessor 724 may assess the validated data and corresponding for the ability to accurately predict a shutdown of chiller 702 and/or other similar equipment. Prediction performance assessor 724 can also perform similar assessment functions for data and corresponding models received from cluster model validator 720 and assess the ability of said data and models to predict events in chiller cluster 706 and/or other similar equipment clusters, such as shutdowns.

[0115] Model manager 710 is shown to include a cluster assigner 726. Cluster assigner 726 is shown to be in communication with prediction performance assessor 724, according to some embodiments. Additionally, chiller model validator 718 is also shown to be in communication with cluster assigner 726, according to some embodiments. Cluster assigner 726 can operate so as to assign single chillers, which may include chiller 702 and/or other similar equipment, to a cluster such as chiller cluster 706 and/or other similar equipment cluster. For example, in the event that chiller model validator 718 determines that data and the corresponding model for a chiller such as chiller 702 and/or other similar equipment is invalid, said chiller can be assigned by cluster assigner 726 to a cluster such as chiller cluster 706 and/or other similar equipment clusters. The assignment of said chiller to a cluster by cluster assigner 726 may allow for the chiller to operate as a part of a cluster such as cluster 706 and/or other similar equipment clusters and thus become integrated with a validated cluster model of the

assigned cluster. Cluster assigner **726** can also assign a chiller such as chiller **702** and/or other similar equipment to a plurality of clusters such as chiller cluster **706** and/or other similar equipment clusters.

[0116] Model manager **710** is shown to include a list of bad chillers **728**, a list of good chillers **730**, and a list of good clusters **732**. It should be noted that, in some embodiments, good chillers and good clusters such as those of list of good chillers **730** and list of good clusters **732** can be referred to as accurately predicting chillers, accurately predicting clusters, normal chillers, and normal clusters, which is to say that accurately predicting chillers and accurately predicting clusters can be synonymous to good chillers and good clusters. Further, accurately predicting chillers and accurately predicting clusters can also correspond to the generation of accurately predicting models, according to some embodiments. It should also be noted that good chillers and good clusters can also be referred to as non-fault chillers and non-fault clusters, according to some embodiments. Similarly, it should be noted that, in some embodiments, bad chillers such as those of list of bad chillers **728** can be referred to as inaccurately predicting chillers and abnormal chillers, which is to say that inaccurately predicting chillers and abnormal chillers can be synonymous to bad chillers. It should also be noted that inaccurately predicting chillers can also be referred to as fault chillers, according to some embodiments. Further, inaccurately predicting chillers can correspond to the generation of inaccurately predicting models, according to some embodiments. List of bad chillers **728**, list of good chillers **730**, and list of good clusters **732** are also shown to be in communication with prediction performance assessor **724**, according to some embodiments. List of bad chillers **728** can correspond to a list of chillers, such as chiller **702** and/or other similar equipment, that has been determined to fail to meet established criteria. For example, if the data collected from a chiller such as chiller **702** and/or other similar equipment is deficient in that it does not accurately represent behavior of the chiller and/or is unable to aid in predicting future events of said chiller, then that chiller may be added to list of bad chillers **728**, according to some embodiments. List of bad chillers **728** can include a plurality of chillers such as chiller **702** and/or other similar equipment and can also be dynamic in that chillers may be added and removed from the list periodically. List of good chillers **730** can also be in communication with prediction performance assessor **724**, according to some embodiments. A chiller such as chiller **702** and/or other similar equipment may be added to list of good chillers **730** because the chiller may provide sufficient data that accurately represents the behavior of the chiller and/or it may aid in the prediction of future events of said chiller and/or other similar chillers. For example, if a chiller such as chiller **702** and/or other similar equipment may produce data indicative of normal behavior of the chiller that may contribute to efforts to predict chiller events such as possible future shutdowns, said chiller may be added to list of good chillers **728**. Similar to list of bad chillers **728**, list of good chillers **730** is dynamic in that chillers can be added and/or removed depending on circumstances and embodiments. List of good clusters **732** is also shown to be in communication with prediction performance assessor **724**, according to some embodiments. A cluster such as chiller cluster **706** and/or other similar equipment cluster may be added to list of good clusters **732** because the cluster can provide sufficient data

that accurately represents the behavior of the cluster and/or it may aid in the prediction of future events of said cluster and/or other similar clusters. Similar to both list of bad chillers **728** and list of good chillers **730**, list of good clusters **732** is dynamic in that clusters may be added and/or removed according to some embodiments.

[0117] Model manager **710** is shown to include a connected chiller service **734**. Connected chiller service **734** is shown to be in communication with list of good chillers **730** as well as list of good clusters **734**, according to some embodiments. Connected chiller service **734** can include all chillers and clusters that have been assessed by prediction performance assessor **724**, and subsequently added to list of good chillers **730** and list of good clusters **732**, according to some embodiments. Connected chiller service **734** can be connected to a plurality of chillers and clusters such as chiller **702** and chiller cluster **706**, respectively, with the connected chillers and clusters potentially being in a plurality of locations. Connected chiller service **734** can also compile data and corresponding models for both chillers and clusters that have been validated by chiller model validator **718** and cluster model **720**, respectively. Data and models compiled by connected chiller service **734** can include models for specific chillers and/or clusters of which the performance has been assessed by prediction performance assessor **724**, according to some embodiments. Assessed data and models of connected chiller service **734** may be assessed for the ability to predict possible future events of chillers and/or clusters, such as shutdowns. Connected chiller service **734** can also include data and models that have been assessed and determined to be good models for specific chillers and/or clusters, as well as data and models that may have been assessed and determined to be good models for general chillers and/or clusters.

[0118] Model manager **710** is shown to include a periodic evaluator **736**. Periodic evaluator **736** is shown to be in communication with list of bad chillers **728**, as well as connected chiller service **734**, according to some embodiments. Periodic evaluator **736** can perform a number of functions, which can depend on the chiller and/or cluster and the corresponding model and data to be evaluated. For example, periodic evaluator **736** can evaluate a specific set of chiller data and the corresponding model for the ability to accurately predict events of one or more chillers, such as a shutdown. Periodic evaluator **736** can also incorporate other operational data into such an evaluation, such as data from previous shutdowns as well as other available chiller operational data. Periodic evaluator **736** can also contribute to the dynamic nature of list of bad chillers **728**, list of good chillers **730**, and list of good clusters **732**, according to some embodiments. For example, if periodic evaluator **736** is configured to evaluate chiller and/or cluster models on a monthly basis, such evaluations may account for operational data for relevant chillers and clusters that has been collected since the most recent period evaluation. As such, periodic evaluator **736** can determine that data and the corresponding model collected from a previously good chiller has not recently been effective in predicting future events of that chiller and/or similar chillers and may reassign such a chiller to list of bad chillers **728**. Similarly, a chiller with data and a corresponding model that was previously ineffective in predicting future events of said chiller and/or other chillers may now produce new data and a new corresponding model that is now effective in predicating future events of the

chiller and/or other chillers, with said chiller being re-assigned by periodic evaluator **736** to list of good chillers **730**. It should also be noted that periodic evaluator **736** can evaluate chillers and clusters based on a number of different parameters, as well as based on chiller data and cluster data collected from chiller sensors **704** of chiller **702** and chiller cluster sensors **708** of chiller cluster **706**, according to some embodiments. Such parameters that may be evaluated by periodic evaluator **736** can include input and output quantities, consumption quantities, efficiency metrics, as well as other possible parameters that may indicate performance and activity of chillers and clusters such as chiller **702** and chiller cluster **706**.

[0119] Referring now to FIG. 7B, a process **740** of chiller evaluation is shown. Steps of process **740** may be performed by some and/or all of the components of FIG. 7A, which shows system **700**. Process **740** can be applied to a single chiller and/or piece of equipment, such as chiller **702**, and/or multiple pieces of equipment similar to and/or thereof. Similarly, process **740** can also be performed for a chiller cluster and/or equipment cluster, such as chiller cluster **706** of FIG. 7A, and/or multiple equipment clusters similar to and/or thereof. It should be noted that process **740** may include additional steps, different steps, or otherwise modified steps from those seen in the process **740** in FIG. 7B.

[0120] In step **742**, chiller sensors **704** and chiller cluster sensors **708** collect chiller data and chiller cluster data, respectively. Additionally, step **742** may include both chiller data and chiller cluster data, only chiller data, or only cluster data, according to some embodiments. Data collected in step **742** may also vary, but may include input and output quantities, consumption quantities, and efficiency metrics, as well as other possible data. Collected data of step **742** can also be evaluated by a component the same as or similar to chiller cluster evaluator **712** in some embodiments. Chiller cluster evaluator **712** can serve to sort and otherwise organize data, as well as perform other functions in some embodiments.

[0121] In step **744**, process **740** compiles the chiller data and the chiller cluster data of step **742** to be prepared for validation. Compilation of chiller data and chiller cluster data in step **744** can be performed by data compiler **714** of FIG. 7A and/or a similar component, according to some embodiments. Additionally, similar to step **742**, step **744** can include compiling solely chiller data, solely chiller cluster data, or a combination of the two. Further, the preparation of the aforementioned chiller data and/or chiller cluster data for validation can be done in various ways. For example, in some embodiments preparation for validation may include chronologically organizing data, while some embodiments may include organizing the data according to source, while some embodiments may include alternative data preparation.

[0122] In step **746**, process **740** trains and validates generated models. In some embodiments, models can be trained and validated based on chiller data and chiller cluster data and can also incorporate historical operational data **722**. The training and validation of the models can be the same or different based on the incorporation of chiller data and/or cluster data. For example, step **746** may include training and validating single chiller models using data that has been collected from single chillers rather than chiller clusters, with the same being true for and historical operational data **722** that may be implemented. Similarly, step **746** may

include training and validating chiller cluster models using data that has been collected from chiller clusters rather than single chillers, with the same being true for any historical operational data **722** that may be incorporated in the training and validating processes. Training processes can be different for single chiller models and chiller cluster models in some embodiments. For example, machine learning, deep learning, and transfer learning techniques implemented in the training of single chiller models may be the same as, similar to, or different from the machine learning, deep learning, and transfer learning techniques used in training chiller cluster models. Further, various validation methods can be used for both validating single chiller models and chiller cluster models.

[0123] In step **748**, process **740** assesses performance of models and categorizing chillers and chiller clusters and making cluster, according to some embodiments. Step **748** can include prediction performance assessor **724**, as well as cluster assigner **726** of FIG. 7A, according to some embodiments. Performance of models in step **742** can be analyzed using various means, methods and techniques, which may vary in some embodiments. For example, models associated with chiller training data can be analyzed differently than models associated with chiller cluster training data, according to some embodiments. Further, assignment of chillers to clusters can vary depending on various factors including user and/or operator preferences. In some embodiments, performance assessment techniques may be variable, and as such so too may list of bad chillers **728**, list of good chillers **730**, and list of good clusters **732** of FIG. 7A.

[0124] In step **750**, process **740** is shown to include sending training data and corresponding models for good chillers and good clusters to connected chiller service. As mentioned with step **748** above, chillers and chiller clusters designated as good chillers and good chiller clusters can vary. Similarly, the models thereof included in step **750** can also vary according to some embodiments in terms of which chillers and/or chiller clusters can have corresponding training data and models sent to a component the same or similar to connected chiller service **734** of FIG. 7A, according to some embodiments. As such, connected chiller service of step **750** can vary, for example including solely chiller training data and models in some embodiments, including solely chiller cluster training data and models in some embodiments, and including both chiller and chiller cluster training data and models in some embodiments.

[0125] In step **752**, process **740** is shown to include periodically evaluating all bad chillers, good chillers, and good chiller clusters according to some embodiments. Chillers and chiller clusters that can be designated as good or bad can vary. Additionally, all bad chillers, good chillers, and good chiller clusters can be periodically evaluated and/or re-evaluated using steps similar to those of process for chiller evaluation **740**. Further, the methods and/or techniques that can be used for periodic evaluation and re-evaluation of bad chillers, good chillers, and good chiller clusters, as well as their associated training data and models, can vary according to some embodiments and can also vary according to user and/or operator preference. Also, step **752** of process **740** can be the final step of process for chiller evolution **740**, according to some embodiments. In some embodiments, process **740** may be cyclical, which is to say that some or all of the steps seen in FIG. 7B may be repeated multiple times.

[0126] Referring now to FIG. 7C, a flow diagram of a process 760 for model training and evaluation is shown. It should be noted that the flow diagram seen in FIG. 7C can include one or more steps and/or processes that may be similar to and/or performed by the components of FIGS. 7A-B, according to some embodiments. FIG. 7C is shown to include both single chillers and cluster chillers, as well as an initial evaluation to determine, identify, and/or sort single chillers from cluster chillers, according to some embodiments. In some embodiments, all existing chillers of interest are known, with said chillers then identified as being either single chillers or part of a chiller cluster. Depending on whether chillers are identified as single chillers or cluster chillers, model training and evaluation can occur differently.

[0127] Process 760 is shown to include all existing chillers of interest 762, for which process for model training and evaluation 760 also includes single chiller training and model evaluation, similar to steps 742 and 744 of FIG. 7B, according to some embodiments. In the instance that single chillers are identified in the initial step of model evaluation, a determination can then be made as to whether training data is available by a data identification 764, which is similar to step 746 of FIG. 7B, according to some embodiments. If a determination can be made that training data is not available, a single chiller may be allocated to a cluster for which one or more models are available, similar to step 748 of FIG. 7B, according to some embodiments. Should training data be available, machine learning, deep learning and/or transfer learning may be applied by a chiller model training 772 in order to generate and/or train one or more models for a single chiller, with historic chiller operational data 770 being incorporated as well according to some embodiments. Once one or more models for a single chiller have been generated, model performance may then be evaluated by a chiller performance and prediction capabilities of the one or more models is deemed to be acceptable, and similar to step 750 of FIG. 7B, the one or more models can then undergo a good chiller identification 782 and can be stored with other identified good models as part of a connected chiller service 778, which can be similar to connected chiller service 734 of FIG. 7A. Other chillers may undergo a bad chiller identification 784. In some embodiments, good models for single chillers can be stored in connected chiller service 778 and used to a shutdown prediction 790, according to some embodiments. Alternatively, in the instance that the performance and prediction capabilities of the one or more models is not deemed to be acceptable, the one or more models can then undergo a periodic evaluation 780, similar to step 752 of FIG. 7B, and the corresponding chillers may be allocated to a list of bad chillers. In the instance that the single chiller is deemed to be a bad chiller, it may be subject to periodic evaluation 780 as such.

[0128] Process 760 is shown to include cluster chiller training and model evaluation as well as single chiller model training and evaluation. Following the identification of cluster chillers rather than a single chiller, a cluster evaluation 766 can implement a clustering algorithm, similar to steps 744 and 746 of FIG. 7B. A new chiller addition 788 may also be incorporated into a cluster, according to some embodiments. Machine learning, deep learning, and/or transfer learning may then be implemented by a cluster model training 768 in order to train cluster chiller models. Historic chiller operational data 770 may also be incorpo-

rated, according to some embodiments. Cluster chiller models may then be evaluated by a cluster performance evaluation 776 for acceptable performance and prediction capabilities, similar to steps 748 and 750 of FIG. 7B, according to some embodiments. In the instance that performance and prediction capabilities are not verified and deemed unacceptable, cluster chillers may be listed for cluster membership, and may then undergo periodic re-evaluation as such, according to some embodiments. In the instance that model performance and prediction capabilities are deemed to be acceptable and are verified, cluster chiller models may be stored in connected chiller service 778 and used to generate a shutdown prediction 790 of for shutdown events, similar to step 750 of FIG. 7B, according to some embodiments. Models deemed to have acceptable performance and prediction capabilities may also be subject to periodic evaluation 780 and re-training as well, similar to step 752 of FIG. 7B.

Adaptive Model Selection with Optimal Hyper-parameters for Anomaly Detection of Connected Chillers

[0129] Referring now to FIG. 8A, a system for model training and selection 800 is shown. System 800 may be configured to train a plurality of different models for a variety of different equipment, similar to chiller 702 and chiller cluster 706 of FIG. 7A. Additionally, system 800 may also be configured to continuously and/or adaptively train machine learning and deep learning models corresponding to both chiller 702 and chiller cluster 706 of FIG. 7A, which is to say that system 800 may be configured to train single-chiller and multiple-chiller models based on available historic training data. System 800 may also implement hyper-parameter tuning of machine learning and/or deep learning models in order to identify and select well-performing models for deployment for connected chiller services, which may be similar to connected chiller service 734 of FIG. 7A. Also, system 800 may also be configured to continuously and/or adaptively re-evaluate and re-train well-performing models as well as poorly performing models, according to some embodiments. This re-evaluating and re-training may involve the implementation and analysis of cluster models, as well as the use of hyper-parameters for tuning purposes. Recently accumulated historic training data may also be a component of the re-evaluation and re-training process that may be performed by system 800, which may also include updating event detection models. That is to say that as operational data is accumulated, this data may be implemented in re-evaluation and re-training of models in order to improve the prediction of future events of components such as chiller 702 and/or chiller cluster 706 such that shutdowns, faults, and other possible events may be more accurately predicted. Additionally, system for model training and selection may also include transfer learning as one possible means of best training models. For example, system 800 may allow for data and corresponding single-chiller models to be applied to clustered-chiller models, and vice-versa. System 800 may also implement machine learning and/or deep learning practices in conjunction with transfer learning practices, for example applying weights and biases of a deep neural network model.

[0130] System 800 may also lead to an increased accuracy of fault prediction models for chiller 702 and chiller cluster 706 of FIG. 7A, which may include connected chillers and/or connected chiller services such as connected chiller service 734 also of FIG. 7A, according to some embodi-

ments. An increase in prediction accuracy for events that may occur for any possible connected chillers and/or other equipment may include many benefits to operators. For example, accurately predicting an event such as a shutdown for a component such as a chiller may allow operators to reduce equipment down time, thus reducing operational costs. Additionally, this may lead to a reduction in possible maintenance costs in the instance that operators may be able to optimize the scheduling of chiller maintenance services. Prediction of equipment events including shutdowns may also allow for reduced repair costs, with appropriately scheduled maintenance possibly reducing the likelihood of emergency maintenance being required. Prediction of possible equipment events may also allow for operators to improve customer service, as well as potentially extend the possible lifespan of equipment. Another possible advantage that may result from the implementation of system 800 is an overall increased revenue for the operator of the equipment, with accurate predictions allowing for ideal operation of subsequent equipment.

[0131] System 800 is shown to include a chiller 802, which is in turn shown to include chiller sensors 804. Chiller 802 is also shown to be in communication with a current operational data compiler 808, which is included in a trainer 806, according to some embodiments. Chiller sensors 804 of chiller 802 may be positioned, on, within, or adjacent to chiller 802, and may also be configured to collect a variety of data. For example, chiller sensors 804 may collect data that includes input and output values, duration of operations, efficiency metrics, as well as other possible data pertaining to chiller 802 and/or the operation thereof. Chiller 802 may be a single unit that operates a system, or may be part of a collection of components, possibly a cluster, with chiller 802 possible being the same, similar to, or different than other components of said collection. Chiller 802 may also have specific attributes that vary. For example, chiller location, orientation, size, capacity, age, and other factors may vary and as such, data collected by chiller sensors 804 may also vary depending on both the specific attributes and characteristics of chiller 802. Chiller sensors 804 may collect data, and may also store, sort, and communicate data with other components that may exist within system 800.

[0132] System 800 is shown to include trainer 806. Trainer 806 is shown to include several components, according to some embodiments. Trainer 806 can also aid in the implementation of some and/or all of the deep learning and machine learning techniques discussed above that components of system 800 collectively implement. For example, while chiller sensors 804 collect data for chiller 802, the analysis of collected data as well as subsequent re-evaluation and re-training of existing models may be performed at least in part by trainer 806. Trainer 806 may operate iteratively and may also operate on a predetermined and/or variable cycle for occur. Trainer 806 may also include some, all, and/or different components than those seen in FIG. 8A, and such variations in components may depend on variables including specifics of chiller 802, chiller sensors 804, as well as other possible factors.

[0133] Trainer 806 can implement different training methods and techniques, which may depend on a variety of factors. For example, training a single chiller model may involve processes that may be the same and/or different than processes used in training a chiller cluster model. Further, training practices and procedures may vary according to the

incorporation of chiller data. For example, training a single chiller model may not involve the use of any chiller cluster data, while training a chiller cluster model may require the implementation of single chiller data. In some embodiments, historical operational data 722 or similar may be incorporated into the processes of trainer 806. For example, historical operational data 722 or similar may contain data from a critical event for which a single chiller model and/or chiller cluster models must be trained to predict. Trainer 806 may also implement historical operational data 722 or similar for a variety of different time frames and scenarios. For example, some historical operational data 722 or similar may become obsolete by means of installation of new chillers, new chiller and/or chiller cluster models, as well as other means. Trainer 806, in some embodiments, can determine historical operational data 722 or similar relevant to specific chillers, as well as specific chiller and/or chiller cluster models in order to train models as accurately as possible. Further, the training processes of trainer 806 can include machine learning, deep learning, and transfer learning techniques similar to those used by various models for prediction.

[0134] Trainer 806 may also implement hyper-parameters in the training of both single chiller and chiller cluster models. In some embodiments, each model for single chillers and/or chiller clusters has a set of hyper-parameters that can be tuned in order to adjust model accuracy. As such, trainer 806 may train models with multiple sets of hyper-parameters in order to evaluate the predictions of the models with the various different sets of hyper-parameters applied. In some embodiments in which machine learning and/or deep learning and/or transfer learning models may be used in the generation of predictions such as the application of single chiller models and cluster chiller models, hyper-parameter iteration can be critical to the accuracy of model predictions. As such, trainer 806 can be configured to perform iterative hyper-parameter processes in order to maximize the accuracy of the predictions generated by the machine learning, deep learning, and transfer learning models of the single chiller and chiller cluster models, according to some embodiments.

[0135] Trainer 806 is shown to include current operational data compiler 808. Current operational data compiler 808 is shown to be in communication with a daily shutdown predictor 810, according to some embodiments. Current operational data compiler 808 may also be in communication with chiller 802 and chiller sensors 804 thereof, which may allow for data collected by chiller sensors 804 relating to chiller 802 to be communicated to current operational data compiler 808. Such data that may be communicated from chiller 802 to current operational data compiler 808 may include usage details, input and output quantities, efficiency metrics, as well as other possible data pertaining to chiller 802 as it operates. That is to say that the data communicated from chiller 802 to current operational data compiler 808 may not impact the immediate re-evaluation and re-training of models that may be performed as a part of trainer 806 and subsequently system for model training 808.

[0136] Trainer 806 is shown to include daily shutdown predictor 810. Daily shutdown predictor 810 is shown to be in communication with current operational data compiler 808, as well as a data updater 826 and a period updater 828, according to some embodiments. Daily shutdown predictor 810 can, through communication with current operational

data compiler **808**, receive current operational data so as to provide a basis for the generation of a daily shutdown prediction, according to some embodiments. For example, daily shutdown predictor **810** may receive current operational data from current operational data compiler **808**, which can impact a daily shutdown prediction generated by current operational data compiler **808**, according to some embodiments. Daily shutdown predictor **810** can generate various daily shutdown predictions, with said predictions variable based on current operational data received from current operational data compiler **808**, as well as the specifics of chiller **802** and the data collected by chiller sensors **804**, which may vary according to some embodiments.

[0137] Trainer **806** is also shown to include data updater **826**. Data updater **826** is also shown to be in communication with daily shutdown predictor **810**, according to some embodiments. As such, data updater **826** can receive daily shutdown predictions that have been generated by daily shutdown predictor **810**, and may update data accordingly, according to some embodiments. Data updater **826** can update a variety of data upon the receipt of daily shutdown predictions from daily shutdown predictor **810**, with the data that may be updated to include data and corresponding models that may be analyzed and implemented in the re-evaluation and re-training of chillers such as chiller **802**, for example. Data updater **826** can also update general data with daily shutdown predictions, which may subsequently allow for analysis of daily shutdown predictions in relation to data collected by chiller sensors **804**. That is to say that data updater **826** may update a variety of data, which may allow for the comparison and accuracy evaluation of daily shutdown predictions generated by daily shutdown predictor **810** in relation to data collected by chiller sensors **804**. Data updater **826** may also update other stored data to include the most recently collected data that may enable further analysis of models in use for the training, possibly through transfer learning, machine learning, and/or deep learning, of chillers such as chiller **802**, for example.

[0138] Trainer **806** is shown to include period updater **828**. Period updater **828** is shown to be in communication with data updater **826**, according to some embodiments. Period updater **828** may update a variety of components of trainer **806** relative to new data that has been collected and subsequently updated by data updater **826**, according to some embodiments. For example, after current operational data compiler **808** has collected data from chiller sensors **804**, with the data having then been processed by daily shutdown predictor **810** and updated by data updater **826**, period updater **828** may update data used in the generation, training, and evaluation of new models as well as the re-evaluation and re-training of existing models. Further, period updater **828** may also provide an indication to components of trainer **806** that recently collected data relates to a specified, period. For example, if a shutdown event were to occur in chiller **802**, period updater **828** may indicate that data collected during the shutdown event of chiller **802** pertains to a specific period and may also indicate specific start and end points to such a period. It should also be noted that the components updated by period updater **828** may include components of trainer **806** seen in some embodiments.

[0139] Trainer **806** is shown to include a past operational data compiler **812**. Past operational data compiler **812** is shown to be in communication with a model generator **814**,

according to some embodiments. Past operational data compiler **812** may compile operations data that spans a set time period, with said time period depending on other attributes of trainer **806**, as well as system **800**. For example, in the instance of a relatively recently installed chiller such as chiller **802**, chiller sensors **804** may not have collected sufficient data to be compiled by past operational data compiler **812**, in which case past operational data compiler **812** may include data collected from other chillers the same as and/or similar to chiller **802**. Past operational data compiler **812** may also prepare compiled data for analysis and application by model generator **814**, such as creating data sets that may be for specific time periods, or sorting data based on possible chiller events that may have occurred during the time period for which the data was collected, according to some embodiments. In some instances, a customer and/or operator of system **800** may select specific occurrences which may be found in past operational data that may be identified by past.

[0140] Trainer **806** is shown to include a hyper-parameter iterator **814**. Hyper-parameter iterator **814** is shown to be in communication with past operational data compiler **812**, according to some embodiments. As such, hyper-parameter iterator **814** can be configured to receive an input from past operational data compiler **812**. In some embodiments, chiller shutdown prediction models can be generated, tuned, and/or trained based on a choice of hyper-parameter. Hyper-parameter iterator **814** can be configured to select, determine and/or identify a hyper-parameter from a set of multiple hyper-parameters through which hyper-parameter iterator **814** may iterate. According to the selection of hyper-parameter by hyper-parameter iterator **814**, appropriate corresponding models may be generated by model generator **816**, according to some embodiments. Generated models may each have a set of hyper-parameters that require tuning in order for the model to operate optimally. As such, developing machine learning, deep learning, and transfer learning models useful in prediction can be dependent on the corresponding sets of hyper-parameters for the models. In some embodiments, hyper-parameter iterator **814** can train multiple models with different sets of hyper-parameters in order to determine which set of hyper-parameters produces the most accurate predictions for each model. Further, in some embodiments this may be an iterative process. Determination of hyper-parameters may be involve analyzed function of equipment such as chiller **802** and may also include user and/or operator preferences. Hyper-parameters implemented by hyper-parameter iterator **814** may vary in some embodiments. In some embodiments hyper-parameters may incorporate data collected from chiller sensors **804** of chiller **802**, or another similarly configured chiller with sensing equipment. For example, if W indicates a hyper-parameter, then W can take values such as $W=30$ days, $W=365$ days, and/or $W=$ 'All Data'. It should be also noted that multiple hyper-parameters and models can exist for a single chiller such as chiller **802**, according to some embodiments.

[0141] Trainer **806** is shown to include model generator **816**. In some embodiments, model generator **816** may be configured to receive one or more hyper-parameters from hyper-parameter iterator **814**, which can be in communication with model generator **816**. Model generator **816** can be dependent on a number of factors, including chiller data and hyper-parameters, that impact the generation of models that may apply machine learning and deep learning concepts,

according to some embodiments. In some embodiments, for example, model generator **816** may generate multiple models for each set time period for which data is available. These models generated may be done so based on one or more hyper-parameters which may be made available to model generator **816**, which may be a component of and made available to trainer **806**, according to some embodiments. Models generated by model generator **816** may be based on training data, with the data possibly having been collected by chiller sensors **804** of chiller **802**, or a similar structure. Model generator **816** may generate models just relevant to chiller **802** in some embodiments or may generate a model or models applicable to a plurality of chillers the same as and/or similar to chiller **802**. Some models generated by model generator **816** can include machine learning models, which more specifically can include Multivariate Gaussian Modeling (MVG) and Graphical Gaussian Modeling (GGM). Similarly, some models generated by model generator **814** can include deep learning models, which more specifically can include Long Short Term Memory (LSTM) with Autoencoder and Variational Autoencoder (VAE) configurations. Additionally, model generator **816** may also construct different models for different scenarios, with possible preferences and specifications for desired models to be indicated by an operator with model generator **816** thus accounting for preferred variables and other potential factors.

[0142] Trainer **806** is shown to include a model evaluator **818**. Model evaluator **818** is shown to be in communication with model generator **816**, according to some embodiments. Model evaluator **818**, through communication with model generator **816**, may receive one or more models that have been trained using one or more hyper-parameters, according to some embodiments. Model evaluator **818** may receive multiple models from model generator **816** that have been trained for a given time period. Also, model evaluator **818** may employ one or more algorithms to evaluate models, according to some embodiments. Algorithms implemented by model evaluator **818** on models received from model generator **816** may be configured to evaluate said models without using any test data. In some instances, the use of test data (artificially generated data that has not been collected from real equipment) may impair the training and eventual operation of models. Model evaluator **818** may ultimately evaluate a plurality of models using a variety of techniques and may also use said techniques and other algorithms to ultimately select one or more models from the set of received models based on evaluation results.

[0143] Trainer **806** is shown to include a model selector **820**. Model selector **820** is shown to be in communication with model evaluator **818**, according to some embodiments. Model selector **820** may select models based on varying statistics calculated depending on the nature of the data received from model evaluator **818**, according to some embodiments. For example, if training data includes a fault such as a shutdown, model selector **820** may select based on a measure of area under the curve (AUC) calculated under receiver operating characteristics (ROC) for the given model. In some instances, a larger area under the curve may correspond to better model performance than a lesser area under the curve. Further, in the instance that faults such as shutdowns were not present in training data, model selector **820** can use a measure of a probability density function (PDF). The measure of the probability density function can

indicate how well a model has learned normal behavior and can be calculated in multiple ways. Some calculation methods can include assigning equal weight to all points of a given training period for which a model is trained, while other calculation methods can assign weights to different points within the time period of the training data so as to prioritize a specific time. The measure of the probability density function may then have a standard deviation for which an inverse thereof can be calculated, with the standard deviation normalized by mean. The calculated values for both the area under the curve of the receiver operating characteristics and the inverse of the standard deviation normalized by mean of the measure of the probability density function can be indicative of how well a model has learned normal behavior. Also, it should be noted that for models trained with data deficient of a shutdown, ROC may not be able to be calculated, thus necessitating the determination of the measure of the probability density function. Additionally, in the selection of models performed by model selector **820**, probability measures may be analyzed regardless of the presence or lack thereof of a shutdown in training data.

[0144] Model selector **820** can be configured to analyze the calculations performed by model evaluator **818** in order to determine and select the best model. In some embodiments, the calculated value of the area under the curve of the receiver operator characteristic which can be calculated for data including shutdowns can be compared to the calculated value of the inverse of the standard deviation normalized by mean for the measure of the probability density function which can be calculated for data that does not include shutdowns. Model selector **820** can then compare the calculated values of the area under the curve of the receiver operating characteristics and the inverse of the standard deviation normalized by mean of the measure of the probability density function. In some embodiments, model selector **820** can be configured to select the model with the greatest value when the calculated values of the area under the curve of the receiver operating characteristics and the inverse of the standard deviation normalized by mean of the measure of the probability density function are compared. That is to say that, according to some embodiments, model selector **820** is configured to select the greatest calculated value of either the area under the curve of the receiver operating characteristics or the inverse of the standard deviation normalized by mean of the measure of the probability density function, with the selection of the greatest value corresponding to the selection of a best performing model.

[0145] Trainer **806** is shown to include a threshold selector **822**. Threshold selector **822** is shown to be in communication with daily shutdown predictor **810**, according to some embodiments. Threshold selector **822** is also shown to receive one or more selected models from model selector **820**, according to some embodiments. The threshold selection of threshold selector **822** may be performed on one or more models to have been determined to have the best hyper-parameter set. Additionally, the selected threshold of the aforementioned selection process of threshold selector **822** may involve a p-measure needed to achieve a desired operating point. In some embodiments, the p-measure for any point in a training period can be a function of the probability of that point being anomalous (non-normal), as calculated by a chosen model. The operating point to be

selected may vary according to user and/or operator preference, among other variable factors. The selection of a threshold by threshold selector **822** may be expressed in terms of a false positive, i.e. false positive should be below 2/365. It should also be noted that in the event that calculation of the ROC is possible, a threshold may be able to be selected by threshold selector **822** directly on the ROC to achieve a given operating point. Conversely, in the event that ROC may not be available, estimation may be performed by threshold selector **822** in terms of probability distribution of the p-measure with the threshold possibly being expressed as a percentile point on the distribution, i.e. the 95th percentile of the p-measure evaluated. Further, time-series modeling can also be performed on the p-measure in order to forecast p-measures for all time points within a given test window, and a threshold can be determined based on a predictive series. It should also be noted that threshold selector **822** may employ a variety of means, including those details above as well as other possible means and methods in order to determine and possibly set a desired threshold which may be determined by a number of inputs and/or factors.

[0146] Trainer **806** is shown to include an operating point generator **824**. Operating point generator **824** is shown to be in communication with threshold selector **822**, according to some embodiments. Operating point generator **824** may be subject to user and/or operator preference. For example, a user and/or operator may have specific preferences in terms of the operation of chiller **802**, for example. As such, models generated, trained, and selected by components of trainer **806** may have constraints and/or operating points tied to the models implemented in the form of thresholds, such as thresholds determined by threshold selector **822**. That is to say that, in the process of threshold selector **822** determining, selecting, and implementing a threshold, operating point generator **824** may influence the aforementioned process of threshold selector **822**, according to some embodiments. Operating point generator **824** may also vary, for example the preferences of one user and/or operator may differ greatly from another, in which case some embodiments may carry substantially different activity of operating point generator **824** and subsequently different activity and results of threshold selector **822**. It should also be noted that both operating point generator **824** and threshold selector **822** may operate simultaneously and may also operate so as to determine multiple operating points and thresholds simultaneously, depending on circumstances.

[0147] Referring now to FIG. **8B**, a process for model training **840** is shown. It should be noted that process **840** and its component steps may be performed by and/or related to components of FIG. **8A**, according to some embodiments. Further, process **840** may be performed for one or more chillers. Additionally, process **840** may be performed iteratively and/or cyclically, which is to say that one or more chillers and their associated training data and corresponding models may experience parts of or process **840** in its entirety multiple times. Process **840** may also be performed with additional steps or with slight modifications according to some embodiments.

[0148] In step **842**, process **840** is shown to include collecting data from chiller sensors and separating past operational data from current operational. Both current operation data and past operational data of step **842** may have been collected by chiller sensors **804** of chiller **802**.

Further, both past operational data and current operational data may include various data such as input and output quantities, efficiency metrics, usage values, among other data and as such may be organized, sorted, and or separated in various ways. Additionally, past operational data and current operational data may be compiled and or otherwise acted upon by past operational data compiler **812** and/or current operational data compiler **808** or similar, both of FIG. **8A**, according to some embodiments.

[0149] In step **844**, process **840** is shown to include generating models based on past operational data. Step **844** of process **840** can be performed by model generator **814** of FIG. **8A** or another similar component, according to some embodiments. Further, step **844** may be performed differently depending on various factors. For example, past operational data for which a model is to be generated may contain data for different time periods (i.e. epochs), according to some embodiments. As such, models may be generated using different methods and or techniques based on the time period for which the data was generated, among other variables such as contents of past operational data. Additionally, step **844** can also include generating models that implement various methods and techniques in predicting future events, with those methods and techniques including machine learning, deep learning, and transfer learning.

[0150] In step **846**, process **840** is shown to include determining hyper-parameters for given models. Step **846** may vary according to user and/or operator preferences that may be incorporated into the determination of hyper-parameters. For example, in the instance that efficiency may be a primary concern for a user and/or operator, such concerns may correspond to a range for which hyper-parameters may exist, according to some embodiments. Further, determination of hyper-parameters may be performed by a component the same as or similar to hyper-parameter iterator **814** of FIG. **8A**, according to some embodiments. In some embodiments, generated models may have one or more sets of hyper-parameters that can be tuned in order to optimize accuracy of the models. In some embodiments, applying hyper-parameters correctly to machine learning and/or deep learning and/or transfer learning models can be a critical component of model performance. As such, step **846** may be performed iteratively, with different sets of hyper-parameters used to train multiple models in order to ultimately select hyper-parameters for each model that produce the most accurate predictions.

[0151] In step **848**, process **840** is shown to include training generated models. Training generated models as seen in step **848** may include components the same as or similar to hyper-parameter iterator **814** and model generator **816** of FIG. **8A**. Additionally, training of generated models may vary according to some embodiments. For example, given that past operational data on which models are based may vary, so too may techniques for model generation. In the instance that past operational data includes several large data sets and subsequently corresponds to several large models, the models may be trained using a different method or technique than for past operational data that includes shorter data sets, according to some embodiments. Further, generated models may be trained differently in some embodiments according to preferences of users and/or operators. That is to say that, in some embodiments, models may be trained according to user and/or operator preferences. For example, if a user and/or operator expressed a

desire to optimize specific parameters for a system such as efficiency or output, models training of step **848** may be trained accordingly.

[0152] In step **850**, process **840** is shown to include evaluating model performance and selecting best performing models. Step **850** may, in some embodiments, include components the same as or similar to model evaluator **818** and model selector **824** of FIG. **8A**. Further, step **850** may vary in both the evaluation process as well as the selection process for generated and trained models. For example, in some instances models may have been generated using past operational data that included shutdown events, which may call for a specific evaluation method that may not be applicable to or effective for models generated using past operational data deficient of shutdowns. Step **850** may also vary according to embodiment in terms of model selection. In some embodiments, models may be evaluated mathematically. Further, models may be evaluated according to the process described for model evaluator **818** in which calculations are performed in order to evaluate models with the specific calculations dependent on the corresponding training data for a model. In some embodiments, a model with one or more shutdowns present in a training period can have a corresponding area under the curve found for generated receiver operating characteristics. Models that do not contain shutdowns in training data may be evaluated by calculating an inverse standard deviation normalized by mean for a measure of a probability density function, according to some embodiments. In some embodiments, the calculated area under the curve of the receiver operating characteristics and the standard deviation normalized by mean for the measure of the probability density function may be compared, with the greatest value indicating the best performing model. In some embodiments in which a user and/or operator has preferences for model performance, such preferences can be incorporated into the model selection process in various ways.

[0153] In step **852**, process **840** is shown to include generating and applying operating points to the selecting models. Step **852** can include operating point generator **824** of FIG. **8A** or similar, according to some embodiments. Step **852** may also vary according to various factors, among them application of user and/or operator preferences. For example, a user and/or operator may have defined performance expectations for chillers, in which case step **852** and the generation of operating points may be impacted by and also incorporate such preferences. Additionally, generation of operating points may depend on past operational data used in generating models, specific chillers that may be impacted, as well as other variables.

[0154] In step **854**, process **840** is shown to include selecting thresholds based on calculations performed on models and corresponding data. Step **854** can include threshold selector **822** of FIG. **8A** or similar, according to some embodiments. Step **854** may also vary in terms of the methods and/or techniques implemented in selection of thresholds. For example, some models may have been generated using past operational data that included shutdowns, for which evaluation of projected model performance must be conducted differently than for models that may have been generated using past operational data that did not include shutdowns, according to some embodiments. Further, models for different chiller types may differ in configuration and may thus require different thresholds and

thus different calculations of their associated models. Step **854**, as with other previous steps, may also incorporate user and/or operator preferences into the selection of thresholds.

[0155] In step **856**, process **840** is shown to include applying thresholds and current operational data to daily shutdown predictions. Step **856** can include daily shutdown predictor **810** of the FIG. **8A** or another similar component, according to some embodiments. Additionally, daily shutdown predictions can incorporate current operational data as well as past operational data. For example, both past operational data and current operational data may have been collected from chiller sensors **804** of chiller **802** in FIG. **8A**, or another similar component. Further, past operational data may be from various previous time periods, while current operational data may serve as an indication of more recent activity of chiller **802** or similar, according to some embodiments. Additionally, daily shutdown predictor may be configured to account for factors not included in past operational data, current operational data, or input from threshold selections. Such factors may include geographic location, planned activity for chiller **802** or similar, as well as other possible user and/or operator preferences that may further influence a possible shutdown.

[0156] In step **858**, process **840** is shown to include updating data and corresponding period with shutdown predictions. Step **858** can include daily shutdown predictor **810**, as well as data updater **826** and period updater **828** or other similar components, according to some embodiments. Step **858** may also vary according to terms of the updating of data and period. For example, time periods for data collection and corresponding model generation may vary according to variables such as specifics of chiller **802** of FIG. **8A** or similar, as well as other possible user and/or operator preferences. Additionally, data updating may vary according to factors such as technical capabilities of equipment involved including memory and hard drive or cloud storage, for example, among other factors. Updating of shutdown predictions may also vary by embodiment for reasons the same as or similar to those above that may possible impact period updating and data updating, as well as other possible reasons such as network configuration, among other possible concerns. It should also be noted that, as the final step in process **840**, step **858** may not necessarily serve as a definitive end to such a process. In some embodiments, process **840** may be cyclical, iterative, or otherwise repeat some steps or all steps of process **840**.

[0157] Referring now to FIG. **8C**, a flow diagram of a process for model training and evaluation **870** is shown. Process **870** can be similar to process **840**, according to some embodiments. The flow diagram of FIG. **8C** and process **870** therein may pertain to systems the same as and/or similar to those for which process **840** may apply. It should also be noted that, according to some embodiments, process **870** may be applied to systems such as that of FIG. **8A** in conjunction with process **840**.

[0158] Process **870** is shown to include a receiving chiller operational data **872**. According to some embodiments, receiving chiller operational data **872** may be done for varying time periods (i.e. epochs), and also may contain a variety of different data. Receiving chiller operational data **872** may include collecting data from chiller **802** and chiller sensors **804** of FIG. **8A**, according to some embodiments. Receiving chiller operational data **872** may also be the same as or similar to that of steps **842** and **844** of FIG. **8B**,

according to some embodiments. Receiving chiller operational data **872** may then include sending data for a hyper-parameter iteration **874**, which may include components similar to hyper-parameter iterator **814** of FIG. **8A** and also similar to step **846** of process **840**, according to some embodiments. Further, process **870** is shown to include a model training **876** and a model evaluation **878**. Model training **876** and model evaluation **878** may receive include an input of data from receiving chiller operational data **872** and may also operate similar to model evaluator **818** of FIG. **8A**. Additionally, model training **876** and model evaluation **878** may correspond to steps **848** and **850** of process **840**, according to some embodiments.

[0159] Process **870** is shown to include a model selection **880**. Model selection **880** may be similar to model selector **820** of FIG. **8A**, according to some embodiments, and may also relate to step **850** of process **840** of FIG. **8B**. Process **870** is also shown to include a threshold selection **882**, according to some embodiments. Threshold selection **882** may include components similar to threshold selector **822** and operating point generator **824** of FIG. **8A**. Further, threshold selection **882** may be similar to step **854** of process **840** of FIG. **8B**, according to some embodiments. Process **870** is also shown to include a shutdown prediction **884**. In some embodiments, shutdown prediction **884** may relate to daily shutdown predictor **810** of FIG. **8A**. Additionally, shutdown prediction **884** may also relate to step **856** of process **840** as seen in FIG. **8B**. Process **870** is also shown to include a data update **886**. Data update **886** can relate to data updater **826** as well as period updater **828** of FIG. **8A** and may also correspond to step **858** of process **840** of FIG. **8B**, according to some embodiments. Data update **886** can serve as the final step in process for model training an evaluation in some embodiments. However, process **870** may be dynamic and may also operate cyclically with some blocks and other components thereof involved in process **870** multiple times.

Automatic Model Threshold Selection for Anomaly Detection of Connected Chillers

[0160] Referring now to FIG. **9A**, a threshold system **900** is shown. System **900** may include a number of other components, such as those seen in FIG. **9A**, although components of system **900** may vary according to some embodiments. System **900** can also operate in series, in parallel, and/or in conjunction with the components of system **700** of FIG. **7A**, as well as system **800** of FIG. **8A**, according to some embodiments. It should also be noted that system **900** may operate in conjunction with a variety of different equipment including but not limited to chiller **702** and chiller cluster **706** of FIG. **7A**, as well as chiller **802** of FIG. **8A**. In some embodiments, system **900** may also operate in conjunction with other equipment other than the aforementioned components of previous figures.

[0161] System **900** is shown to include a threshold selector **902**. Threshold selector **902** may include some or all of the components seen in FIG. **9A** but may also be deficient of some components seen in FIG. **9A** or may also include additional components not seen in FIG. **9A**, according to some embodiments. Threshold selector **902** may also perform similarly to threshold selector **822** of FIG. **8A**. Additionally, threshold selector **902** may also work in conjunction with machine learning, deep learning, and/or transfer learning techniques, such as those involved in systems of FIG. **7A** and FIG. **8A**, as well as models and other products

generated through the implementation of such techniques. Generally, threshold selector may be configured to aid in the generation of a selection of one or more thresholds to be applied to one or more models that may be implemented in order to govern behavior of equipment that may include chillers, such as chiller **802** of FIG. **8A**.

[0162] Threshold selector **902** can employ multiple methods and techniques in order to select a threshold for one or more models. In some embodiments, threshold selector **902** may evaluate training data for models for which a threshold is to be selected. For example, if training data for a given model includes a shutdown, threshold selector **902** can perform certain calculations in order to determine a threshold. Further, if training data for a model does not include a shutdown, threshold **902** can perform different calculations in order to determine a threshold. Threshold selector **902** can implement the calculation of receiver operating characteristics for models with training data that included a shutdown, according to some embodiments. Further, an operating point can be obtained on the receiver operating characteristics by threshold selector **902**, with the operating point allowing for the identification of a corresponding threshold. Additionally, probability measures can also be incorporated in the calculation of the receiver operating characteristics, and the operating points on the receiver operating characteristics can serve as false positive constraints. In some embodiments, threshold selector **902** can be configured to estimate a probability distribution. Further, threshold selector **902** can then obtain an operating point and corresponding threshold as a percentile point of the probability distribution. In some embodiments, threshold selector **902** can then implement the selected threshold from either calculation method that accounted for the presence of a shutdown event in the training data for a model or a lack thereof.

[0163] Thresholds selected by threshold selector **902** can act as a reference point in order for anomalous data to be identified, according to some embodiments. For example, if a model (machine learning, deep learning, transfer learning, etc.) is operating with a threshold selected by threshold selector **902**, then said threshold can act to define data analyzed by the model as non-anomalous (normal) or anomalous. Further, if activity occurs that is analyzed by the model and shows data exceeding the threshold selected by threshold selector **902**, then that data can be identified as anomalous. In some embodiments, the identification of anomalous data can lead to a prediction of a future fault, such as a shutdown.

[0164] Threshold selector **902** is shown to include shutdown evaluator **906**. Shutdown evaluator **906** is shown to be in communication with model training data **904**, according to some embodiments. Model training data **904** can be the same as or similar to data collected from chiller **702**, chiller cluster **706**, and/or other similar components. Further, model training data may correspond to a variety of different epochs, according to some embodiments. Shutdown evaluator **906** can be configured to receive an input of data from model training data **904** and may also be configured to process and/or evaluate said data. Shutdown evaluator **906** may also be configured to receive a variety of data, which is to say that any input to shutdown evaluator **906** from model training data **904** may include data of varying size, type, content, and relevant equipment, among other factors, according to some embodiments. Shutdown evaluator **906** may also be configured to analyze data that may be received as an input from

model training data **904** for specific content and/or events, such as shutdown events, for example. Further, in the instance that shutdown evaluator **906** may receive an input of data from model training data **904**, shutdown evaluator **906** may use one or more of a variety of techniques to process said inputted data in order to identify and shutdown events. If shutdown evaluator **906** may identify one or more shutdown events in data that may have been received from model training data **904**, shutdown evaluator **906** may sort and/or otherwise categorize data according to the presence of a shutdown event of a lack thereof. Additionally, in the event that shutdown evaluator **906** identifies a shutdown in any inputted data, shutdown evaluator **906** may output said data including the shutdown event differently than data for which no shutdown events were identified by shutdown evaluator **906**. For example, in the event that shutdown evaluator **906** may receive an input of data that is determined to contain a shutdown event, shutdown evaluator **906** may output that data to an ROC calculator **910**. Conversely, in the event that shutdown evaluator **906** may receive an input of data that is determined to be deficient of any shutdown events, shutdown evaluator **906** may output that data to a distribution estimator **908**, according to some embodiments.

[0165] Threshold selector **902** is shown to include distribution estimator **908**. Distribution estimator **908** is shown to be in communication with shutdown evaluator **906**, according to some embodiments. Distribution estimator **908** can be configured to estimate a probability distribution of a p-measure and/or a threshold based on a percentile point of a distribution. Additionally, distribution estimator **908** may also be configured to estimate a threshold based on a percentile point which may be evaluated under a selected model, according to some embodiments. For example, a specific model which may have been created by and/or implement in operation the practices of machine learning, deep learning, and/or transfer learning may have a distribution calculated by distribution estimator **908** based on a percentile point of the distribution, i.e. the 95 thpercentile of p-measure evaluated under a selected model. In some instances, for data received by distribution estimator **908** that is determined to contain a shutdown event, other threshold determination methods may not yield threshold values of those determined using a method including distribution estimator **908**. It should also be noted that the method and or methods of distribution estimator **908** may include an input from a probability generator **912**, according to some embodiments. Probability generator **912** may be configured to determine relevant probability measures which may vary according to data type, content of data, among other factors, and provide probability data to distribution estimator **908**. That is to say that a received input from probability generator **912** may be a component of a method implemented by distribution estimator **908** in order to generate an estimated distribution.

[0166] Threshold selector **902** is shown to include a threshold calculator **918**. Threshold calculator **918** is shown to be in communication with distribution estimator **918**, according to some embodiments. Threshold calculator **918** may also be configured to obtain a threshold in terms of a percentile point configuration, according to some embodiments. Additionally, it should be noted that threshold calculator **918** can be configured to determine one or more thresholds, according to some embodiments. For example, threshold calculator **918** may receive one or more distribu-

tions from distribution estimator **908**, and in turn may calculate one or more thresholds for each of the one or more models. In some embodiments, threshold calculator **918** is also shown to be in communication with a percentile point constraint **920**. According to some embodiments, percentile point constraint may use one or more methods to configure an operating point, which may then be communicated to threshold calculator **918**. Percentile point constraint **920** can also be configured, in some embodiments, to define a range for which threshold calculator **918** may calculate a threshold for data.

[0167] Threshold selector **902** is shown to include ROC calculator **910**. ROC calculator **910** is shown to be in communication with shutdown evaluator **906**, according to some embodiments. ROC calculator **910** can be configured to calculate receiver operating characteristics. ROC calculator **910** may also be configured to receive an input from shutdown evaluator **906**, according to some embodiments. For example, in the event that shutdown evaluator **906** were to determine that specific data contained a shutdown event, said data may be inputted to ROC calculator **910**. Further, ROC calculator **910** can then calculate an ROC for one or more data sets. In some embodiments for which data may be determined to include a shutdown event, ROC calculations such as those that may be performed by ROC calculator **910** may enable the determination of a threshold. If an ROC has been calculated for a data set, a threshold may be able to be determined and/or selected directly on the calculated ROC in order to achieve a given operating point, according to some embodiments. It should also be noted that ROC calculator **910** may be configured to calculate one or more receiver operating characteristics for one or more data sets.

[0168] Threshold selector **902** is shown to include an operating point calculator **916**. Operating point calculator **916** is shown to be in communication with ROC calculator **910**, according to some embodiments. Operating point calculator **916** may be configured to determine and/or identify an operating point based on an input received from ROC calculator **910**, according to some embodiments. For example, upon receiving an input that may include receiver operating characteristics from ROC calculator **910**, operating point calculator **916** may then determine one or more operating points based on one or more data sets received from ROC calculator **910**. Further, operating point calculator **916** may be configured to select a threshold directly on receiver operating characteristics of inputted data, with said threshold corresponding to a desired operating point, according to some embodiments. Additionally, operating point calculator **916** is shown to be in communication with a false positive constraint **914**, according to some embodiments. False positive constraint **914** may be configured to input data, for example a false positive constraint, to operating point calculator **916**. In some embodiments, a false positive constraint inputted to operating point calculator **916** from false positive constraint **914** may define a range and/or interval for which operating points may be selected on given receiver operating characteristics by operating point calculator **916**.

[0169] Threshold selector **902** is shown to include a threshold identifier **922**. Threshold identifier **922** is shown to be in communication with operating point calculator **916**, as well as threshold calculator **918**, according to some embodiments. Threshold identifier **922** can be configured to receive one or more inputs from both operating point calculator **916**

and threshold calculator **918**. For example, threshold identifier **922** may receive an input from threshold calculator **918** in the event that a given data set was deficient of any shutdowns, while threshold identifier **922** may receive operating points (corresponding to thresholds) on receiver operating characteristics from operating point calculator **916** in the event that a given data set did include one or more shutdowns. Further, threshold identifier **922** may be configured to evaluate and identify one or more optimal thresholds for both instances in which a data set included one or more shutdowns, as well as instances for which one or more data sets was deficient of shutdowns. That is to say that given an input of one or more operating points and/or thresholds for one or more given models, threshold identifier **922** may determine one or more optimal thresholds for one and/or both different possibilities in terms of the contents of inputted data sets.

[0170] Referring now to FIG. 9B, process for model evaluation **930** is shown. Process **930** may include multiple steps, according to some embodiments. Process **930** may also be structured to evaluate training data collected from chillers such as chiller **802** and corresponding chiller sensors **804** of FIG. 8A and/or other equipment in some embodiments. Further, the steps of process **930** can be performed by some and/or all of the components of system **900**, according to some embodiments. For example, with reference to chiller **802** and corresponding chiller sensors **804** of FIG. 8A, chiller sensors **804** may collect data from chiller **802**. Further, data collected by chiller sensors **804** from chiller **802** may be used to generate models with machine learning, deep learning, and/or transfer learning. Collected data used in the generation of those models, and by association the models themselves, may be evaluated using process **930**.

[0171] In step **932**, process **930** is shown to include gathering training models generated for epoch *n*. Step **932** may include model training data **904** of system **900**, according to some embodiments. Training models may be generated according to training data, and training data may be collected from sources such as chiller sensors **804** of chiller **802** as seen in FIG. 8A. Further, training models and corresponding training data gathered in step **932** may be for a single time period, noted as epoch *n* in step **932**. That is to say that models gathered in step **932** may all have been generated and trained using data from a single epoch, which is also to say that models gathered in step **932** may be representative of a single epoch. It should also be noted that models gathered in step **932** may be the same and/or different. For example, some models gathered in step **932** may use machine learning techniques and/or deep learning techniques and/or transfer learning techniques.

[0172] In step **934**, process **930** is shown to include determining if shutdowns are present in the training period. Training data and corresponding training models may be evaluated using one or more methods to determine if the training period for which training data was collected and training models were generated includes shutdowns. Step **934** can include shutdown evaluator **906** of system **900**, according to some embodiments. This evaluation may produce an outcome that yes, training data and model does include a shutdown, or no, training data and model does not include a shutdown. In the instance that such a determination may be made, different steps may be taken in terms of further analysis and calculations for training data and models. In some embodiments, training periods evaluated as part

of step **934** may vary. Additionally, so too may shutdown activity. That is to say that training data used for different models may vary, and so too may the appearance of a shutdown in that training data. As such, step **934** may implement multiple methods in order to analyze data for a training period and determine if a shutdown occurred during said period.

[0173] In step **936**, process **930** is shown to include calculating receiver operating characteristics. In calculating the receiver operating characteristics, various probability measures may also be considered. Step **936** can include ROC calculator **910** of system **900**, according to some embodiments. In some embodiments, receiver operating characteristics may only be possible to calculate for models trained with data that included at least one shutdown. Additionally, receiver operating characteristics calculation of step **936** may vary according to model and training data. Receiver operating characteristics of step **936** can vary according to training data and model parameters such as duration of training period and model techniques (machine learning, deep learning, transfer learning, etc.). For example, the receiver operating characteristics may be calculated differently for training data from a longer time period (epoch) than for a shorter time period according to some embodiments.

[0174] In step **938**, process **930** is shown to include obtaining operating points on the receiver operating characteristics. Step **938** can include operating point calculator **916** of system **900**, according to some embodiments. In some embodiments, an operating point may be selected according to preferences of a user and/or operator. For example, if a user and/or operator desires optimized efficiency, step **938** can be configured to select an operating point that aligns with optimized efficiency. Further, step **938** can also include obtaining an operating point according to other factors such as maximized performance as well as other preferences. In step **938**, the operating point obtained can correspond to a threshold, which can then be selected.

[0175] In step **940**, process **930** is shown to include, in the event that shutdowns are present in the training period, estimating a probability distribution. Step **940** can include distribution estimator **908** of system **900**, according to some embodiments. Models for which the training data does not include a shutdown may not allow for calculation of receiver operating characteristics. As such, the alternative method of step **940** is used in which the probability distribution is estimated. The probability distribution of step **940** can also be a measure of a probability density function, which may vary according to training data and model parameters such as duration of training period and model techniques (machine learning, deep learning, transfer learning, etc.).

[0176] In step **942**, process **940** is shown to include obtaining an operating point configured as a percentile point on the probability distribution of step **940**. In some embodiments, the operating point may be restricted and/or selected according to preferences of a user and/or operator. Step **942** can include threshold calculator **918** of system **900**, according to some embodiments. For example, if a user desired maximized output, the operating point of step **942** can be selected accordingly. Additionally, the operating point of step **942** can correspond to a threshold on the probability distribution. In some embodiments, the selection of the operating point can correspond to a threshold to be applied to a model.

[0177] In step 944, process 940 is shown to include the selection of a threshold. Depending on training data for models, operating points and thresholds may be determined differently. In some embodiments, thresholds obtained for models based on the operating points obtained in step 938 and/or step 942 may be determined by similar and/or the same means in step 944. Step 944 can include threshold identifier 922 of system 900, according to some embodiments. For example, thresholds determined based on calculating receiver operating characteristics and identifying operating points thereon can be similar to or the same as thresholds determined by estimating a probability distribution and identifying operating points as percentile points thereon. Further to the previous example, thresholds determined in step 944 can be comparable whether determined by the same or similar methods or different methods, such as the two differing methods of process 940.

[0178] Referring now to FIG. 9C, a flow diagram of a process for threshold calculation and selection 950 is shown. Process 950 can be similar to process 930, according to some embodiments. It should also be noted that process 950 can be applied to various training data and models generated thereof for chillers. In some embodiments, the models for which process 950 is conducted can employ machine learning, deep learning, and/or transfer learning techniques. Further, the training data associated with the aforementioned models can also vary in time period as well as contents of the data. Additionally, components seen in FIG. 9A can be used to perform some and/or all of process 950, according to some embodiments.

[0179] Process 950 is shown to include collecting training data for selected models 952. Training data of collecting training data for selected models 952 can be from a single epoch, according to some embodiments. Further, training data of collecting training data for selected model 952 can also vary by time period, contents, and other factors. Similarly, models generated by training data of collecting training data for selected model 952 may also vary according to techniques implemented which can include machine learning, deep learning, and/or transfer learning. Process 950 is also shown to include identification of shutdowns 954. Identification of shutdowns can involve analyzing training data of collecting training data for selected model 952, according to some embodiments. Training data of collecting training data for selected model 952 can also vary according to time duration for data as well as contents of the data, which can result in multiple data analysis techniques being used in identification of shutdowns 954 in order to detect any shutdowns present in data.

[0180] Process 950 is shown to include calculating receiver operating characteristics (ROC) 956 in response to identification of shutdowns present in training data. In some embodiments, calculating receiver operating characteristics 956 can include the incorporation of various probability measures. Depending on the contents of training data of process 950, the result of the receiver operating characteristics can vary. Process 950 is shown to include obtaining operating points on receiver operating characteristics 958. Operating point of obtaining operating points on receiver operating characteristics 958 can also correspond to a threshold and can also serve as a false positive constraint, according to some embodiments. Process 950 is shown to include estimating a probability distribution 960 in response to determining that training data is free of shutdowns. In

some embodiments, probability estimation of estimating a probability distribution 960 can correspond to a measure of a probability density function and can vary according to the contents of training data for which the probability distribution was estimated. Process 950 is shown to include obtaining an operating point as a percentile point 962 on the estimated probability distribution of p-measures, according to some embodiments. Operating point of obtaining an operating point as a percentile point 962 can, in some embodiments, correspond to a threshold.

[0181] Process 950 is shown to include selecting a threshold for an epoch 964. Threshold of selecting a threshold for an epoch 964 can, in some embodiments, be similar and/or the same independent of the calculation means implement in order to determine said threshold. Further, threshold of selecting a threshold for an epoch 964 can be determined to allow for optimal operation of one or more models, according to some embodiments. Thresholds determined in selecting a threshold for an epoch 964 can also vary according to any user and/or operator preferences. For example, if a user and/or operator desired maximized output rather than efficiency, thresholds selected in selecting a threshold for an epoch 964 can be modified to reflect such preferences. Additionally, thresholds of selecting a threshold for an epoch 964 are selected and implemented in order to provide a reference point by which anomalous data can be recognized, which is to say that data detected by operating models that exceeds thresholds of selecting a threshold for an epoch 964 can be identified as anomalous.

[0182] Referring now to FIG. 10, a time plot 1000 is shown. Time plot 1000 shows an automatic threshold selection, according to some embodiments. Time plot 1000 can be generated by stitching together adjacent test windows for a given chiller. Time plot 1000 can correspond to chiller 802 of FIG. 8A or can be another similar and/or different chiller. Multiple epochs can be seen in plot 1000, which shows data spanning a 15-month time frame.

[0183] Time plot 1000 is shown to include a y-axis 1010. Y-axis 1010 is shown to include a measure of calculated p-measures, spanning in value from 0 to -250, according to some embodiments. In some embodiments, p-measure of y-axis 1010 can be calculated as a natural log of a probability that data at a time point is not anomalous (normal). Time plot 1000 is also shown to include an x-axis 1020. X-axis is shown to include a measure of time, scaled every other month for a total of 15 months. As seen on x-axis 1020, data shown on time plot 1000 corresponds to a total time period of September of 2015 to November of 2016, according to some embodiments.

[0184] Time plot 1000 is shown to include four sets of data, which includes normal data 1030, anomaly data 1040, shutdown data 1050, and threshold data 1060. Normal data 1030 can be seen as a p-measure selected every epoch for a given model. On time plot 1000, it can be seen that normal data 1030 fluctuates between approximate values of 0 and -50 according to the p-measure on y-axis 1010. Normal data 1030 can also be considered to be non-anomalous data, which is to say that normal data 1030 is free of anomalies. Normal data 1030 can be indicative of standard and expected chiller operation. Time plot 1000 is also shown to include anomaly data 1040, according to some embodiments. Anomaly data 1040 can also be considered non-normal data, according to some embodiments. Anomaly data 1040 is seen to fluctuate between approximate values of 0

and -250 and is seen to have the greatest approximate range of values, according to some embodiments. Anomaly data 1040 can correspond to a time period in close proximity to a fault event for a chiller, for example a shutdown or other event that can be classified as a chiller fault. Time plot 1000 is also shown to include shutdown data 1050. In some embodiments, shutdown data 1050 can be indicative of a shutdown or other chiller fault that occurred. As such, shutdown data 1050 can have an approximate p-measure value of 0. Further, in some embodiments shutdown data 1050 can be indicative of a lack of chiller activity which can be due to factors including a shutdown due to loss of power, necessary maintenance, malfunction, and other factors. Time plot 1000 is also shown to include threshold data 1060. In some embodiments, threshold data 1060 can correspond to one or more thresholds that have been determined by a component such as threshold identifier 922 of FIG. 9A. Further, threshold data 1060 can be dynamic in some embodiments, as seen on time plot 1000 in which threshold data fluctuates between p-measures of approximately 0 and -70. In some embodiments, threshold data 1060 can be dependent on other data as well as various calculations, identifications and determinations.

[0185] Time plot 1000 is also shown to include an anomaly area 1070. Anomaly area 1070 is shown to include a concentration of anomaly data 1040, according to some embodiments. Anomaly data 1040 can, in some embodiments, be indicative of a possible shutdown. Anomaly area 1070 includes a concentration of anomaly activity, which can correspond to shutdown activity as seen by shutdown data 1050, according to some embodiments. The time period of time plot 1000 defined on x-axis 1020 as July 2016 to November 2016 includes anomaly area 1070, according to some embodiments. Within this time period, a plurality of shutdown data 1050 is present, which can correspond to a plurality of anomaly data 1040, according to some embodiments. It should be noted that, in some embodiments, the fluctuation of normal data 1030 is lesser than that of anomaly data as seen in anomaly area 1070. Time plot 1000 indicates, according to some embodiments, dynamic threshold activity as seen by threshold data 1060, with anomaly activity including anomaly data 1040 having p-measure values exceeding p-measure values of threshold data 1060, as well as shutdown data 1050 with p-measure values of approximately 0. In some embodiments, establishment of a threshold as seen by threshold data 1060 a presence of anomaly data 1040 exceeding the p-measure values of threshold data 1060 can lead to generation of a fault and/or shutdown prediction.

Cost Savings from Fault Prediction and Diagnosis

[0186] Referring now to FIG. 11, system 1100 for identifying faults and generating cost savings is shown, according to an exemplary embodiment. In various embodiments, system 1100 is similar to chiller analysis system 602. For example, system 1100 may implement one or more machine learning models as described above in reference to FIGS. 6-10 to generate predictions. In various embodiments, system 1100 includes fault prediction and diagnosis system 1112. Fault prediction and diagnosis system 1112 may communicate with building subsystems 428 and/or equipment 1190 via network 446. Equipment 1190 may include one or more pieces of building equipment associated with a building and/or a space. For example, equipment 1190 may include HVAC equipment such as boilers, chillers, fans,

dampers, humidifiers, rooftop units, and/or the like. In various embodiments, equipment 1190 includes equipment sensors 1192. For example, a fan may include a speed sensor configured to measure a speed of the fan in Hertz. As another example, a supply unit may include a temperature sensor. Equipment sensors 1192 may include temperature sensors, speed sensors, airflow sensors, humidity sensors, and/or the like. In various embodiments, equipment 1190 and/or equipment sensors 1192 transmit information to fault prediction and diagnosis system 1112. For example, equipment 1190 may transmit a cooling valve setpoint and a cooling valve value.

[0187] Fault prediction and diagnosis system 1112 is shown to include processing circuit 1110 having processor 1114 and memory 1116. Processor 1114 may be a general purpose or specific purpose processor, an application specific integrated circuit (ASIC), one or more field programmable gate arrays (FPGAs), a group of processing components, or other suitable processing components. Processor 1114 is configured to execute computer code or instructions stored in memory 1116 or received from other computer readable media (e.g., CDROM, network storage, a remote server, etc.).

[0188] Memory 1116 may include one or more devices (e.g., memory units, memory devices, storage devices, or other computer-readable medium) for storing data and/or computer code for completing and/or facilitating the various processes described in the present disclosure. Memory 1116 may include random access memory (RAM), read-only memory (ROM), hard drive storage, temporary storage, non-volatile memory, flash memory, optical memory, or any other suitable memory for storing software objects and/or computer instructions. Memory 1116 may include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described in the present disclosure. Memory 1116 may be communicably connected to processor 1114 via processing circuit 1110 and may include computer code for executing (e.g., by processor 1114) one or more processes described herein.

[0189] Memory 1116 is shown to include machine learning models 1120, analysis circuit 1130, cost savings circuit 1140, control circuit 1150, and training circuit 1160. Machine learning models 1120 may include one or more machine learning models. For example, machine learning models 1120 may include a machine learning model trained on historical operational data from a HVAC system. Machine learning models 1120 is shown to include univariate outlier machine learning circuit 1122 and multivariate outlier machine learning circuit 1124. Univariate outlier machine learning circuit 1122 and/or multivariate outlier machine learning circuit 1124 are configured to generate one or more fault predictions based. For example, fault prediction and diagnosis system 1112 may execute univariate outlier machine learning circuit 1122 on predicted operational parameters to identify one or more predicted faults. In some embodiments, machine learning models 1120 include or otherwise implement the machine learning systems and methods described above with reference to FIGS. 6-10. In various embodiments, machine learning models 1120 include a machine learning model configured to generate future operational parameters for a BMS system and/or a component thereof (e.g., an HVAC system, etc.). For example, fault prediction and diagnosis system 1112 may

train a model on historical operating parameters of a HVAC system and may execute the trained model (e.g., using recent operating data, etc.) to generate predicted operating parameters (e.g., future operating parameters, etc.). In various embodiments, machine learning models **1120** are iteratively updated. For example, machine learning models **1120** may be updated based on user feedback regarding the predicted faults. As another example, machine learning models **1120** may generate predicted operating parameters and/or predicted faults and may update one or more machine learning models based on comparing the predicted operating parameters and/or the predicted faults to corresponding measured operating parameters and/or detected faults (e.g., faults that occurred, etc.). In various embodiments, fault prediction and diagnosis system **1112** dynamically updates machine learning models **1120**.

[0190] Analysis circuit **1130** may receive fault predictions and/or context data from machine learning models **1120** and/or other systems/devices (e.g., equipment **1190**, building subsystems **428**, etc.) and may classify and/or diagnose fault predictions using the received data. For example, analysis circuit **1130** may classify faults as either high zone temperature faults or low zone temperature faults (e.g., based on whether a predicted supply air temperature is higher or lower than a predicted supply air temperature setpoint, etc.). In some embodiments, machine learning models **1120** generate a binary fault determination. In some embodiments, analysis circuit **1130** analyzes the binary fault determinations to generate a fault diagnosis for each predicted fault. For example, machine learning models **1120** may generate one or more binary outputs (e.g., fault (1), no fault (0), etc.) that populate a fault diagnosis matrix and the fault diagnosis matrix may be used to diagnose a predicted fault (e.g., by mapping a combination of binary outputs to a fault diagnosis, etc.). In some embodiments, analysis circuit **1130** diagnosis predicted faults according to the table below:

Univariate Result of Supply Air Temperature	Multivariate Result of All Parameters	Fault Diagnosis
No Fault (0)	No Fault (0)	No Fault
Fault (1)	No Fault (0)	Supply Air Temperature Sensor Fault
No Fault (0)	Fault (1)	Other than Supply Air Temperature Sensor Fault
Fault (1)	Fault (1)	Fault of Supply Air Temperature because of Other Feeding Equipment

[0191] Cost savings circuit **1140** may calculate cost savings associated with predicted faults. For example, cost savings circuit **1140** may calculate the cost savings of a predicted fault based on a rule engine execution time. In various embodiments, cost savings circuit **1140** calculates the cost savings based on energy consumption associated with the predicted fault corresponding to a time period until the predicted fault is identified by a rules engine. In various embodiments, cost savings circuit **1140** calculates energy consumption using predicted functional parameters (e.g., as generated by machine learning models **1120**, etc.). In various embodiments, cost savings circuit **1140** calculates energy consumption associated with the predicted fault. For example, cost savings circuit **1140** may calculate an amount of energy (e.g., measured in Watts, British-thermal-units

(BTUs), etc.) associated with overheating a space (e.g., as in a high zone temperature fault, etc.) based on the degree of overheating and/or a length of time the space was overheated. In some embodiments, cost savings circuit **1140** calculates a cost savings based on the energy consumption. For example, cost savings circuit **1140** may determine a dollar value associated with the energy consumption that was avoided due to predicting and/or correcting the fault before it occurred (e.g., by multiplying a price of energy times the energy consumption, etc.).

[0192] Control circuit **1150** may perform one or more actions related to the predicted fault. For example, control circuit **1150** may automatically generate a work order ticket. As another example, control circuit **1150** may adjust a temperature setpoint. As another example, control circuit **1150** may generate an alert associated with the predicted fault and/or a suggestion on how to avoid the predicted fault (e.g., suggested maintenance, suggested repairs, etc.). In some embodiments, control circuit **1150** may operate one or more BMS components such as a HVAC system. For example, control circuit **1150** may disable a HVAC system for a period of time related to the predicted fault to avoid wasting energy.

[0193] Training circuit **1160** may train the one or more models of machine learning models **1120**. For example, training circuit **1160** may retrieve historical data from a BMS and/or component thereof and train a machine learning model using the historical data. In some embodiments, training circuit **1160** iteratively trains the one or more models. For example, training circuit **1160** may train a model of machine learning models **1120** using historical fault information and may update the trained model based on user feedback regarding one or more predicted faults generated by the trained model.

[0194] Referring now to FIG. 12, method **1200** for generating a fault prediction is shown, according to an exemplary embodiment. In various embodiments, system **1100** performs method **1200**. At step **1202**, fault prediction and diagnosis system **1112** may receive equipment sensor data. For example, fault prediction and diagnosis system **1112** may receive sensor data including a supply air temperature associated with supply air for a space supplied by a HVAC system. In various embodiments, the sensor data includes operating parameters associated with a HVAC system. In various embodiments, the sensor data is current sensor data for a space. Additionally or alternatively, the sensor data may include historical sensor data. At step **1204**, fault prediction and diagnosis system **1112** may generate a model based on past equipment sensor data. For example, fault prediction and diagnosis system **1112** may retrieve historical operating data for a HVAC system and may train a machine learning model using the historical operating data.

[0195] At step **1206**, fault prediction and diagnosis system **1112** may generate one or more functional parameters using a machine learning algorithm. In various embodiments, the machine learning algorithm includes the machine learning model trained in step **1204**. In various embodiments, the machine learning algorithm includes a neural network. In some embodiments, the machine learning algorithm includes a linear regression system. In various embodiments, the one or more functional parameters include operational parameters for a HVAC system. For example, fault prediction and diagnosis system **1112** may generate one or more predicted operational parameters for a HVAC system, the

one or more predicted operational parameters corresponding to a future time period. At step **1208**, fault prediction and diagnosis system **1112** may analyze the one or more functional parameters to generate a fault prediction. For example, fault prediction and diagnosis system **1112** may execute a univariate one class SVM on one or more of the one or more functional parameters. As another example, fault prediction and diagnosis system **1112** may execute a multivariate one class SVM on one or more of the one or more functional parameters. In various embodiments, the fault prediction includes context information. For example, the fault prediction may include a fault type, one or more associated pieces of equipment, a fault category, a fault diagnosis, a root cause, and/or the like.

[0196] At step **1210**, fault prediction and diagnosis system **1112** may compare the fault information associated with the fault prediction to generate a fault diagnosis and fault distinguishing information. In various embodiments, step **1210** includes classifying the predicted fault and/or diagnosing the predicted fault. For example, fault prediction and diagnosis system **1112** may analyze one or more binary determinations generated by a univariate SVM and/or a multivariate SVM to determine a fault diagnosis. In various embodiments, the fault diagnosis includes at least one of (i) no fault, (ii) supply air temperature sensor fault, (iii) other than supply air temperature fault, or (iv) fault of supply air temperature because of other feeding equipment. In some embodiments, the fault distinguishing information includes context information such as a listing of one or more pieces of equipment associated with the fault and/or a root cause of the fault.

[0197] At step **1212**, fault prediction and diagnosis system **1112** may compute cost savings associated with the predicted fault. In various embodiments, step **1212** includes calculating an energy consumption associated with the predicted fault. For example, fault prediction and diagnosis system **1112** may determine an amount of electricity associated with a high temperature fault (e.g., an amount of energy used to heat a space above what would be required to achieve a temperature setpoint for the space, etc.). In various embodiments, fault prediction and diagnosis system **1112** generates the cost savings based on the energy consumption. For example, fault prediction and diagnosis system **1112** may retrieve prices associated with various resources (e.g., utilities such as electricity) and may calculate a cost savings based on the retrieved prices and the energy consumption. In various embodiments, step **1212** includes transmitting an indication of the cost savings to a user (e.g., displaying the cost savings on a user interface, etc.). At step **1214**, fault prediction and diagnosis system **1112** may perform preventative maintenance associated with the predicted fault. For example, fault prediction and diagnosis system **1112** may automatically generate a work order ticket for repair of a piece of equipment indicated as faulty by the predicted fault. In various embodiments, step **1214** includes preventing the predicted fault from occurring.

[0198] Referring now specifically to FIG. 13, method **1300** for generating a predicted fault and cost savings is shown, according to an exemplary embodiment. In various embodiments, system **1100** performs method **1300**. At step **1302**, fault prediction and diagnosis system **1112** may generate a predicted supply air temperature and/or additional functional parameters corresponding to a designated time period using a machine learning model. In various embodi-

ments, step **1302** includes executing a machine learning model on HVAC operational data to generate predicted functional parameters such as a supply air temperature for a designated time period.

[0199] At step **1304**, fault prediction and diagnosis system **1112** may execute at least one of (i) a first univariate one class SVM using the predicted supply air temperature, (ii) a multivariate one class SVM using at least one of the additional functional parameters, or (iii) a second univariate one class SVM using at least one of the additional functional parameters to predict a fault associated with the designated time period. In various embodiments, the predicted fault corresponds to a difference between a predicted operational parameter and a predicted operational parameter setpoint. In various embodiments, the predicted fault includes context information. For example, the predicted fault may include a source of the fault or a cause of the fault.

[0200] At step **1306**, fault prediction and diagnosis system **1112** may compare the predicted supply air temperature to a predicted supply air temperature setpoint to determine whether a fault exists. In various embodiments, step **1306** includes verifying and/or validating the predicted fault. In some embodiments, step **1306** includes classifying the predicted fault. At step **1308**, fault prediction and diagnosis system **1112** may classify the predicted fault based on a result of the first univariate one class SVM and/or the multivariate one class SVM to generate a fault diagnosis. In various embodiments, the fault diagnosis may include a type of fault. In some embodiments, the fault diagnosis includes a source of the fault. In various embodiments, step **1308** includes performing a lookup (e.g., using one or more binary determinations, etc.) in a table as described above in reference to analysis circuit **1130**. At step **1310**, fault prediction and diagnosis system **1112** may calculate a cost savings associated with the predicted fault. In various embodiments, step **1310** includes calculating an energy consumption associated with the predicted fault. In some embodiments, step **1310** includes transmitting an indication of the calculated cost savings to a user. Additionally or alternatively, fault prediction and diagnosis system **1112** may perform one or more actions. For example, fault prediction and diagnosis system **1112** may compare the cost savings to a threshold (e.g., associated with a cost of preventing the predicted fault, etc.) and may take one or more actions based on the comparison (e.g., automatically generate a ticket to fix the predicted fault if doing so would save money or preemptively silence a number of alarms associated with the predicted fault if attempting to fix the predicted fault would cost more than allowing the fault to occur, etc.). In various embodiments, a work order ticket may include instructions for an individual to repair and/or otherwise interact with a piece of equipment to achieve a desired outcome. For example, a work order ticket may be submitted to a work order ticketing system that instructs maintenance personnel to repair a defective blower fan in a HVAC system to solve a low zone temperature fault.

[0201] Referring now to FIG. 14A, diagram **1410** illustrating predicted supply air temperature is shown, according to an exemplary embodiment. In various embodiments, system **1100** generates diagram **1410**. For example, fault prediction and diagnosis system **1112** may generate one or more predicted operational parameters such as predicted supply air temperature corresponding to a period of time. Diagram **1410** is shown to include actual (measured) supply

air temperature **1412** and predicted supply air temperature **1414**. In various embodiments, actual supply air temperature **1412** corresponds to a measured supply air temperature within a space and/or building. In various embodiments, predicted supply air temperature **1414** corresponds to a predicted supply air temperature generated by fault prediction and diagnosis system **1112**. In various embodiments, fault prediction and diagnosis system **1112** may execute one or more machine learning model on predicted supply air temperature **1414** to generate a predicted fault. For example, FIG. **14B** illustrates diagram **1410** having actual supply air temperature **1422** and predicted supply air temperature fault **1424**, according to an exemplary embodiment. In various embodiments, predicted supply air temperature fault **1424** corresponds to predicted supply air temperatures that are associated with a fault (e.g., a deviation from a predicted setpoint, etc.).

Configuration of Exemplary Embodiments

[0202] The construction and arrangement of the systems and methods as shown in the various exemplary embodiments are illustrative only. Although only a few embodiments have been described in detail in this disclosure, many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.). For example, the position of elements can be reversed or otherwise varied, and the nature or number of discrete elements or positions can be altered or varied. Accordingly, all such modifications are intended to be included within the scope of the present disclosure. The order or sequence of any process or method steps can be varied or re-sequenced according to alternative embodiments. Other substitutions, modifications, changes, and omissions can be made in the design, operating conditions and arrangement of the exemplary embodiments without departing from the scope of the present disclosure.

[0203] The present disclosure contemplates methods, systems and program products on any machine-readable media for accomplishing various operations. The embodiments of the present disclosure can be implemented using existing computer processors, or by a special purpose computer processor for an appropriate system, incorporated for this or another purpose, or by a hardwired system. Embodiments within the scope of the present disclosure include program products comprising machine-readable media for carrying or having machine-executable instructions or data structures stored thereon. Such machine-readable media can be any available media that can be accessed by a general purpose or special purpose computer or other machine with a processor. By way of example, such machine-readable media can comprise RAM, ROM, EPROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code in the form of machine-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer or other machine with a processor. Combinations of the above are also included within the scope of machine-readable media. Machine-executable instructions include, for example, instructions and data which cause a general-purpose computer, special purpose computer, or special purpose processing machines to perform a certain operation or group of operations.

[0204] Although the figures show a specific order of method steps, the order of the steps may differ from what is depicted. Also, two or more steps can be performed concurrently or with partial concurrence. Such variation will depend on the software and hardware systems chosen and on designer choice. All such variations are within the scope of the disclosure. Likewise, software implementations could be accomplished with standard programming techniques with rule-based logic and other logic to accomplish the various connection steps, processing steps, comparison steps and decision steps.

What is claimed is:

1. A fault prediction system for a building or space, comprising a processing circuit including a processor and memory, the memory having instructions stored thereon that, when executed by the processor, cause the processing circuit to:

receive data relating to a plurality of components, the data indicating performance of the plurality of components; generate, based on the received data, a univariate prediction model and a multivariate prediction model; generate, using the received data, one or more predicted operational parameters for the plurality of components corresponding to a future time period; and

execute at least one of the univariate prediction model or the multivariate prediction model on the one or more predicted operational parameters to predict a fault associated with at least one of the plurality of components to occur during the future time period.

2. The fault prediction system of claim 1, wherein the instructions further cause the processing circuit to calculate an amount of energy consumption associated with the predicted fault based at least in part on the one or more predicted operational parameters.

3. The fault prediction system of claim 2, wherein the instructions further cause the processing circuit to calculate a cost savings associated with the predicted fault based on the amount of energy consumption calculated.

4. The fault prediction system of claim 3, wherein the instructions further cause the processing circuit to display the predicted fault to a user based on a calculated cost savings associated with the predicted fault, where predicted faults with high cost savings are emphasized over predicted faults with low cost savings.

5. The fault prediction system of claim 1, wherein the instructions further cause the processing circuit to perform an action comprising at least one of (i) generate a suggestion for a user regarding how to prevent the predicted fault, the suggestion including a cause of the predicted fault or (ii) automatically generate a work order ticket including an indication of a piece of equipment associated with the predicted fault.

6. The fault prediction system of claim 5, wherein the instructions further cause the processing circuit to compare a cost associated with the action to a cost associated with the predicted fault to determine whether to generate the suggestion or automatically generate the work order ticket.

7. The fault prediction system of claim 1, wherein the instructions further cause the processing circuit to populate a fault diagnosis matrix using the univariate prediction model and the multivariate prediction model and map one or more entries of the fault diagnosis matrix to a fault diagnosis to determine a cause of the predicted fault.

8. The fault prediction system of claim 1, wherein the instructions further cause the processing circuit to compare a predicted setpoint of the one or more predicted operational parameters to a predicted operational parameter associated with the predicted setpoint of the one or more predicted operational parameters to classify the predicted fault as at least one of a high zone temperature fault or a low zone temperature fault.

9. One or more non-transitory computer-readable storage mediums having instructions stored thereon that, when executed by a processor, cause the processor to:

- receive data relating to a plurality of components associated with operations in a building or space, the data indicating performance of the plurality of components, the components being energy consuming components;
- generate, based on the received data, a univariate prediction model and a multivariate prediction model;
- generate, using the received data, one or more predicted operational parameters for the plurality of components corresponding to a future time period; and
- execute at least one of the univariate prediction model or the multivariate prediction model on the one or more predicted operational parameters to predict a fault associated with at least one of the plurality of components to occur during the future time period.

10. The one or more non-transitory computer-readable storage mediums of claim 9, wherein the instructions further cause the processor to calculate an amount of energy consumption associated with the predicted fault based at least in part on the one or more predicted operational parameters, wherein the data comprises usage time, efficiency metrics, or input and output quantities.

11. The one or more non-transitory computer-readable storage mediums of claim 10, wherein the instructions further cause the processor to calculate a cost savings associated with the predicted fault based on the amount of energy consumption calculated.

12. The one or more non-transitory computer-readable storage mediums of claim 11, wherein the instructions further cause the processor to display the predicted fault based on a calculated cost savings associated with the predicted fault, where predicted faults with high cost savings are emphasized over predicted faults with low cost savings.

13. The one or more non-transitory computer-readable storage mediums of claim 9, wherein the instructions further cause the processor to perform an action comprising at least one of (i) generate a suggestion for a user regarding how to prevent the predicted fault, the suggestion including a cause of the predicted fault or (ii) automatically generate a work order ticket including an indication of a piece of equipment associated with the predicted fault.

14. The one or more non-transitory computer-readable storage mediums of claim 13, wherein the instructions further cause the processor to compare a cost associated with the action to a cost associated with the predicted fault to determine whether to generate the suggestion or automatically generate the work order ticket.

15. The one or more non-transitory computer-readable storage mediums of claim 9, wherein the instructions further cause the processor to populate a fault diagnosis matrix

using the univariate prediction model and the multivariate prediction model and map one or more entries of the fault diagnosis matrix to a fault diagnosis to determine a cause of the predicted fault.

16. The one or more non-transitory computer-readable storage mediums of claim 9, wherein the instructions further cause the processor to compare a predicted setpoint of the one or more predicted operational parameters to a predicted operational parameter associated with the predicted setpoint of the one or more predicted operational parameters to classify the predicted fault as at least one of a high zone temperature fault or a low zone temperature fault.

17. A building management system (BMS), comprising a processing circuit including a processor and memory, the memory having instructions stored thereon that, when executed by the processor, cause the processing circuit to:

- receive data relating to a plurality of components, the data indicating performance of the plurality of components;
- generate, based on the received data, a univariate prediction model and a multivariate prediction model;
- generate, using the received data, one or more predicted operational parameters for the plurality of components corresponding to a future time period;

- execute at least one of the univariate prediction model or the multivariate prediction model on the one or more predicted operational parameters to predict a fault associated with at least one of the plurality of components to occur during the future time period;

- calculate an amount of energy consumption associated with the predicted fault based at least in part on the one or more predicted operational parameters; and

- perform an action comprising at least one of (i) generate a suggestion for a user regarding how to prevent the predicted fault, the suggestion including a cause of the predicted fault or (ii) automatically generate a work order ticket including an indication of a piece of equipment associated with the predicted fault.

18. The building management system (BMS) of claim 17, wherein the instructions further cause the processing circuit to calculate a cost savings associated with the predicted fault based on the amount of energy consumption calculated, wherein the data comprises shutdown data.

19. The building management system (BMS) of claim 18, wherein the instructions further cause the processing circuit to display the predicted fault based on a calculated cost savings associated with the predicted fault, where predicted faults with high cost savings are emphasized over predicted faults with low cost savings.

20. The building management system (BMS) of claim 17, wherein the data includes chiller data indicating performance of a plurality of chillers, wherein the univariate prediction model and the multivariate prediction model include a single chiller prediction model and a cluster chiller prediction model, and wherein predicting the fault includes predicting a chiller fault with at least one of the single chiller prediction model or the cluster chiller prediction model.

* * * * *